

MOLTEN SALT REACTOR MICROSCOPIC DELAYED NEUTRON PRECURSOR GROUP
FORMULATION AND EVALUATION

BY

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DISSERTATION

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Abstract

Fluid-fueled molten salt reactors (MSRs) have garnered interest since the Generation IV International Forum in the early 2000s, where the MSR was designated as one of six advanced reactors of interest. A unique consideration in MSRs are delayed neutron precursors (DNPs), which move from areas of high importance in the core to areas of lower importance in the core and ex-core regions. This causes the effective delayed neutron fraction to decrease, leading to changes in reactivity based on the salt flow rate. DNPs are important for modeling and simulation of MSRs.

Many studies have investigated how the effective delayed neutron fraction changes based on the movement of fuel salt in MSRs. However, these studies do not account for the effect of fuel salt movement and reprocessing on the DNP group parameters used in the transient simulations. The effect of neglecting these MSR-specific phenomena is reduced accuracy in the DNP group parameters in MSR simulations.

These phenomena are important to consider, as simulations use the DNP group parameters to enable safety, security, and safeguards by design. By using DNP group parameters with greater accuracy, models can more accurately reflect physical phenomena. Having greater accuracy in predictive models is key for developing an advanced reactor pipeline.

The DNP group parameters for MSRs are resolved using the microscopic, also known as aggregate or summation, method. This method entails using individual DNP data to reconstruct the DNP groups by summing them together in simulated irradiation experiments. Pulse irradiation experiments occur at very short time scales relative to MSR physical phenomena and are used to generate the short lived DNP group parameters. Saturation, or infinite, irradiation experiments occur at time scales several times longer than the longest DNP half-life and are used to generate the longer lived DNP group parameters. Because the saturation irradiation lasts on the order of minutes, the thermal-hydraulic and chemical effects in MSRs have a non-negligible impact. Therefore, the longer lived DNP group parameters will vary, while the shorter lived DNP group parameters will remain constant.

A non-linear least squares solver is used to generate an improved set of DNP group parameters. Verification is conducted via comparison with total delayed neutron yields and average half-lives from the literature.

The methodology used to generate these parameters can be replicated for other MSR conditions, enabling the formation of more accurate sets of DNP group parameters for MSRs. This improves the accuracy of MSR simulations, which is important for safety, security, and safeguards by design as well as for developing an advanced reactor pipeline. Proposed work includes the following: development and evaluation of a more sophisticated model for generating the DNP group parameters, validation via comparison with the Molten Salt Reactor Experiment (MSRE) during pump start-up and coast-down, generation of DNP group spectra, evaluation of various safety, security, and safeguards related transients using improved DNP group parameters, and evaluation of various datasets on DNP group parameters.

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Table of contents

Chapter 1	Introduction	1
Chapter 2	Background and Motivation	5
Chapter 3	Methodology	36
Chapter 4	Preliminary Results	51
Chapter 5	Proposed Work	71
Appendix A	Derivation of the Delayed Neutron Count Rate in the 0D Flow Model	80
Appendix B	Derivation of the Equilibrium Concentration from the Cumulative Fission Yield	83

Chapter 1

Introduction

One of the generation IV reactor designs is the liquid fueled molten salt reactor, or fluid-fueled molten salt reactor (MSR) [89]. MSRs are advanced reactors which are intended to improve safety and efficiency over previous reactor designs. MSRs, instead of a solid fuel, use a liquid fuel composed of a molten salt mixed with some fissile nuclide, such as ^{235}U . There are a wide variety of benefits to MSRs: high operating temperature, low operating pressure, and high thermal conductivity and capacity of the salts. Other benefits include a high coefficient of thermal expansion for negative temperature coefficient of reactivity, high resource utilization through higher burn-up, reduced cost of fabricating and transporting fuel elements, and the removal of fission products to improve the neutron economy [163].

MSRs have several barriers to deployment which must be addressed before they are commercialized. Corrosion and material degradation is an issue for MSRs, as the high temperature salts are highly corrosive. Another barrier to deployment is in reactor safety, security, and safeguards by design, as MSRs have distinct designs compared to traditional light water reactors. To properly design MSRs, we must have models which are able to simulate them correctly. One specific phenomena which is of critical importance in MSRs are the delayed neutron precursors (DNPs).

DNPs are nuclides which decay via β^- decay, producing a daughter, called an emitter, which emits a neutron [11]. Properly modeling the DNPs in reactors is important, otherwise simulations may predict incorrect power responses, reactivity worth, and reactor controllability [43, 44, 95]. The DNPs are typically not modeled explicitly in simulations, but are instead represented using DNP group parameters, which are intended to have the same physical effect at a much lower computational cost. A brief historical timeline of DNP group parameter modeling advancements is given in Tables 1.1 and 1.2.

The progress in DNP modeling has progressed rapidly thanks to experimental data available for individual DNP, which allowed for more accurate group parameter formulation. However, these formulations of DNP group parameters do not account for certain physics unique to MSR. Although there has also been much analysis on DNPs in MSRs, that work has also not investigated this gap [197, 201, 99, 170, 94]. Most of these studies investigate movement of DNPs during transients, not how the fuel movement and chemical reprocessing affects the formulation of DNP group parameters. However, to have accurate transient models, it is necessary that the DNP group parameters are accurate. Thus, a new step forward for group parameter development is for DNP group parameters to accurately reflect the physical phenomena in MSRs.

This physical phenomena can be incorporated at varying levels of fidelity, with associated differences in cost and accuracy. The least accurate and fastest is a 0D scaled model. This model neglects time dependence and

Table 1.1: Timeline of advancements in DNP modeling up to 1983.

1939	Roberts et al. discovers delayed neutrons [145]; Bohr and Wheeler explains delayed neutrons [21].
1946	Burgy et al. conducts initial spectral measurements [31].
1947	Hughes et al. initial use of six DNP groups at Argonne heavy water pile [76].
1957	Keepin et al. generates improved DNP group parameters [87, 191].
1965	Amiel identifies seven DNPs that account for 80% of delayed neutrons [6].
1969	Marmol compiles experimental data on emission probability and half-life for 34 DNPs [45].
1972	Tomlinson maps 38 DNPs onto six groups [181].
1977	Saphier et al. constructs delayed neutron spectra from 20 DNPs [160].
1982	Rudstam generates DNP group parameters and spectra using 67 DNPs based on half-lives [152].
1983	England et al. generates DNP group parameters and spectra using 105 DNPs based on half-lives [53].

decay chains while scaling the flux and chemical removal rates based on the salt fractions in different regions of the primary loop. The intermediate approach is a 0D flow model. This model accounts for time dependence and decay chains, while applying the flux and chemical removal rates only in the relevant parts of the primary loop. Finally, the most accurate and slowest approach is a spatially resolved model. This model accounts for time and space dependence over the entire primary loop.

The preliminary results of this work cover important DNPs, sensitivity studies of the DNP group parameters, and verification of the 0D scaled model. The results show that the total delayed neutron yield and average half-life values are within uncertainty bounds between the raw DNP data and the DNP group parameters. This verifies the

Table 1.2: Timeline of advancements in DNP modeling from 1983 onward.

1983	England et al. generates DNP group parameters and spectra using 105 DNPs based on half-lives [53].
1988	Brady generates DNP group parameters and spectra using 271 DNPs using least squares [25]. Group parameters in ENDF/B-VI.8 set and remain unchanged through ENDF/B-VIII.0 [192, 29, 151, 23, 52, 24].
1999	Parish et al. states DNP group parameters in Evaluated Nuclear Data File (ENDF) are in need of revision [126].
2014	International Atomic Energy Agency (IAEA) Coordinated Research Project (CRP) on Development of a Reference Database for β -delayed neutron emission data [117].
2020	Leconte et al. uses more than 400 DNPs to combine data with macroscopic experimental measurements [98].
2025	This study develops DNP group parameters for use in MSRs.

process of turning DNP data into DNP group parameters. The sensitivity studies reveal that each DNP can multiple groups, not only the ones that are closest to them. Finally, the verification with results from the literature reveals a main weakness of the 0D scaled model: it does not include pulse irradiation data. This results in a poor set of DNP group parameters for short-lived groups.

The proposed work starts with development of the 0D flow model, which is followed by a verification stage to ensure the improved DNP group parameters are correct for a static case. This is then followed by validation of the model with a transient simulation of the Molten Salt Reactor Experiment (MSRE) with experimental data to ensure the improved DNP group parameters are physically accurate. The improved group spectra are then included in the formulation of the DNP group parameters, and verified for a static case. Finally, transients of the MSRE for various safety, security, and safeguards-related scenarios are simulated using the improved DNP group parameters and pre-existing group parameters to study their impact. This work is considered the critical work for a full analysis, though there is also non-critical work presented in Section 5.4. The non-critical work includes development of

the spatially resolved model, uncertainty tracking of MSR physical phenomena in the DNP group parameters, and investigation into the effects of varying the incident neutron energy spectrum on the DNP group parameters.

Chapter 2

Background and Motivation

This chapter provides the background and motivation for generation of more accurate DNP group parameters for MSRs and implementation of those parameters for safety, security, and safeguards design purposes. The background information includes information on DNPs and MSR fuel depletion, and the motivation includes safety, security, and safeguards by design. The DNP group parameters in MSRs differ because fuel circulates and undergoes online chemical reprocessing. These effects can be captured in spatially resolved MSR fuel depletion simulations. The existing methodology for DNP group parameter generation does not account for the physical phenomena present in MSR fuel depletion [87, 182, 110, 14, 189, 25, 76, 107, 108, 134, 42, 134, 154, 192, 23, 98]. More accurate DNP group parameters are generated by accounting for these physical phenomena, which is particularly important for safety, security, and safeguards related concerns.

2.1 Delayed Neutron Precursors

Before delving deeper into the DNP group parameters, it is useful to first understand DNPs conceptually. DNPs are nuclides which decay via β^- decay, producing a daughter, called an emitter, which emits a neutron [11]. DNPs are an important consideration in nuclear reactors [43]. In this work, the term DNP will be solely used rather than referring to emitters, as the delay of the neutron emission is effectively only due to the β decay [21, 125]. Classifying between the emitter and the DNP is not relevant for this work.

The delay in neutron emission significantly affects the reactor period and transient behavior. Equation (2.1) shows the time-dependent rate of delayed neutron emission from a single DNP as:

$$\dot{n}_{d,i}(t) = P_{n,i} \lambda_i N_i(t), \quad (2.1)$$

where,

$\dot{n}_{d,i}$ = the delayed neutron activity, or count rate, from DNP i [$n \cdot s^{-1}$]

$P_{n,i}$ = the average number of delayed neutrons emitted from
decay of nuclide i , or the emission probability [-]

λ_i = the decay constant of DNP i [s^{-1}]

$N_i(t)$ = the number of DNP i [*atoms*].

In MSRs, the movement of the fuel salt necessitates spatial resolution for tracking DNP distributions and concentrations. In a solid fuel reactor, such as a light water reactor (LWR), spatial resolution is not necessary to properly model the DNPs, as the fuel remains stationary. Modeling space and time together largely increases computational cost, making problems involving DNPs in MSRs more challenging than handling DNPs in stationary fuels.

Properly modeling the DNPs in reactors is important, otherwise simulations may predict incorrect power responses, reactivity worth, and reactor controllability [43, 44, 95]. Section 2.1.3 shows how the power response and reactivity worth depend on the DNPs. The reactor controllability varies based on the available number of DNPs. If there are fewer delayed neutrons available due to movement of the fuel salt, such as in a MSR, then the reactor period becomes shorter and makes the reactor harder to control.

2.1.1 Measurement

Macroscopic and microscopic approaches are used for measuring DNPs [126, 154]. The macroscopic approach, discussed in detail in Section 2.1.1, uses a fissile sample irradiation to get the delayed neutron count rate, which can be approximated using groups. Conversely, the microscopic approach, discussed in detail in Section 2.1.1, uses the data from many experiments combined together to simulate a fissile sample irradiation and generate the delayed neutron count rate. The microscopic approach is more challenging, as it depends on the work of many experimentalists. However, it offers more detail on the delayed neutron count rate.

Macroscopic Measurement of DNPs

The macroscopic approach entails irradiating a sample containing a single fissile nuclide at or near a given neutron energy. This delayed neutron count rate from this sample can then be measured and used to formulate DNP group parameters. Analysis and collection of macroscopic experiment data is common in the literature [87, 182, 110, 14, 189, 25, 76, 107, 108, 40].

Small samples are used to minimize neutron multiplication within the sample to avoid spectral effects [87]. The spectral effects are due to the differing neutron energy spectra between each DNP group and the incident irradiating neutrons [87]. Another reason for using a small sample is because of physical challenges associated with the rabbit system [30].

Two different types of irradiation are used: pulse and infinite. Pulse irradiation has an irradiation time shorter than the shortest half-life, and it captures the shorter lived DNPs with more accuracy than the longer lived DNPs [25]. Infinite irradiation has an irradiation time longer than several times the longest half-life, and it captures the longer lived DNPs with more accuracy than the shorter lived DNPs [25]. After irradiation, the delayed neutrons from the sample are counted as a function of time. The delayed neutron count rate as a function of time after irradiation, $n_d(t)$, is given as:

$$\dot{n}_d(t) = \epsilon \sum_{i=1}^I P_{n,i} \lambda_i N_i(t), \quad (2.2)$$

where,

$\dot{n}_d(t)$ = the delayed neutron count rate [$n \cdot s^{-1}$]

I = the total number of DNPs [-]

ϵ = the detector efficiency [-].

Once this delayed neutron count rate is generated, the data is used to generate DNP group and kinetics parameters for that fissile nuclide and neutron energy. Three different approaches have been used to generate DNP group parameters from macroscopic data: exponential stripping (or peeling) [30, 86], least squares [25, 87], and using half-lives of specific DNPs [160]. Some approaches have even used a combination, such as least squares fitting with specific half-lives [173]. The exponential stripping, or peeling, method is essentially a more subjective form of least squares fitting, where exponential curves are subtracted away from the raw data from right to left [138]. The use of specific half-lives to form the DNP groups is less subjective and offers a more physical interpretation of the groups. The least squares approaches are the most objective approach and generate groups which physically do not embody a single DNP. Instead, they group together the DNPs and effectively capture the overall physical effect, minimizing the objective function. For this work, the least squares approach is used, as it offers an objective formulation that can be generically applied. The specifics of the least squares approach used to form the DNP parameters is discussed in Section 2.1.2. Figure 2.1 (recreated from [87]) shows an experimental setup for collecting delayed neutrons.

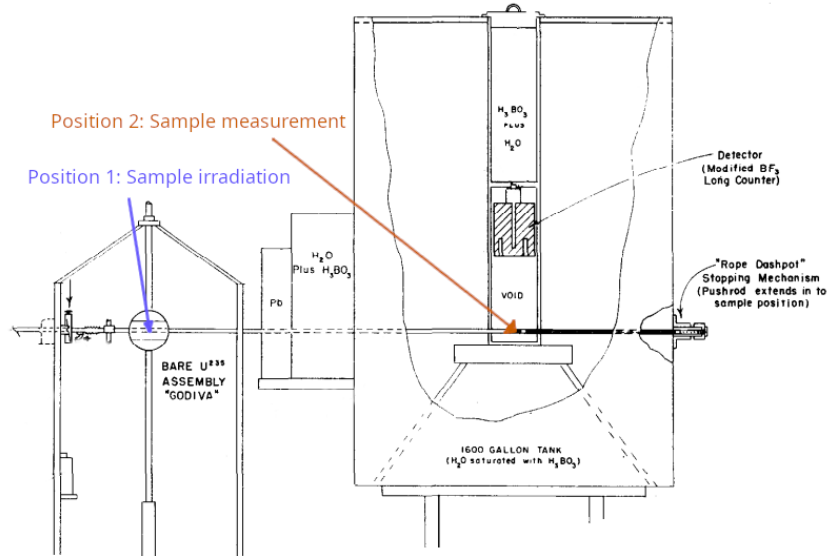


Figure 2.1: Simplified experimental setup for measuring delayed neutron emission from irradiation of a 3 gram sample in Godiva referenced from [87].

Challenges associated collecting this experimental data [11] include experimental noise and signal obfuscation

due to moderation. Experimental noise becomes significant after long periods of time relative to the half-life of the longest-lived DNP, around 1 minute from ^{87}Br , as the delayed neutron counts decrease. Additionally, the short lived groups cannot be measured directly due to moderated neutrons from irradiation, so there is a gap between irradiation and when the sample can be moved to a delayed neutron counter. Even after a pulse irradiation, the long-lived groups form a background that interfere with measurement of the short-lived groups [11]. Due to these effects, measurement of delayed neutrons from the sample is less accurate initially, becomes increasingly accurate, and then becomes less accurate again as the delayed neutron counts decrease.

Similarly, uncertainty arises from the measurement of the number of fissions. Although the number of fission events can be measured somewhat accurately using a fission chamber after irradiation or other techniques, the fission chamber does not account for source multiplication within the irradiated sample. Keepin et al. suggests a technique to correct for the sample self-multiplication. The sample self-multiplication require a correction factor of up to 3% in the number of fissions.

By the year 2000, there were already 245 individual sets of DNP group parameters for 20 different fissile nuclides [172]. Spriggs and Campbell go into a high level of detail, discussing the various analysis methods, neutron energies, fissile nuclides, counting intervals, and sample compositions for many of these DNP group parameter formulations [172]. Tuttle also provides a detailed review of many macroscopic DNP group parameters pre-1974, including details on the sample, reactor, counter, and travel time of the sample between irradiation and measurement [182]. Tables 2.1 through 2.4 show some macroscopic DNP group parameters from the literature for thermal fission of ^{235}U [126].

Table 2.1: Macroscopic formulation of group yields from thermal fission of ^{235}U (recreated from [126]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [87]	0.00052	0.00346	0.00310	0.00624	0.00182	0.00066
Charlton et al. [36]*	0.00054	0.00344	0.00319	0.00607	0.00186	0.00066
Saleh et al. [158]	0.00055	0.00363	0.00297	0.00593	0.00168	0.00063
Synetos and Williams [175]*	0.00051	0.00328	0.00214	0.00583	0.00270	0.00063
Tuttle [182]	0.00065	0.00362	0.00320	0.00692	0.00218	0.00044
Waldo et al.[189]	0.00055	0.00366	0.00327	0.00660	0.00192	0.00070

* Modified to include a sixth group by Parish et al. [126].

Table 2.2: Macroscopic formulation of group yield uncertainties from thermal fission of ^{235}U (recreated from [126]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [87]	0.00006	0.00025	0.00041	0.00047	0.00019	0.00013
Charlton et al. [36]*	0.00003	0.00005	0.00024	0.00013	0.00014	0.00014
Saleh et al. [158]	0.00005	0.00032	0.00027	0.00104	0.00015	0.00012
Synetos and Williams [175]*	0.00004	0.00020	0.00036	0.00037	0.00029	0.00012
Tuttle [182]	0.00007	0.00013	0.00041	0.00019	0.00021	0.00007
Waldo et al.[189]	0.00006	0.00041	0.00041	0.00038	0.00017	0.00009

* Modified to include a sixth group by Parish et al. [126].

There is reasonable agreement between the different DNP group parameters, though there is variation among

Table 2.3: Macroscopic formulation of group half-lives from thermal fission of ^{235}U (recreated from [126]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [87]	55.89897	22.72614	6.24457	2.30281	0.60802	0.23028
Charlton et al. [36]*	55.45177	22.43195	6.02737	2.18658	0.49159	0.23028
Saleh et al. [158]	55.45177	22.65187	6.24457	2.31049	0.63013	0.23028
Synetos and Williams [175]*	56.35343	22.35959	7.00149	2.72893	0.77016	0.23028
Tuttle [182]	54.57852	21.86584	6.02737	2.22877	0.49511	0.17911
Waldo et al.[189]	54.57852	21.86584	6.02737	2.22877	0.49511	0.17911

* Modified to include a sixth group by Parish et al. [126].

Table 2.4: Macroscopic formulation of group half-life uncertainties from thermal fission of ^{235}U (recreated from [126]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [87]	1.35239	0.67061	0.22503	0.09181	0.08000	0.02219
Charlton et al. [36]*	0.44361	0.50817	0.10482	0.06208	0.03138	0.02219
Saleh et al. [158]	3.99253	0.14805	0.39380	0.03851	0.01719	0.02219
Synetos and Williams [175]*	0.45816	0.28851	0.77794	0.21488	0.09413	0.02219
Tuttle [182]	1.28926	0.82773	0.20965	0.08600	0.04244	0.02545
Waldo et al.[189]	1.28926	0.82773	0.20965	0.08600	0.04244	0.02545

* Modified to include a sixth group by Parish et al. [126].

them. Not only did each approach use a slightly different experimental methodology via different energy spectra, data collection methods, sample transfer time between irradiation and measurement, sample size, and sample composition, but there are also differences in how the data was processed. There are several ways that the data was processed from delayed neutron count rates into DNP group parameters among the works in these tables: simultaneous non-linear least squares fitting of saturation and pulse irradiation data, separate non-linear least squares fitting of pulse and saturation irradiation data and subsequent combination, non-linear least squares of an intermediate amount of time between a pulse and saturation, and exponential stripping.

Keepin et al. used both a simultaneous fitting of DNP group parameters from pulse and saturation irradiation data as well as a separate fitting of four groups from each irradiation. The separate fitting of four groups were normalized at the second group, and Keepin et al. reports that this method gave better results than the simultaneous fit [87].

Charlton et al. use the equation for an intermediate amount of time that is not infinite or a pulse, as shown in Equation (2.5). They also only use five groups, as the sample took half a second to move from irradiation to the detector where delayed neutron counts could be measured. The shortest lived group six DNPs were not included, but Parish et al. added group six values for their own analysis [126]. Charlton et al. use a least squares solve to determine the DNP group parameters, according to Spriggs and Campbell [172].

Saleh et al. use the equation for an intermediate amount of time, similar to Charlton et al., where the equation is described in Equation (2.5). However, instead of using a least squares solve, they use exponential stripping [158].

Synetos and Williams use the same approach as Charlton et al. [175, 173].

Tuttle formulates suggested values based on reviewing the literature and applying empirical correlations [182].

Waldo et al. apply a transit time correction using an empirical correlation, and the DNP group parameters are fitted using a least squares solve [189]. However, it is not made clear if pulse or saturation irradiations are used. Because the conclusions indicate that the DNP group parameters are less applicable for short times or short irradiations, this implies that a saturation irradiation was used.

Microscopic Measurement of DNPs

The other approach used to measure DNPs is the microscopic – also referred to in the literature as the aggregate or summation – approach, which uses data from individual nuclides summed together to form a simulated delayed neutron count rate. This count rate can then be used to generate DNP group parameters, the same as in the count rate from the macroscopic method.

This approach was investigated in detail in the 1990-2000's [134, 42, 154, 192, 23], with progress through the 2010's with more data [117, 79]. Leconte et al. used this approach as recently as 2020 [98]. This approach requires experiments on individual DNP nuclides, from which emission probabilities and half-lives can be measured. Typically, this involves rapid chemical extraction or isotope separation and magnetic tape to rapidly measure short lived nuclides, such as in the study by Björnstad et al. where an aluminized mylar tape is used [19]. The magnetic tape is used to transport samples or remove longer-lived decay products [167, 155]. Experimental methods include the neutron- β coincidence technique, separate counting of neutrons and β particles, separate counting of neutrons and γ rays, ion counting, fission yield from a set number of fissions, gamma counting of a long-lived daughter, and normalization of the neutron counter using a known precursor [155].

Other experimental methods used to study DNPs include on-line mass spectrometry, helium-3 spectrometry, proton-recoil detectors, the time-of-flight method, and neutron scintillators [43]. On-line mass spectrometry combined with a helium-3 ionization chamber and a proton-recoil detector allows for measurement of the delayed neutron energy spectrum [166, 18]. The time-of-flight technique consists of measuring the flight time of a neutron over a known distance to calculate the velocity, including a neutron scintillator [18, 39, 63, 38].

This microscopic measurement data is exceptionally useful, as it allows for treatment of DNPs on a nuclide-by-nuclide basis. This is particularly important for MSRs, where the spatial DNP concentration depends on half-lives, yields, thermo-chemistry, and thermal-hydraulics. With data on individual nuclides, higher fidelity treatment of DNPs becomes possible.

There have been many new measurements in the past several decades on these nuclides. The IAEA has compiled and evaluated this data in a publicly accessible form [49, 48]. This data, when combined with cumulative fission yields and neutronics data, can be used to model the DNPs explicitly rather than using groups. This is particularly useful for understanding what nuclides have the greatest impact, as well as for simulating experimental configurations which have yet to be thoroughly investigated for DNP group fitting.

Tables 2.5 through 2.8 show some microscopic DNP group parameters from the literature for thermal fission of ^{235}U [126]. There is reasonable agreement between the different DNP group parameters, though it should be noted that Brady and England may be closer to ENDF/B-VI.8 because ENDF/B-VI.8 uses the Brady and England results in its formulation.

ENDF/B-VI.8 pulls from Brady and England, as well as other works by England [151, 23, 52, 24]. This work and Brady and England's approach entails collecting 271 DNPs, generating a delayed neutron count rate curve, and then using the two four group aggregation approach described by Keepin et al. [87, 23].

Mills et al. use two different approaches: fitting all parameters simultaneously as described by Keepin et al. and using the Keepin et al. half-lives while fitting only the yields [118, 87]. The values presented here are from the approach where all parameters are fit simultaneously for all DNP group parameters.

Table 2.5: Microscopic formulation of group yields from thermal fission of ^{235}U (recreated from [126]).

Source	Group					
	1	2	3	4	5	6
ENDF/B-VI.8 [151]	0.00058	0.00302	0.00288	0.00646	0.00265	0.00111
Mills et al. [118]	0.00056	0.00325	0.00196	0.00658	0.00287	0.00122
Brady and England [23]	0.00068	0.00341	0.00292	0.00611	0.00310	0.00158

* Modified to include a sixth group by Parish et al. [126].

Table 2.6: Microscopic formulation of group yield uncertainties from thermal fission of ^{235}U (recreated from [126]).

Source	Group					
	1	2	3	4	5	6
ENDF/B-VI.8 [151]	-	-	-	-	-	-
Mills et al. [118]	0.00004	0.00023	0.00020	0.00051	0.00035	0.00012
Brady and England [23]	0.00004	0.00019	0.00016	0.00034	0.00017	0.00009

* Modified to include a sixth group by Parish et al. [126].

Table 2.7: Microscopic formulation of group half-lives from thermal fission of ^{235}U (recreated from [126]).

Source	Group					
	1	2	3	4	5	6
ENDF/B-VI.8 [151]	51.72740	21.19716	5.73797	2.28913	0.81595	0.24295
Mills et al. [118]	55.27489	22.80089	7.67860	2.77148	1.07381	0.28177
Brady and England [23]	52.11633	21.32761	5.68620	2.18727	0.70114	0.23462

* Modified to include a sixth group by Parish et al. [126].

Table 2.8: Microscopic formulation of group half-life uncertainties from thermal fission of ^{235}U (recreated from [126]).

Source	Group					
	1	2	3	4	5	6
ENDF/B-VI.8 [151]	-	-	-	-	-	-
Mills et al. [118]	0.09918	0.11723	0.37861	0.11259	0.08521	0.01020
Brady and England [23]	-	-	-	-	-	-

* Modified to include a sixth group by Parish et al. [126].

Comparison of Macroscopic and Microscopic Approaches

Huynh and Jouanne offer a very useful comparison between the macroscopic and microscopic approaches via different datasets [79]. Specifically, they compare the total delayed neutron yield from various data sources from macroscopic measurements to the calculated total delayed neutron yield from microscopic measurements. This data can also be compared to the previously shown tables, from which the total delayed neutron yield can be calculated. The total data compilation is presented in Table 2.9.

Table 2.9 shows that the total delayed neutron yield for thermal fission of ^{235}U ranges from 0.01477 to 0.01905

Approach	Data	Delayed Yield
Microscopic	JEF-2.2 [11]	1.708 ± 0.12
Microscopic	JEF-3.1.1 [159]	1.477 ± 0.08
Microscopic	ENDF/B-VII.1 [34]	1.905 ± 0.10
Microscopic	ENDF/B-VI.8 [151]	1.670
Microscopic	Mills et al. [118]	1.645 ± 0.07
Microscopic	Brady and England [23]	1.780 ± 0.05
Macroscopic	JEF-2.2 [11]	1.655
Macroscopic	JEF-3.1.1 [159]	1.620
Macroscopic	ENDF/B-VII.0 [33]	1.585
Macroscopic	Keepin et al. [87]	1.580 ± 0.07
Macroscopic	Charlton et al. [36]	1.577 ± 0.03
Macroscopic	Saleh et al. [158]	1.645 ± 0.11
Macroscopic	Synetos and Williams [175]	1.510 ± 0.06
Macroscopic	Tuttle [182]	1.700 ± 0.05
Macroscopic	Waldo et al.[189]	1.670 ± 0.07

Table 2.9: Total delayed neutron yields of 100 fissions from thermal fission of ^{235}U (based on [79]).

(or 100 times that for 100 fissions, which makes the data more legible). The microscopic values range from 1.477 to 1.905, capturing the minimum and maximum of the range of values seen in the table. The macroscopic values range from 1.510 to 1.700, presenting a tighter range of values. Another point of interest is that there does not appear to be a consistent difference between the microscopic and macroscopic values for the same dataset. For instance, JEF-2.2 has a larger microscopic yield, but JEF-3.1.1 has a larger macroscopic yield.

2.1.2 DNP Group and Kinetics Parameters

In Section 2.1.1, the various methods for experimentally understanding delayed neutron precursors were discussed. The next stage of understanding these precursors is to understand how they are used in kinetics models, as well as the various ways they are represented. The microscopic and macroscopic experimental approaches provide the necessary data to create more easily handled forms. By summing the contributions from each DNP in the microscopic approach, the macroscopic and microscopic approaches can be said to provide data in the form of a delayed neutron count rate over time. Section 2.1.2 discusses the least squares approach to convert this data into several DNP groups.

The DNPs are commonly represented in six groups. These groups were developed originally for computational efficiency and due to lacking data for individual precursors. Brady and England found that increasing the number of groups for point kinetics calculations from six to nine caused no significant differences [23]. Their study shows that increasing the number of DNP groups alone does not provide an increase in accuracy for point kinetics simulations.

Representing the DNPs in group form also simplifies the calculation of various kinetics parameters. Various kinetics parameters are introduced in Section 2.1.3, though without detail on the purpose of each variable. This section provides more information on what these kinetics and delayed neutron group parameters physically represent, while Sections 2.1.2 and 2.1.2 detail how they are calculated. The DNP group parameters directly affect many of the kinetics parameters, and both sets of data are used in kinetics and dynamics problems. Table 2.10 goes through each of the terms of importance for this section.

Table 2.10: DNP group parameters and kinetics parameters along with their definitions.

Parameter	Description
λ_k / λ_i	The decay constant for DNP group k / nuclide i
$\bar{\nu}_{d,k} / \bar{\nu}_{d,i}$	The delayed neutron yield for DNP group k / nuclide i
χ_k / χ_i	The spectra of DNP group k / nuclide i
$P_{n,i}$	The delayed neutron emission probability for nuclide i
$\bar{\tau}$	The mean half-life
Y	The total delayed neutron yield
β_k / β_i	The delayed neutron fraction of DNP group k / nuclide i
β	The total delayed neutron fraction
β_{eff}	The effective delayed neutron fraction
Λ	The mean neutron generation time
Λ_{eff}	The effective mean neutron generation time

The DNP groups come with associated parameters, such as the group decay constants, λ_k , and the group yields, $\bar{\nu}_{d,k}$. The group yields are also known as group abundances, and can be found represented using a_k , b_k , ν_{dk} , and α_k in the literature [87, 191, 125, 23]. These group yields can be converted to group delayed neutron fractions, β_k , by dividing the group yields by the average number of neutrons emitted per fission event, as shown in Equation (2.23). The group yields represent the number of delayed neutrons produced per fission in group i , and when summed yield the total delayed neutron yield per fission. The β_k and λ_k terms are the defining parameters for the DNP groups, along with the group spectra, χ_k , for DNP group k . With this data, the other group parameters can be derived, such as the total delayed neutron yield, Y , and delayed neutron fraction β .

Although the groups are useful, there are times when individual DNPs are needed. For example, in 1986, Perry et al. demonstrates that kinetics calculations yield different results when using individual nuclides compared to the six-group parameters [132]. In 1989, Brady and England also use the individual nuclides to compare with group data [23], and updated the six-group parameters based on the explicit DNPs. Updating the six-group parameters enabled greater consistency between experimental results and simulations.

Moving from the group parameters to the kinetics parameters, there is a convenient analogy which can be used to better understand some of the kinetics parameters. The effective multiplication factor, k_{eff} , is an important term in reactor physics. This term represents the scaling of the number of neutrons over each generation in a finite reactor system. The infinite multiplication factor, k_{∞} , represents the multiplication without considering leakage terms. This is analogous to the delayed neutron fraction and mean generation time terms, where the effective form of the term requires information on the specific reactor geometry.

The adjoint flux is important for calculating the effective form of many kinetics parameters. The adjoint flux is not a physical quantity, but rather serves as an importance, or weighting, function, describing the importance of the neutron flux in the reactor. For example, areas further from the center of the core are less important than those closer to the center, which is reflected by a decrease in the effective delayed neutron fraction in MSRs as DNPs move away from the center of the core due to fuel salt advection. The adjoint flux can be calculated by taking the adjoint of the Boltzmann transport equation, or by using the iterated fission probability (IFP) method, which provides an importance term proportional to the adjoint flux [178, 121].

The delayed neutron fraction, β , represents the ratio of delayed neutron production over all neutron production [116]. Related to the delayed neutron fraction is the effective delayed neutron fraction, β_{eff} . This term is more sophisticated, as rather than being a physical property of a fissile nuclide, it depends on the composition and

geometry of the reactor. For MSRs, it also accounts for the movement of the DNP in different parts of the reactor.

A straightforward way to interpret the effective delayed neutron fraction is that it is the delayed neutron fraction corrected for the actual composition and geometry of the reactor. More specifically, the effective delayed neutron fraction represents the ratio between the delayed-neutron induced fission to the total number of fission events weighted by the spectrum and adjoint neutron flux [150]. Alternatively, it can be thought of as the ratio of the spectrum weighted and adjoint weighted delayed neutron production over all spectrum weighted and adjoint weighted neutron production [116]. For MSRs, the movement of the fuel salt also affects the effective delayed neutron fraction, as the integrated adjoint neutron flux is reduced due to movement of the DNPs out of the core and into less neutronically-important regions.

Another term, the mean neutron generation time, Λ , is the mean time between generation of a neutron and fission induced by that neutron [150]. This term also has an effective form, Λ_{eff} , which is the adjoint weighted form. The effective mean neutron generation time can vary significantly, on the order of hundreds of percent, from the mean neutron generation time [184]. In contrast, the effective delayed neutron fraction varies by on the order of 20% from the delayed neutron fraction [184].

Calculation of DNP Group Parameters

The DNP group parameters are important for simulating transients. The DNP group parameters are formulated based on the fitting of the delayed neutron count rate, as discussed in Section 2.1.1. Equation (2.2) shows the exact formulation for the delayed neutron count rate as a function of time measured by a detector for I individual DNPs for any number of fissile nuclides, recreated here as:

$$\dot{n}_d(t) = \epsilon \sum_{i=1}^I P_{n,i} \lambda_i N_i(t).$$

Equation (2.3) instead gives the K group delayed neutron count rate by the same detector for a pulse irradiation as:

$$\dot{n}_d(t) \approx \epsilon F_s \sum_{k=1}^K \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}. \quad (2.3)$$

For an infinite irradiation (also referred to as a saturation irradiation), Equation (2.4) is used rather than Equation (2.3), given as [25]:

$$\dot{n}_d(t) \approx \epsilon \dot{F}_s \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t}. \quad (2.4)$$

Equation (2.5) shows the intermediate case, where the irradiation time, T , is neither infinite nor a pulse [192]. This equation gives the delayed neutron count rate as:

$$\dot{n}_d(t, T) \approx \epsilon \dot{F}_s \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t} (1 - e^{-\lambda_k T}). \quad (2.5)$$

The terms in Equations (2.3), (2.4), and (2.5) are defined as follows:

$$\begin{aligned}
K &= \text{the number of DNP groups [-]} \\
\lambda_k &= \text{the decay constant of DNP group } k \text{ [s}^{-1}\text{]} \\
F_s &= \text{the total number of fission events in the pulse irradiation [-]} \\
\dot{F}_s &= \text{the fission rate for a saturation irradiation [s}^{-1}\text{]} \\
\bar{v}_{d,k} &= \text{the group yield for group } k \text{ [-] .}
\end{aligned}$$

More information on the derivation of these equations can be found in Appendix A. These equations are specifically for a single fissile nuclide, and the derivation can be found in the work of Wilson and England [192].

The pulse irradiation approach gives better statistics on the short lives DNPs, while the longer lived DNPs are better represented by saturation irradiation. Brady discusses this, and uses both pulse and saturation irradiation to form groups [25]. Specifically, Brady uses the four long-period groups from the saturation irradiation and the four short-period groups from the pulse irradiation renormalized at the second DNP group.

Once the delayed neutron count rate is acquired, an over-determined non-linear least squares approach is used to calculate the group yields and decays. For example, Brady used STEPIT, an algorithm for finding local minima of a continuous function, to handle the least squares solve with both pulse and infinite irradiation experimental data [25, 35]. Waldo et al. also solve the least squares problem, though the specific approach used is not specified [189]. Using Equation (2.3), (2.4), or (2.5), the general form of the solve is given as:

$$A\vec{x} \approx \vec{b}. \quad (2.6)$$

Based on the irradiation type used, the coefficient matrix will change, as shown in Equations (2.3), (2.4), and (2.5). Equation (2.7) gives the coefficient matrix for a pulse irradiation as:

$$A = \epsilon F_s \begin{bmatrix} \lambda_1 e^{-\lambda_1 t_0} & \lambda_2 e^{-\lambda_2 t_0} & \dots & \lambda_n e^{-\lambda_n t_0} \\ \lambda_1 e^{-\lambda_1 t_1} & \lambda_2 e^{-\lambda_2 t_1} & \dots & \lambda_n e^{-\lambda_n t_1} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1 e^{-\lambda_1 t_m} & \lambda_2 e^{-\lambda_2 t_m} & \dots & \lambda_n e^{-\lambda_n t_m} \end{bmatrix}. \quad (2.7)$$

Note that this term changes for a saturation irradiation, removing the λ_i term and changing the fission term from net fissions to the fission rate.

Within this least squares equation, the solution vector contains only the group yields, given in Equation (2.8) as:

$$\vec{x} = \begin{bmatrix} \bar{v}_{d,1} \\ \bar{v}_{d,2} \\ \vdots \\ \bar{v}_{d,K} \end{bmatrix}. \quad (2.8)$$

The target vector contains the delayed neutron count rate data, shown in Equation (2.9) as:

$$\vec{b} = \begin{bmatrix} n_d(t_0) \\ n_d(t_1) \\ \vdots \\ n_d(t_m) \end{bmatrix}. \quad (2.9)$$

Using ordinary least squares, with a non-relative residual, would lead to early time step count rates dominating due to their larger magnitudes. One way to resolve this is to take the natural logarithm of the target vector $\vec{b}$, such that the count rate evolution is linear in time. Another way to resolve this is to use a relative residual, given by Equation (2.10) as:

$$r = \min_x \left\| \frac{\vec{b} - A\vec{x}}{\vec{b} + \epsilon} \right\|_2^2, \quad (2.10)$$

where,

$$\begin{aligned} \epsilon &= \text{a small value to avoid division by zero} \\ &= 1 \times 10^{-12}. \end{aligned}$$

The Euclidean norm, $\|\cdot\|_2$, is defined in Equation 2.11 as:

$$\|\vec{x}\|_2 = \sqrt{x_1^2 + \dots + x_n^2}. \quad (2.11)$$

By solving this non-linear system of equations, the group decay constants, λ_k , and yields, $\bar{v}_{d,k}$, can be generated. Once this group data is generated, and if there are multiple fissioning nuclides, F , then Equation (2.12) can be used to determine the K group decay constants using [43] as:

$$\lambda_k = \frac{\sum_F \lambda_k^{(F)} \beta_{k,eff}^{(F)}}{\sum_F \beta_{k,eff}^{(F)}}, \quad (2.12)$$

where,

$$\begin{aligned} \lambda_k^{(F)} &= \text{the decay constant of DNP group } k \text{ from fission of nuclide } F \\ \beta_{k,eff}^{(F)} &= \text{the effective delayed neutron fraction of DNP group } k \text{ from fission of nuclide } F. \end{aligned}$$

Alternatively, a fission-weighting approach can be used. The DNP group parameters are generated for a given fuel composition, weighted by the volumetric fission rate fraction of each fissile nuclide, F , similar to Equation (2.12). For example, the delayed neutron yield of group k is given as:

$$\bar{v}_{d,k}^{(FW)} = \left(\sum_F \int \Sigma_f^{(F)} \phi dE d\vec{r} \right)^{-1} \sum_F \left(\bar{v}_{d,k}^{(F)} \int \Sigma_f^{(F)} \phi dE d\vec{r} \right), \quad (2.13)$$

while the decay constant of group k is given as:

$$\lambda_k^{(FW)} = \left(\sum_F \int \Sigma_f^{(F)} \phi dE d\vec{r} \right)^{-1} \sum_F \left(\lambda_k^{(F)} \int \Sigma_f^{(F)} \phi dE d\vec{r} \right), \quad (2.14)$$

where

$\bar{v}_{d,k}^{(F)}$ = the DNP group k delayed neutron yield from nuclide F [-]

$\lambda_k^{(F)}$ = the DNP group k decay constant from nuclide F [s^{-1}]

ϕ = the neutron flux [$cm^{-2} \cdot s^{-1}$]

$\bar{v}_{d,k}^{(FW)}$ = the fission-weighted DNP group k delayed neutron yield [-]

$\lambda_k^{(FW)}$ = the fission-weighted DNP group k decay constant [s^{-1}]

$\Sigma_f^{(F)}$ = the macroscopic fission cross section for nuclide F [cm^{-1}].

The fission-weighted values can also be multiplied by an importance term to account for the difference between the delayed and prompt neutron fission spectra [141]. Equation (2.13) becomes:

$$\bar{v}_{d,k} = \bar{v}_{d,k}^{(FW)} \frac{\int \chi_{d,k}(E) \phi^*(E) dE}{\int \chi(E) \phi^*(E) dE}, \quad (2.15)$$

and Equation (2.14) becomes:

$$\lambda_k = \lambda_k^{(FW)} \frac{\int \chi_{d,k}(E) \phi^*(E) dE}{\int \chi(E) \phi^*(E) dE}, \quad (2.16)$$

where

$\chi_{d,k}$ = the delayed neutron fission spectrum for DNP group k [-]

χ = the total fission spectrum [-]

ϕ^* = the adjoint flux [$cm^{-2} \cdot s^{-1}$].

These equations thus far have not included the energy spectra, but instead have focused on the delayed neutron count rate. Including the energy in the delayed neutron count rate requires the energy spectra, $\chi_{d,k}(E)$, of the relevant group to be included, as shown in Equation (2.17) as:

$$\dot{n}_d(E, t) \approx \epsilon \tilde{F}_s \sum_{k=1}^K \chi_{d,k}(E) A_k e^{-\lambda_k t}, \quad (2.17)$$

where,

$\tilde{F}_s$ = the generic fission term [- or s^{-1}]

A_k = the generic matrix term [s^{-1} or -]

$\chi_{d,k}(E)$ = the delayed neutron energy spectrum of DNP group k [-].

This is the general form of the saturation and pulse irradiation equations, where A_k represents $\bar{v}_{d,k} \lambda_k$ for a pulse irradiation or $\bar{v}_{d,k}$ for saturation irradiation. $\tilde{F}_s$ is the total number of fissions for a pulse irradiation or the fission rate for saturation irradiation.

Calculation of DNP Group Spectra and Concentrations

Another consideration in calculating group parameters is constructing group concentrations from individual nuclide data. This is important for situations where reducing the problem from several hundred DNPs to a few DNP groups

reduces computational cost. A specific use case for this is explicitly modeling all DNPs spatially and then using those spatially resolved concentrations with an existing six-group structure. In such a case, the DNPs must be combined in such a manner to preserve their contributions to each DNP group. The reverse is also true, where a given six-group spatially resolved concentration can be separated into individual DNPs spatially resolved concentrations.

Two different methods are considered for separating and combining the DNP groups into individual nuclides and vice versa. These methods have been used to generate DNP group spectra, so the group spectra can also be calculated using these methods:

- Grouping based on half lives [53, 152]
- Grouping based on fractional contribution to adjacent groups [25]

The grouping based on half lives requires comparing each nuclide's half-life to a constant half-life bin structure formulated from analysis of individual DNPs. The groupings based on half lives are given in the top two sets in Table 2.11. The bottom set of half lives is from ENDF/B-VIII.0 and is used with the fractional contribution approach [29].

Table 2.11: Half-lives of DNP groups from various sources.

Source	τ_1 [s]	τ_2 [s]	τ_3 [s]	τ_4 [s]	τ_5 [s]	τ_6 [s]
[53]	$\tau > 31$	$31 \geq \tau > 9$	$9 \geq \tau > 3$	$3 \geq \tau > 0.8$	$0.8 \geq \tau > 0.3$	$0.3 \geq \tau$
[152] [*]	$\tau > 30$	$30 \geq \tau > 10$	$10 \geq \tau > 4$	$4 \geq \tau > 1.4$	$1.4 \geq \tau > 0.4$	$0.4 \geq \tau$
[151]	$\tau = 51.98$	$\tau = 21.17$	$\tau = 5.74$	$\tau = 2.29$	$\tau = 0.82$	$\tau = 0.24$

^{*} Rudstam uses $\tau_1 = 55.6$, but to ensure all DNPs are included, this has been modified.

England and Wilson chose their half-life bins based on the half-lives given in ENDF/B-V, while specifying that of the reasonable choices for bounding values, "No one was completely satisfactory" [53]. Rudstam selected half-life bins into six groups, although it is clarified that "...the grouping is somewhat arbitrary" [152]. The half-life grouping method thus results in a series of comparisons; if a nuclide is within a given group's bounds, the concentration of that nuclide is added to that group.

For the fractional grouping, the half-life selection is less arbitrary and is instead based on the DNP group parameters. The ENDF/B-VI.8 data comes from the ^{235}U half-lives from the (n, ν_d) reaction [151, 23, 52, 24]. These delayed neutron data have remained unchanged through ENDF/B-VIII.0 [29, 151]¹.

The fractional grouping is more complex, but assigns nuclides to groups in a more rigorous manner. The method uses two groups, i and $i + 1$, for each nuclide k . These groups are determined via Equation (2.18) as:

$$\lambda_i < \lambda_k < \lambda_{i+1} \quad (2.18)$$

Nuclides which don't fit in these bounds will have the entire fractional contribution go to the closest group. Otherwise, the nuclide will have some fraction of its concentration or spectra go to each DNP group. The fractional contribution, or weight, of nuclide k to each group is represented with $f_{k,i}$ and $f_{k,i+1}$. The total fraction must be equal to 1 as shown in Equation (2.19) as:

$$f_{k,i} + f_{k,i+1} = 1 \quad (2.19)$$

¹The ENDF/B-VIII.1 data is also the same as ENDF/B-VI.8, but it is currently unpublished.

To solve for the fractions, the least squares minimization problem is minimized [25]:

$$\int_0^\infty \left(\lambda_k e^{-\lambda_k t} - f_{k,i} \lambda_i e^{-\lambda_i t} - f_{k,i+1} \lambda_{i+1} e^{-\lambda_{i+1} t} \right)^2 dt \quad (2.20)$$

Solving this equation for where the partial derivatives are zero for each fraction, $f_{k,i}$ and $f_{k,i+1}$, combined with Equation (2.19), yields the overdetermined system of equations as:

$$A = \begin{bmatrix} \lambda_i & \frac{2\lambda_i \lambda_{i+1}}{\lambda_i + \lambda_{i+1}} \\ \frac{2\lambda_i \lambda_{i+1}}{\lambda_i + \lambda_{i+1}} & \lambda_{i+1} \\ 1 & 1 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} \frac{2\lambda_i \lambda_k}{\lambda_i + \lambda_k} \\ \frac{2\lambda_k \lambda_{i+1}}{\lambda_k + \lambda_{i+1}} \\ 1 \end{bmatrix}, \quad \vec{x} = \begin{bmatrix} f_{k,i} \\ f_{k,i+1} \end{bmatrix}. \quad (2.21)$$

Because the number of rows is greater than the number of columns in the matrix, the left-inverse is used. This system of equations is then solved as shown in Equation (2.22) as:

$$\vec{x} = (A^T A)^{-1} A^T \vec{b} \quad (2.22)$$

With the fractional contribution terms, $\vec{x}$, the concentration of a given nuclide can then be multiplied by each fraction to show how much it contributes to that group. With a summation of each nuclides contribution, the group concentrations can thus be calculated and vice versa. Alternatively, the spectra can be calculated by using the same approach but with individual DNP spectra instead of concentrations.

Calculation of Kinetics Parameters

The other relevant parameters to be calculated are simpler and do not require solving least squares problems. The k^{th} DNP group delayed neutron fraction is calculated as:

$$\beta_k = \frac{\bar{v}_{d,k}}{\bar{v}}. \quad (2.23)$$

A useful term for comparing DNP group parameters is the average half-life [135, 136, 98]. Parish et al. also use the average half-life, but with a non-standard formulation which does not include the natural log of two [126]. Equation (2.24) gives the average half-life as:

$$\bar{\tau} = \ln(2) \sum_k^K \frac{\bar{v}_{d,k}}{\lambda_k}. \quad (2.24)$$

Equation (2.25) shows how the total delayed neutron yield, or average number of delayed neutrons per fission event, $\bar{v}_d$, can be calculated [67]:

$$\bar{v}_d = \sum_k^K \bar{v}_{d,k}. \quad (2.25)$$

This yield can also be approximately represented for an individual nuclide by integrating Equation (2.1) over all time and normalizing to a single fission, where the concentration over time is assumed to be:

$$N_i(t) = N_i(0) e^{-\lambda_i t}. \quad (2.26)$$

This yields Equation (2.27), where the total number of delayed neutrons emitted from a single nuclide i is given as:

$$\int_0^\infty \dot{n}_{d,i}(t) dt = P_{n,i} N_i(0). \quad (2.27)$$

After dividing by the total number of fission events, Equation (2.28) shows that the total delayed neutron yield can be written as [79, 112]:

$$\bar{v}_d = \sum_{i=1}^I P_{n,i} CFY_i, \quad (2.28)$$

where,

CFY_i = the cumulative fission yield of nuclide i .

This also means that the delayed neutron yield of each contributing DNP can be calculated.

The delayed neutron fraction can be written as:

$$\beta = \frac{\bar{v}_d}{\bar{v}} \quad (2.29)$$

The mean neutron generation time can be calculated as:

$$\Lambda = \frac{l}{k}, \quad (2.30)$$

where,

l = the prompt neutron lifetime [s]

k = the neutron multiplication factor [-].

(2.31)

This can be calculated using a time-dependent version of a Monte Carlo code along with point reactor kinetics (PRK) [150].

Finally, there are the effective parameters: the effective delayed neutron fraction, β_{eff} , and the effective mean neutron generation time, Λ_{eff} . β_{eff} is the spectrum and adjoint weighted ratio of the delayed neutron induced fission to the induced fissions from all neutrons [150]. This is given in Equation (2.32) as:

$$\beta_{eff} = \frac{\int \Phi^\dagger(\vec{r}, E, \vec{\Omega}) \chi_d(E) \beta \bar{v}(E) \Sigma_f(\vec{r}, E) \Phi(\vec{r}, E, \vec{\Omega}) dE d\vec{\Omega} d\vec{r}}{\int \Phi^\dagger(\vec{r}, E, \vec{\Omega}) \chi(E) \bar{v}(E) \Sigma_f(\vec{r}, E) \Phi(\vec{r}, E, \vec{\Omega}) dE d\vec{\Omega} d\vec{r}}, \quad (2.32)$$

where,

$$\begin{aligned}
\Phi^\dagger &= \text{the adjoint neutron flux } [n \cdot cm^{-2} \cdot s^{-1}] \\
\chi_d &= \text{the delayed neutron energy spectrum } [-] \\
\chi &= \text{the neutron energy spectrum } [-] \\
\beta &= \text{the delayed neutron fraction } [-] \\
\bar{\nu} &= \text{the average number of neutrons released per fission } [-] \\
\Sigma_f &= \text{the macroscopic fission cross section } [cm^{-1}] \\
\Phi &= \text{the neutron flux } [n \cdot cm^{-2} \cdot s^{-1}].
\end{aligned}$$

Λ_{eff} can also be calculated explicitly by its definition, which is the "...time between the birth of a neutron and the subsequent absorption that induces fission." This is shown in Equation (2.33) as [184]:

$$\Lambda_{eff} = \frac{\int \Phi_0^\dagger \frac{1}{\nu} \Phi_0 dE d\vec{\Omega} d\vec{r}}{\int \Phi_0^\dagger F_0 \Phi_0 dE d\vec{\Omega} d\vec{r}} \quad (2.33)$$

where,

$$\begin{aligned}
\Phi_0^\dagger &= \text{the critical adjoint neutron flux } [n \cdot cm^{-2} \cdot s^{-1}] \\
\Phi_0 &= \text{the critical neutron flux } [n \cdot cm^{-2} \cdot s^{-1}] \\
\nu &= \text{the average neutron speed } [cm \cdot s^{-1}] \\
F_0 &= \text{the total fission operator } [cm^{-1}].
\end{aligned}$$

Because only the absorptions that induce fission are of interest, the fission operator is used.

There are other methods used to calculate β_{eff} and Λ_{eff} in the literature [184]. These include the direct integration, or adjoint weighted, method [88]; the eigenvalue, or perturbation, method [72]; and the iterated fission probability method [20]. Λ_{eff} can also be calculated using the pulsed method [150]. For β_{eff} , there are also the prompt, Meulekamp, and Spriggs methods which can be used [150, 116]. Aufiero et al. have conducted a comparison of the Meulekamp, iterated fission probability, and prompt methods, which show that the iterated fission probability and prompt methods both yield similar results while the Meulekamp method is 200 pcm higher for the Molten Salt Fast Reactor (MSFR) [9].

2.1.3 Kinetics and Dynamics Methods

There are several different methods which have been used for analyzing transients in MSRs. These methods include, as of 2018, "...fluid-flow-velocity field coupled multi-group, time-resolved, neutron diffusion; ... point reactor kinetics schemes; and quasi-statics; generally all coupled, tightly or loosely, to a thermal-hydraulics solver" [193]. In general, although these methods have varying accuracy and costs, none of them include MSR specific phenomena when generating the DNP group parameters.

Multi-Group Diffusion

Multi-group diffusion kinetics methods are among the most accurate, and computationally expensive, approaches applied to transient problems, surpassed only by transport methods. Hurwitz and Ehrlich provide a standard

formulation of multi-group diffusion written in 1953 [78].

Examples of software that use fluid-flow-velocity field coupled multi-group include GenFOAM and Grif-fin/Pronghorn coupling using Multiphysics Object-Oriented Simulation Environment (MOOSE) [58, 176]. Other examples can be found in the review by Wooten et al. [92, 59, 91] and elsewhere in the literature [197, 195, 93, 122, 128]. These studies typically use Picard iteration with multi-group neutron diffusion coupled with thermal-hydraulics to accurately capture the multi-physics effects of the reactor. However, this accuracy also comes with a higher computational cost than simpler methods. The general form of the coupling is given as [193]:

$$\begin{aligned} \frac{1}{\bar{v}_g} \frac{\partial \phi_g}{\partial t} = & \nabla \cdot D_g \nabla \phi_g + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} \\ & + (1 - \beta) \chi_{p,g} \sum_{g'=1}^G (\nu \Sigma_f)_{g'} \phi_{g'} \\ & \sum_{i=1}^I \chi_{d,g,i} \lambda_i C_i - \Sigma_{a,g} \phi_g - \sum_{g' \neq g}^G \Sigma_{s,g \rightarrow g'} \phi_g, \end{aligned} \quad (2.34)$$

and,

$$\frac{\partial C_i}{\partial t} = \beta_i \sum_{g=1}^G (\bar{v}_g \Sigma_f)_g \phi_g - \lambda_i C_i - \nabla \cdot (C_i \vec{u}) + \nabla \cdot \frac{\nu_T}{Sc_T} \nabla C_i, \quad (2.35)$$

where,

- $\bar{v}_g$ = the average number of neutrons produced by fission of group g [-]
- ϕ_g = the scalar neutron flux of group g [$n \cdot cm^{-2} \cdot s^{-1}$]
- D_g = the group g diffusion term [cm]
- G = the total number of energy groups [-]
- $\Sigma_{s,g' \rightarrow g}$ = the macroscopic scattering cross section from group g' to group g [cm^{-1}]
- $\chi_{p,g}$ = the number of prompt neutrons born in group g [-]
- $\chi_{d,g,i}$ = the number of delayed neutrons from precursor group i born in group g [-]
- C_i = the concentration of DNPs in group i [$atoms \cdot cm^{-3}$]
- $\vec{u}$ = the velocity profile [$cm \cdot s^{-1}$]
- $\Sigma_{a,g}$ = the macroscopic absorption cross section for group g [cm^{-1}]
- ν_T = the eddy viscosity [$cm^2 \cdot s^{-1}$]
- Sc_T = the turbulent Schmidt number [-].

Modified Point Kinetics

PRK are a simpler approach to modeling a transient, with the basic form given as:

$$\frac{dn(t)}{dt} = \frac{\rho(t) - \beta_{eff}}{\Lambda} n(t) + \sum_{i=1}^I \lambda_i C_i(t), \quad (2.36)$$

and,

$$\frac{dC_i(t)}{dt} = \frac{\beta_{eff,i}}{\Lambda} n(t) - \lambda_i C_i(t), \quad (2.37)$$

where,

$$\begin{aligned}
n &= \text{the relative neutron density [-]} \\
\rho &= \text{the reactivity [-]} \\
\beta_{eff} &= \text{the effective delayed neutron fraction [-]}.
\end{aligned}
\tag{2.38}$$

The derivation of the PRK equations can be found in the work by Henry, with a fluid-fueled derivation in the work by Lapenta et al. [72, 96]. Modified point kinetics (MPK) are used for MSRs, as standard PRK do not properly handle the movement of the fuel salt.

PRK in general struggle to capture spatial effects, such as flow variation, which is important for MSRs. PRK do, however, give a correct prediction for general evolution trends [193]. Wooten et al. provide four different publications which use MPK in MSR analysis [68, 115, 169, 196].

The work by Guo et al. analyzes unprotected loss of heat sink (ULOHS) and unprotected loss of flow (ULOF) transients in the Molten Salt Actinide Recycler and Transmuter reactor (MOSART). The PRK equations are modified into the delayed point reactor kinetics (DPK), which model precursor concentrations in-core and ex-core, with the primary set of equations given as:

$$\frac{dn(t)}{dt} = \frac{\rho - \tilde{\beta}_{eff}}{\Lambda} n(t) + \sum_{i=1}^I \lambda_i C_{i,c}(t), \tag{2.39}$$

$$\begin{aligned}
\frac{dC_{i,c}(t)}{dt} &= \frac{\beta_i}{\Lambda} n(t) - \left(\lambda_i + \frac{1}{\tau_c} \right) C_{i,c}(t) \\
&\quad + \frac{1}{\tau_l} \left(\frac{V_l}{V_c} \right) C_{i,l}(t - \tau_l),
\end{aligned}
\tag{2.40}$$

$$\frac{dC_{i,l}(t)}{dt} = \frac{1}{\tau_c} \left(\frac{V_c}{V_l} \right) C_{i,c}(t) - \left(\lambda_i + \frac{1}{\tau_l} \right) C_{i,l}(t). \tag{2.41}$$

These equations, shown in Equations (2.39) through (2.41) (recreated from [68, 169]), are also used by Shi et al. in the Reactor Excursion and Leak Analysis Program (RELAP). Shi et al. simulate a load demand change, primary flow transient, secondary flow transient, and reactivity transient in the Molten Salt Breeder Reactor (MSBR) [169]. This approach enables modeling of ex-core decay of DNPs, capturing effects at power and for loss of flow, but fails to capture all oscillatory effects [193].

Intermediate terms used in Equations (2.39) through (2.41) are shown in Equations (2.42) through (2.45) as:

$$\tilde{\beta}_{eff} = \beta_{static} - \beta_{loss}, \tag{2.42}$$

$$\beta_{loss} = \frac{\Lambda(t)}{n(t)} \sum_{i=1}^I \lambda_i C_{i,l}, \tag{2.43}$$

$$\tau_c = \frac{M_c}{\dot{M}(t)}, \tag{2.44}$$

$$\tau_l = \frac{M_l}{\dot{M}(t)}. \tag{2.45}$$

The terms in Equations (2.39) through (2.45) are defined as follows:

$$\begin{aligned}
V_l &= \text{the volume of the loop [cm}^3\text{]} \\
V_c &= \text{the volume of the core [cm}^3\text{]} \\
\beta_{static} &= \text{the value of } \beta_{eff} \text{ without flow [-]} \\
M_c &= \text{the mass in the core [kg]} \\
M_l &= \text{the mass in the loop [kg]} \\
\dot{M}(t) &= \text{the mass flow rate [kg} \cdot \text{s}^{-1}\text{]}.
\end{aligned}$$

A similar method is also used by Suzuki and Shimazu for analyzing a reactivity initiated accident in the FUJI-12 reactor [174]. That work uses a simpler form of the equations, without dividing the precursor concentrations into core and loop regions.

The work by Merle-Lucotte et al. uses the "IPK method," which may stand for "Improved Point-Kinetic method," though this is not explicitly stated [115]. This method uses a fixed mesh for neutronics and a mobile mesh for fluid properties. Merle-Lucotte et al. found that the PRK method without any modifications was able to match the IPK method for general trends and end-states, but was unable to capture reactivity oscillations properly. The IPK method was able to capture these oscillations, but assumes uniform fission and instantaneous heat transfer [115].

The work by Zanetti et al. extends the TRACE code, a finite volume code used for LWR transients [196]. The DNP concentration as a function of space and time is resolved upwind in space and with backward Euler for time. Adjoint-weighted quantities discretized with the trapezoidal rule are then passed to PRK equation, and the set of DNP concentration and the PRK power, or neutron density, equations are resolved using LU factorization.

Another approach to modify the PRK is to alter the kinetics parameters based on a phase-space integration weighted with the adjoint flux [97]. Alternatively, effective flow-inclusive group yields can be calculated, as shown by Haubenreich and used in PRK equations [71].

Quasi-Statics

Quasi-static methods seek to maintain the accuracy of multi-group diffusion approaches while also having the cost savings of PRK. The general approach of these methods is to update PRK parameters with occasional multi-group diffusion simulations to capture changes in flux and DNP spatial distributions [193]. The general formulation used for quasi-static methods is very similar among the various publications that use it [50, 176]. A complete derivation of the quasi-static method can be found in the work by Henry, who discusses the shape and amplitude flux functions, and the work by Devooght, who discusses the generalized quasi-static method [72, 47]. There are also other studies which discuss the quasi-static method in detail [13].

Wooten et al. introduce several different publications which have used quasi-statics [193]. One of these, from 2005 by Rineiski et al., covers the SIMMER code, which couples neutronics, thermal-hydraulics, and structure codes [144]. The neutron flux is constrained in the work by a term containing the phase-space integration and the weighting function used. The neutron flux is treated as:

$$\Phi(\vec{x}, t) = \phi_A(t)\psi(\vec{x}, t), \quad (2.46)$$

where,

$$\begin{aligned}\Phi &= \text{the neutron flux } [n \cdot cm^{-2} \cdot s^{-1}] \\ \phi_A &= \text{the neutron flux amplitude function [n]} \\ \psi(\vec{x}, t) &= \text{the neutron flux shape function } [cm^{-2} \cdot s^{-1}].\end{aligned}$$

Rineiski et al. solve "shape" steps to handle the neutronics and DNP spatial concentrations. The equations used in the "shape" steps are recreated here in Equations (2.47) and (2.48). These equations are solved by time-discretizing Equation (2.47) into a set of steady-state-like equations, which are then treated using the DANTSYS module of SIMMER [5]. These primary equations are given as:

$$\begin{aligned}\frac{1}{\bar{v}} \frac{\partial \psi(\vec{x}, t)}{\partial t} + \frac{1}{\bar{v} \phi_A} \frac{d \phi_A}{dt} \psi(\vec{x}, t) + M(t) \psi(\vec{x}, t) = \\ (1 - \beta) \chi_p F(t) \psi(\vec{x}, t) + \frac{1}{\phi_A} \chi_d \sum_{i=1}^I \lambda_i C_i(\vec{x}, t) + \frac{1}{\phi_A} Q(\vec{x}, t),\end{aligned}\tag{2.47}$$

and,

$$\frac{\partial C_i(\vec{x}, t)}{\partial t} = \phi_A \beta_i F \psi(\vec{x}, t) - \lambda_i C_i(\vec{x}, t) - \nabla(u(\vec{x}, t) C_i(\vec{x}, t)).\tag{2.48}$$

Within these "shape" steps are also sub-steps referred to as "reactivity" steps. At each of these "reactivity" steps, the PRK parameters and flux amplitude are calculated. The equations used in the "reactivity" steps are recreated here as:

$$\frac{d \phi_A}{dt} = \left(\frac{\rho - \beta_{eff}}{\Lambda} \right) \phi_A + q_{eff} + \sum_{i=1}^I \lambda_i (c_{s,i} + c_{m,i}),\tag{2.49}$$

$$\frac{dc_{s,i}}{dt} = \frac{\beta_{eff}}{\Lambda} \phi_A - \lambda_i C_{s,i}.\tag{2.50}$$

$$\rho = \frac{\langle W(\vec{x}), (\chi F(t) - M(t)) \psi(\vec{x}, t) \rangle}{\langle W(\vec{x}), \chi F(t) \psi(\vec{x}, t) \rangle}.\tag{2.51}$$

$$c_{m,i} = \frac{\langle W(\vec{x}), \chi_d C_{m,i} \rangle}{\Lambda \langle W(\vec{x}), \chi F(t) \psi(\vec{x}, t) \rangle}.\tag{2.52}$$

Intermediate terms used in Equations (2.49) through (2.52) are given as:

$$\frac{\partial C_{m,i}(\vec{x}, t)}{\partial t} = -\nabla(u(\vec{x}, t) C_{s,i}(\vec{x}, t)) - \lambda_i C_{m,i}(\vec{x}, t) - \nabla(u(\vec{x}, t) C_{m,i}(\vec{x}, t)),\tag{2.53}$$

and,

$$\frac{\partial C_{s,i}(\vec{x}, t)}{\partial t} = \phi_A(t) \beta_i F(t) \psi(\vec{x}, t) - \lambda_i C_{s,i}(\vec{x}, t).\tag{2.54}$$

The $C_s(\vec{x}, t)$ term is computed at steady state by setting the left-hand side of both Equation (2.47) and (2.48) to zero as well as the fluid velocity to zero at $t=0$. This allows Equation (2.53) to be solved.

The terms in Equations (2.46) through (2.54) are defined as follows:

- $M(t)$ = the neutron disappearance operator [cm^{-1}]
- $F(t)$ = the neutron production operator [cm^{-1}]
- χ_p = the prompt neutron energy spectra [-]
- $C_i(\vec{x}, t)$ = the concentration of DNP group i [$atoms \cdot cm^{-3}$]
- $Q(\vec{x}, t)$ = the external neutron source [$n \cdot cm^{-3} \cdot s^{-1}$]
- $u(\vec{x}, t)$ = the DNP velocity [$cm \cdot s^{-1}$]
- $C_{s,i}(\vec{x}, t)$ = the "standard" precursor concentration of group i [$atoms \cdot cm^{-3}$]
- $C_{m,i}(\vec{x}, t)$ = the "movable" precursor concentration of group i [$atoms \cdot cm^{-3}$]
- ρ = the reactivity [-]
- $c_{s,i}$ = the PRK "standard" precursor concentration of group i [$atoms$]
- $c_{m,i}$ = the PRK "movable" precursor concentration of group i [$atoms$]
- q_{eff} = the PRK external neutron source [$n \cdot s^{-1}$]
- Λ = the neutron generation time [s]
- $W(x)$ = the weighting function (the default in SIMMER
is an adjoint flux computed at steady-state conditions) [-].

2.2 Molten Salt Reactor Fuel Depletion

Liquid-fueled MSRs are nuclear reactors which have the fissile material dissolved in a circulating molten salt. This fuel salt circulates through the in-core region, where most fissions occur, and through external piping in the ex-core region, where chemical processing occurs, after which it re-enters the in-core region. The salt is both the fuel and coolant in an MSR, as the ex-core region also contains the heat exchanger.

There are many different MSR designs in the literature, as shown in Table 2.12 [37]. These designs feature a circulating salt containing fissile material. In particular, the circulating fuel salt allows for online chemical reprocessing, which causes depletion to vary spatially, leads to movement of DNPs, and requires close coupling of thermal-hydraulics to neutronics for realistic modeling.

MSR Name	Source
FUJI series	[62]
Actinides Molten Salt TransmutER (AMSTER)	[185]
MOSART	[80]
Fast Spectrum Molten Salt Reactor (FS-MSR)	[74]
European MSFR	[114]
Small Mobile Molten Salt Reactor (SM-MSR)	[32]
MSBR	[146]
MSRE	[147]
Accelerator-Driven Neutron Applications (ADNA) Tier-1 reactor	[22]
Single fluid and two fluid denatured molten salt reactors (DMSRs)	[51, 4]
Thorium molten salt reactor (TMSR)	[124, 113]

Table 2.12: A collection of various MSR designs publicly available in the literature.

There are many publications in the literature which have investigated online chemical reprocessing of MSRs, as shown in Table 2.13, which has been recreated from [161]. Table 2.13 shows which MSR depletion simulations have modeled DNPs in the "DNP" column, showing most models do not include DNP movement in depletion simulations. The table is sorted by the "Reprocessing" column, which signifies the method(s) used in the different MSR models, discussed in Section 2.2.1. It is also sorted by the publication year in the "Year" column. The table shows there are many MSR concepts for which depletion was modeled. The study by Valdez et al. investigates two MSR designs, specifically the MSBR and the Molten Salt Demonstration Reactor (MSDR). These models used a variety of spatial refinement levels, shown in the "Model" column, ranging from unit cells to full 3D models.

Table 2.13: Molten salt reactor models analyzed with methods approximating online reprocessing (recreated from [161]).

Publication	Reprocessing	Model	Reactor	DNP	Year
Valdez et al. [83]	Continuous	Unit Cell	Several	No	2020
Zhuang et al. [199]	Continuous	Full 3D	MSFR	No	2020
Xia et al. [194]	Continuous	Full 3D	MSFR	No	2019
Zhou et al. [198]	Continuous	Full 3D	MSBR	Yes	2018
Aufiero et al. [10]	Continuous	Full 3D	MSFR	No	2013
Rodriguez-Vieitez [148]	Continuous	Unit Cell	ADNA	No	2002
Rykhlevskii [157]	Batchwise	Full 3D	MSBR	No	2019
Betzler et al. [17]	Batchwise	Unit Cell	MSBR	No	2017
Ridley and Chvala [143]	Batchwise	Full 3D	Toy	No	2017
Ahmad et al. [4]	Batchwise	Unit Cell	DMSR	No	2015
Park et al. [127]	Batchwise	Full 3D	MSBR	No	2015
Sheu et al. [168]	Batchwise	Full 3D	MOSART	No	2013
Nuttin et al. [123]	Batchwise	Full 3D	MSBR	No	2005
Fiorina et al. [57]	Mixed	2D Slice	MSFR	No	2012
Nagy et al. [120]	Mixed	Unit Cell	Toy	Yes	2008
Merle-Lucotte et al. [113]	Mixed	Full 3D	TMSR	No	2007
Bauman et al. [69]	Mixed	2D Slice	MSBR	Yes	1971

Although these studies incorporate the effects of online chemical reprocessing on depletion, they do not explicitly model the spatial effect on depletion. Instead, a simple approximation is used to handle the in-core and ex-core regions while accounting for the entire primary salt volume. The reason why simulations do not include fully resolved spatial resolution in depletion calculations is due to the computational cost.

2.2.1 Depletion Simulations

There are many approaches to simulating depletion with different computational costs and levels of accuracy. Building a computational tool to handle depletion requires two components: the depletion matrix and a numerical solver.

Depletion is the process of fission, transmutation, decay, and the changes to material compositions this process causes. Simulating depletion requires specific information, such as decay constants, fission yields, and fission and transmutation cross-section data. The fission and transmutation cross-section data can come from a transport calculation, while the other data can be read from a data library [100]. Simulating depletion entails numerically

solving the Bateman equations; a generalized form of which is given as [100]:

$$\begin{aligned} \frac{dN_j}{dt} = & \sum_{i \neq j} [(\gamma_{i \rightarrow j} \sigma_{f,i} \phi + \lambda_{i \rightarrow j} + \sigma_{i \rightarrow j} \phi) N_i] \\ & - (\lambda_j + \sigma_j \phi + \lambda_{r,j}) N_j + \dot{N}_{f,j}, \end{aligned} \quad (2.55)$$

where,

- N_j = the atomic density of isotope j [$atoms \cdot cm^{-3}$]
- $\gamma_{i \rightarrow j}$ = the fractional fission product yield of j from the fission of isotope i
- $\sigma_{f,i}$ = the microscopic fission cross-section of isotope i [cm^2]
- ϕ = the spectrum-averaged scalar flux in the fuel region [$n \cdot cm^{-2} \cdot s^{-1}$]
- $\lambda_{i \rightarrow j}$ = the decay constant of decay $i \rightarrow j$ [s^{-1}]
- $\sigma_{i \rightarrow j}$ = the microscopic transmutation cross-section of reaction $i \rightarrow j$ [cm^2]
- N_i = the atomic density of isotope i [$atoms \cdot cm^{-3}$]
- λ_j = the decay constant of isotope j [s^{-1}]
- $\lambda_{r,j}$ = the removal constant for removal of isotope j [s^{-1}]
- σ_j = the microscopic total transmutation cross-section of isotope j [cm^2]
- $\dot{N}_{f,j}$ = the feed rate for isotope j [$atoms \cdot cm^{-3} \cdot s^{-1}$].

Equation (2.55) includes a continuous reprocessing method, where the chemical removal rate, $\lambda_{r,j}$, is part of the differential equation. There are different ways of modeling this reprocessing, including batchwise methods where material is only removed at certain time intervals [161]. For this work, the continuous reprocessing method is used.

The matrix form of Equation (2.55) is given as:

$$\frac{d\vec{N}}{dt} = A\vec{N}, \quad (2.56)$$

where,

- $\vec{N}$ = the vector of material compositions [$atoms \cdot cm^{-3}$]
- A = the depletion matrix [s^{-1}].

The depletion matrix contains the gain and loss terms for each nuclide in N , which vary continuously as a function of time and depend on the concentrations as well.

There are several approaches to solving this equation, such as the Transmutation Trajectory Analysis (TTA) method and the Chebyshev Rational Approximation Method (CRAM) [101]. Other codes may use external software to compute depletion, such as REM, an in-house code used by Nuttin et al. [123] to implement 4th order Runge-Kutta. Codes may also use a power series approximation of the matrix exponential, as discussed by Aufiero et al. [10].

2.2.2 Stationary and Circulating Fuel Comparison

For stationary fuel, the Bateman equation as shown previously in Equation (2.55) is sufficient to simulate depletion. However, for the purposes of a streamlined comparison with circulating fuel, the Bateman equation can be rewritten

as:

$$\frac{dN_j}{dt} = S_j(t) - \lambda_{net,j} N_j(t), \quad (2.57)$$

where,

S_j = the source term for nuclide j ,

containing fission, decay, and transmutation sources [$atoms \cdot cm^{-3} \cdot s^{-1}$]

$\lambda_{net,j}$ = the loss term for nuclide j ,

containing decay and transmutation losses [s^{-1}].

Over the course of a depletion step, the depletion matrix is assumed to be constant, so the source term varies in time only due to the effects of the changing compositions of other nuclides. The loss term, $\lambda_{net,j}$, does not vary with time in this equation.

This form of the Bateman equation does not have space or energy resolution, which makes it simpler to understand and compare with the circulating fuel version. When solving this equation, the common approach is to have different material groupings, or blocks, in different spatial regions. Each of these material blocks will have different compositions and neutronic properties, such as flux, used for that entire material. This way, the spatial resolution is represented by the material blocks, and the system of ordinary differential equations can be solved rapidly.

Circulating fuel is more complex, as instead of solving a system of ordinary differential equations, a system of partial differential equations must be solved. The simplified static Bateman equation can be modified to include advection and diffusion terms, yielding:

$$\frac{dN_j}{dt} = S(\vec{r}, t) - \lambda_{net,j} N_j(\vec{r}, t) + \nabla \cdot (D \nabla N_j(\vec{r}, t) - \vec{u}(\vec{r}, t) N_j(\vec{r}, t)), \quad (2.58)$$

where,

D = the mass diffusivity [$cm^2 \cdot s^{-1}$].

The depletion matrix is assumed to have no spatial resolution in this equation. Because of this assumption, only the concentrations and material blocks provide spatial resolution to the problem. Spatial refinement can be added to this equation based on the specific approach, such as using a spatially resolved neutron flux.

As previously mentioned, the main challenge faced when solving this problem is the computational cost. However, there are many studies which have attempted to develop methods to get around this problem. One common approach, the scaled flux method, is computationally inexpensive [16]. However, this method does not model the in-core and ex-core regions in a physical manner. The scaled flux method scales the flux by the fraction of fuel salt in-core compared to ex-core. This approach does not model the in-core and ex-core regions explicitly, and instead assumes the salt is perfectly mixed in the primary loop.

Another approach has shown how spatial resolution can be investigated using a system of ordinary differential equations to solve the Bateman equation. This approach can estimate the steady-state concentration of certain nuclides across a flow loop, but it does not account for temporal resolution [164].

Alternatively, a more accurate treatment was recently demonstrated through MOOSE-based coupling of the deterministic neutron transport code Griffin with the thermal fluids code Pronghorn, allowing for spatial and temporal resolution of certain nuclides [177]. Tano et al. present an approach where only insoluble, medium-lived,

and high concentration nuclides are tracked with pre-computed spatial distributions, which only need to be updated if the medium-lived isotopes are not in equilibrium after the thermal-hydraulics perturbation.

Spatially and temporally resolved approaches, such as the one used by Tano et al., are more computationally expensive than those that only require solving a system of ordinary differential equations. However, Tano et al. has shown that including circulation of the fuel salt in depletion directly affects the equilibrium concentration of ^{135}Xe by approximately 45% compared to a spatially unresolved approach without flux scaling [177]. This indicates there may be a need for spatially resolved depletion simulations to get accurate fuel salt compositions.

Tano et al. and Betzler have both described fully modeling advection in depletion as computationally intractable [177, 16]. This is because tracking all nuclides over space and time requires far more computational work than handling a traditional time dependent system of ordinary differential equations (ODEs). This spatially resolved system of partial differential equations (PDEs) requires accuracy in the short time scale flow effects. These small time steps cause multi-year depletion simulations to be exceedingly expensive.

Numerical stability of the problem, when using an explicit-time solving scheme, uses the Courant number:

$$C = \frac{u\Delta t}{\Delta z}, \quad (2.59)$$

where,

Δt =the time step size [s]

Δz =the length between spatial nodes [cm].

The Courant number, C , must be less than or equal to 1 for the Courant–Friedrichs–Lewy condition in an explicit finite differencing scheme [162]. This equation limits the time step size based on the flow rate and spatial mesh size. Although using an implicit time solver can work around this condition, the overall computational cost remains high due to the number of nuclides, the time step size, spatial mesh resolution, and the overall amount of time simulated in depletion.

For a short depletion simulation, such as a saturation irradiation of a fuel sample, capturing these effects is computationally tractable. Because the depletion lasts only for several minutes instead of several years, the computational cost is orders of magnitude smaller. A high-fidelity, spatially resolved depletion simulation of an MSR sample irradiation experiment is feasible to generate improved DNP group parameters.

2.3 Safety, Security, and Safeguards

2.3.1 Safety

An important part of ensuring safe reactor operation is making sure safety-related transient events do not disrupt safe reactor operations. These transients can be caused by postulated initiating events (PIEs), which are defined in the IAEA Safety Standard Series No. NS-R-1 [1]. Of particular importance are PIEs which potentially lead to design basis accidents (DBAs). Although No. SSR-2/1 has superseded No. NS-R-1, the PIEs and DBAs for MSRs have not changed [2]. Although there are not specific events listed for MSRs, guidance is provided on categories of events are of interest.

These events include equipment failures, such as pumps failing or pipes rupturing; human error, such as improper maintenance or faulty operator actions; external events, such as an earthquake; and combinations of events. For combinations of events, the IAEA specifies that these events should make sense in combination and in

the time frame modeled. For example, if a particular event lasts a long time or has a long lasting effect, then that event could possibly overlap with another unrelated event. Alternatively, dependent events could potentially occur at the same time.

Safety-related transients for MSRs include ULOHS, ULOF, unprotected transient over power (UTOP), and fuel salt over cooling (FSOC) [103, 41]. An ULOHS transient is when the heat exchanger experiences a failure, possibly caused by a pump failure in the intermediate loop, causing a loss of cooling of the reactor. For the MSFR, Brovchenko et al. have shown that slow ULOHS transients that take place over a minute or longer minimize the increase in fuel salt temperature [27]. An ULOF transient is when the flow rate in the primary loop becomes zero, caused by pump failure in the primary loop. An UTOP transient is a reactivity insertion event, which could be caused by various means, such as removing poisons, adding fissile material, or temperature changes. A FSOC transient is when the fuel salt temperature decreases, which could be caused by an increased heat removal rate from the heat exchanger.

The current work is important for safety-related transients, as the MSR specific DNP group parameters will impact the results of these transients. Depending on the properties of the specific reactor, which affect the group parameters, the static DNP group parameter transient simulation may generate overly or insufficiently conservative results.

2.3.2 Security

The goal of security, or physical protection (PP), is to prevent actors from acquiring information, stealing nuclear material, or committing sabotage [140]. Security is enforced through physical protection systems, which detect, delay, and neutralize the adversarial actor before they can complete their objective [7, 64]. There are several different categories which are important to understanding security: the actor type, the actor capabilities, their objectives, and the strategies they use to achieve said objectives. The main consideration for security in this work is protection of the public by preventing: theft of hazardous radioactive material and sabotage with the intent to damage facilities, disrupt operations, or release radioactive material. The majority of the information presented in this section comes from The Proliferation Resistance and Physical Protection Evaluation Methodology Working Group of the Generation IV International Forum [140].

The actor type involved in PP can be an individual, a group, a large organization, or a non-host state combined with any of the former. These actors may be composed of some combination of insiders and outsiders. Insiders are employees or contract staff who may be sympathetic with or suborned by the hostile actor. Outsiders can include terrorist groups, agents of proliferant states, or lone individuals. Based on the specifics of the actor type, their capabilities will vary significantly.

There are five categories of actor capabilities: knowledge, skills, weapons and tools, number of actors, and dedication. The knowledge of the actor will vary depending on if they have insider knowledge or not, and how much information that insider knows. The skills of the actor will affect their strategy they use to achieve their goals, though a large organization or non-host state can be assumed to have access to a wide variety of skills. The weapons and tools of the actor will affect their strategy, but access to an insider would grant more tools accessed with less effort. The number of actors will vary with the specific strategy of that actor as well as their tactics, as a more discrete approach would require fewer actors than a forceful approach. The dedication of the actor is important to consider, as an insufficiently motivated actor may be deterred with sufficient security measures.

The objectives of the actor may include: disruption of operations, radiological release, nuclear explosives, Radiological Dispersal Devices (RDDs), and/or information theft. The objectives of the actor will influence their other decisions, particularly their strategies. If an actor's objective requires force, then the capabilities and strategies implemented by that actor will vary from one whose objective can be accomplished covertly. Assuming the actor is

rational, they will select a strategy which optimizes the probability of successfully completing their objective.

The actor's strategies include their modes of attack and tactics. The modes of attack vary significantly, including cyber attacks, waterborne attacks, direct or diversionary attack using light anti-tank weapons, and more. The tactics actors can use include stealth, deceit, and overt force. The stealth and deceit tactics are more covert, including avoiding or deactivating security systems for the stealth tactic or using false identification or authorization for the deceit tactic. The overt force tactic ties more directly into many of the modes of attack, which may itself also serve as a diversion for a different tactic.

For this work, the specific actor capabilities and strategies are not considered. Additionally, only a single type of actor and objective combination is considered – an actor with access to an insider and the objective to disrupt operations through sabotage. This work investigates the formulation of new DNP group parameters, which impact the reactor behavior during a transient. There may be security issues that arise during a sabotage-initiated transient due to the recalculated kinetics and DNP group parameters.

2.3.3 Safeguards

Preventing the spread of nuclear material for weapons is the goal of nuclear safeguards. Safeguards is defined as the nuclear reactor characteristics that impede "...the diversion or undeclared production of nuclear material and the misuse of technology by the Host State seeking to acquire nuclear weapons or other nuclear explosive devices." [140]. This definition agrees with the definition made at the IAEA international workshop in 2002 [81]. In this work, proliferation resistance (PR) is used as a way to understand and measure certain criteria used in safeguards. PR is a spectrum, so it should not be considered necessary to optimize the PR, but rather as a way to evaluate strengths and weaknesses in the design relating to safeguards.

There are six measures for PR: technical difficulty, cost, time, fissile material type, detection probability, and detection resource efficiency. In this work, the measures analyzed include time and detection probability. The time measure is analyzed by conducting diversion scenarios over various time frames to determine the effects on diversion fingerprints. A very low PR time is 0-3 months, while a very high PR time is >30 years [140]. The detection probability measure goes hand-in-hand with time, as a protracted diversion scenario is likely to have a reduced detection probability. A very low PR detection probability is 0-5 percent, while a very high PR detection probability is 95-100 percent [140].

The Department of Energy (DOE) released an updated nuclear material control and accountability (MCA) Order in February of 2023 that cancels DOE Order 474.2 Chg 4, which was last updated in 2016 [75]. Based on the more recent Order, special nuclear material (SNM) is classified as shown in Table 2.14.

Table 2.14: The SNM classification table is recreated from [75].

Material Type	Accountable Quantity [g]	Weight Field Used for Element	Weight Field Used for Isotope
^{235}U	1.0	total uranium	^{235}U
^{233}U	1.0	total uranium	^{233}U
^{242}Pu	1.0	total plutonium	^{242}Pu
$^{239-241}\text{Pu}$	1.0	total plutonium	$^{239}\text{Pu} + ^{241}\text{Pu}$
^{238}Pu	0.1	total plutonium	^{238}Pu
$^{241}\text{Am}^{**}$	1.0	total americium	^{241}Am
$^{243}\text{Am}^{**}$	1.0	total americium	^{243}Am
$^{237}\text{Np}^{**}$	1.0	total neptunium	-

^{**} Americium and neptunium are treated as SNM if they are separated.

Table 2.14 is used along with the graded safeguards table from the updated Order to properly classify the attractiveness level and categorization of SNM. These categories include weapons, pure products, high-grade materials, low-grade materials, and all other materials, ranging from attractiveness level "A" to "E". For this work, only materials of attractiveness level "C" and below are relevant. High-grade materials, with attractiveness level "C", include fuel elements and assemblies. Low-grade materials, with attractiveness level "D", include solutions with target concentrations of 1-25 g/L. All other materials, with attractiveness level "E", are made up of highly irradiated forms, solutions with concentrations <1 g/L, or uranium containing <20% ^{235}U or <10% ^{233}U .

The quantities provided in Table 2.14 show the minimum reportable, or accountable, quantities for each material type. These quantities make up the lower limit of Category IV materials for "B" through "E" attractiveness level materials. The SNM Category depends on the quantity of material as well as the attractiveness level of that material. Higher Category material has more stringent requirements due to their greater strategic importance, with Category I SNM having the most requirements. For example, if material is classified as attractiveness level "C", then any quantity of plutonium or $^{233}\text{U} \geq 6$ kg is considered Category I SNM. Alternatively, if the same material contains ^{235}U , separated ^{237}Np , or ^{241}Am and ^{243}Am , then an amount ≥ 20 kg is considered Category I SNM.

If the material in the reactor is classified as attractiveness level "D", then any quantity ≥ 16 kg is considered Category II, as Category I quantities do not exist for this attractiveness level. Material of attractiveness level "D" do not have a listed Category I quantity for material containing ^{235}U , separated ^{237}Np , or ^{241}Am and ^{243}Am .

Material of attractiveness level "E" does not have Category I, II, or III quantities. Instead, reportable quantities are considered Category IV.

Significant quantities (SQs) of material are also of interest, of which there are two categories: direct use and indirect use. Direct use nuclear material consists of ^{233}U , highly enriched uranium (HEU) (^{235}U enriched over 20%), and any amount of ^{239}Pu with less than 80% ^{238}Pu . Indirect use nuclear material consists of thorium and low enriched uranium (LEU) (^{235}U enriched at less than 20%). SQs of the different material categories are given in Table 2.15 [140, 3, 12].

This information is important for the current work, as it provides values for quantities of diverted material to analyze. The exact classification, materials, and quantities to use will vary based on the specifics of the fuel salt and diversion scenario. However, this information provides a set of data which can be directly referenced to determine

Table 2.15: Significant quantities of various materials are given along with their use category.

Use	Material	Significant Quantity [kg]
Direct	Plutonium	8
Direct	HEU	25
Direct	²³³ U	8
Indirect	²³⁵ U	75
Indirect	Natural Uranium	10,000
Indirect	Depleted Uranium	20,000

the diversion rates.

For example, a fuel salt categorized as direct use with a plutonium diversion would have 8kg of plutonium as a significant quantity. If that fuel salt also had an attractiveness level of "D", then 16kg of plutonium would be Category II. Then, the diversion time could vary over a range of months to years, based on various PR times. A method of detection for these diversion scenarios is via change in reactivity due to the change in delayed neutron precursors and fissile concentration. The detection probability is then estimated based on this measurable reactivity change.

Chapter 3

Methodology

This chapter details the tools and methods used to both generate the improved DNP group parameters and to validate them. Three models are covered in this chapter: the 0D scaled model, which is already complete; the 0D flow model, which is proposed; and the spatially resolved model, which is discussed for potential avenues of future research. Although each model varies in its fidelity and cost, the general methodology is very similar: generate concentrations of DNPs from irradiation of a sample, convert those concentrations into a delayed neutron count rate, and then use non-linear least squares to fit DNP group parameters to the count rate curve. Validation of the DNP group parameters requires solving kinetics problems that make use of the parameters. There are also transient problems of interest that affect safety, security, and safeguards by design.

3.1 Tools

Several external software tools evaluate and generate the improved DNP group parameters. Molten Salt Delayed Neutron (MoSDeN), an in-house developed Python package, collects and combines data, OpenMC is used for data handling and higher fidelity analysis, and the MOOSE app Moltres evaluates and refines them at different fidelity levels [183, 149, 105]. Preliminary results use MoSDeN to generate the improved DNP group parameters, while the proposed work will use OpenMC and Moltres. Each tool is described in detail in its corresponding sub-section.

3.1.1 MoSDeN

MoSDeN is a Python package designed to enable generation of DNP group parameters using the microscopic approach for any given reactor of interest. MoSDeN allows for varying levels of fidelity in the generation, ranging from only using nuclear data and broad assumptions to a high-fidelity multiphysics simulation with neutronics and thermal-hydraulics. The MoSDeN package relies on the SciPy, NumPy, Matplotlib, pandas, and Uncertainties packages [187, 70, 77, 179, 55]. Many features of MoSDeN also require OpenMC to function properly [149]. MoSDeN automates data collection and processing, simulates pulse and/or saturation irradiations, converts the concentrations into a delayed neutron count rate, and generates group parameters.

The preliminary work on MoSDeN uses the 0D scaled model, while the proposed work will make use of the 0D flow model. Both models are discussed in more detail in Section 3.2, as is the spatially resolved model, which is proposed for future work.

3.1.2 OpenMC

OpenMC is an open-source Monte Carlo neutral particle transport code. Its Python Application Programming Interface (API) offers functionality that simplifies interfacing with MoSDeN, an in-house developed open-source Python package. The sample irradiation process is straightforward and configurable, as users can adjust parameters through an input file. OpenMC also offers useful feature in depletion, where a zero-power depletion step is rapidly calculated since only decay is tracked. This feature allows resolving ex-core residence time depletion steps quickly. Additionally, OpenMC's transfer rate functionality during depletion enables simulation of chemical removal in an MSR.

For this work, MoSDeN by default uses the ENDF/B-VII.1 based depletion chains provided by OpenMC and ENDF/B-VII.1 cross section data. The thermal spectrum chain applies to neutron irradiation below 1 *eV*, the fast spectrum chain applies to neutron irradiation above thermal and below 5 *MeV*, and anything above that is not treated in this work [172]. The models by default assume monoenergetic neutrons irradiate the sample.

3.1.3 Moltres

Moltres is an open-source MOOSE application that simulates MSRs by solving the multi-group neutron diffusion and Navier-Stokes equations [105]. MOOSE is an open-source, parallel, finite element framework for multiphysics applications [66]. MOOSE includes many pre-built applications that couple physical phenomena. The HEAT CONDUCTION and NAVIER-STOKES modules are particularly useful in this work, as they can couple with Moltres for general thermal-hydraulic modeling [131, 104, 133, 129]. Moltres will simulate the MSRE with different DNP group parameters. Moltres will also validate the improved DNP group parameters in this work.

The MSRE reactivity data can be used for validation, and was previously modeled with Moltres [105]. The MSRE core consists of flow through channels, simplifying its modeling compared to open-pool designs due to the restricted flow path. Since the initial demonstration, Park et al. have noted general agreement in results, attributing differences to simplifications and assumptions in the geometry [129]. Additionally, Park has published a dissertation detailing 1-D, 2-D, and 3-D neutronic models of the MSRE [128].

There are two main capabilities in Moltres which are of interest. Moltres couples three physical phenomena to model an MSR: neutron flux, DNPs, and thermal-hydraulics. This allows Moltres to model transient simulations of MSRs, which will be used to validate the improved DNP group parameters. Moltres can also simulate transients using the MSR-specific DNP group parameters, and it can generate improved parameters with higher fidelity than the other methods. This is of interest for future work, which may use higher-fidelity multi-physics to generate improved DNP group parameters.

Neutron Flux Modeling

Neutron flux modeling determines the energy of neutrons interacting with the sample and the number of interactions. It determines the fission rate and the resulting fission product concentrations, including DNPs. The time-dependent multi-group neutron diffusion equation appears in Equation (2.34), recreated here:

$$\begin{aligned}
\frac{1}{v_g} \frac{\partial \phi_g}{\partial t} = & \nabla \cdot D_g \nabla \phi_g + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} \\
& + (1 - \beta) \chi_{p,g} \sum_{g'=1}^G (v \Sigma_f)_{g'} \phi_{g'} \\
& - \sum_{i=1}^I \chi_{d,g,i} \lambda_i C_i - \Sigma_{a,g} \phi_g - \sum_{g' \neq g}^G \Sigma_{s,g \rightarrow g'} \phi_g
\end{aligned}$$

Modeling the neutron flux requires accounting for the DNPs. DNPs contribute to the neutron flux and possess their own energy spectra, necessitating their inclusion as a separate term.

DNP Modeling

The time dependent DNP concentration is given in Equation (2.35) and repeated here:

$$\frac{\partial C_i}{\partial t} = \beta_i \sum_{g=1}^G (v \Sigma_f)_g \phi_g - \lambda_i C_i - \nabla \cdot (C_i \vec{u}) + \nabla \cdot \frac{v_T}{S c_T} \nabla C_i$$

This is the primary equation of interest when validating the improved DNP group parameters.

For the future work involving formulation of improved DNP group parameters within Moltres for the spatially resolved model, this equation is less helpful. This is because it requires DNP group parameters, which Moltres should be generating. To generate the DNP group parameters, this equation is replaced by (2.58), recreated here:

$$\frac{dN_j}{dt} = S(\vec{r}, t) - \lambda_{net,j} N_j(\vec{r}, t) + \nabla \cdot (D \nabla N_j(\vec{r}, t) - \vec{u}(\vec{r}, t) N_j(\vec{r}, t))$$

The spatially resolved model will use spatially resolved depletion over the irradiation simulation. The change to the equation results in tracking concentrations of the individual DNPs instead of tracking concentrations of DNP groups.

Thermal-Hydraulic Modeling

Moltres offers time-dependent, incompressible, buoyancy-driven flow [128]:

$$\rho \frac{\partial \vec{u}}{\partial t} = -\rho(\vec{u} \cdot \nabla) \vec{u} - \nabla p + \mu \nabla^2 \vec{u} + \rho \alpha \vec{g} (T - T_{\text{ref}}), \quad (3.1)$$

$$\nabla \cdot \vec{u} = 0, \quad (3.2)$$

where

$$\begin{aligned}
\rho &= \text{fluid density } [kg \cdot m^{-3}] \\
p &= \text{pressure } [Pa] \\
\vec{u} &= \text{flow rate } [m \cdot s^{-1}] \\
\mu &= \text{dynamic viscosity } [Pa \cdot s] \\
\alpha &= \text{coefficient of thermal expansion } [K^{-1}] \\
\vec{g} &= \text{gravitational force vector } [N] \\
T &= \text{fluid temperature } [K] \\
T_{\text{ref}} &= \text{reference temperature at which the nominal} \\
&\quad \text{density is provided } [K].
\end{aligned}$$

The temperature distribution is given by [128] as:

$$\rho c_p \frac{\partial T}{\partial t} = -\rho c_p \vec{u} \cdot \nabla T + \nabla \cdot (k \nabla T) + Q_f - Q_s, \quad (3.3)$$

where the fission heat source term is defined [128] as:

$$Q_f = \sum_{g=1}^G \epsilon_g \Sigma_g^f \phi_g, \quad (3.4)$$

where

$$\begin{aligned}
c_p &= \text{specific heat capacity of molten salt } [J \cdot kg^{-1} \cdot K^{-1}] \\
k &= \text{effective thermal conductivity of molten salt } [W \cdot m^{-1} \cdot K^{-1}] \\
Q_f &= \text{fission heat source } [W \cdot m^{-3}] \\
\epsilon_g &= \text{average fission energy released by neutrons in group } g [J] \\
Q_s &= \text{heat sink/removal } [W \cdot m^{-3}].
\end{aligned}$$

With the combination of the flux modeling, precursor modeling, and thermal-hydraulic modeling, Moltres can almost simulate MSRs. However, there is one additional MSR phenomenon required to complete the model, the flow of the fuel salt between the in-core to ex-core regions.

Ex-core and In-core Coupling

Moltres treats the ex-core region as a tightly-coupled 1D pipe geometry with coupling using Picard iterations and the inlet/outlet boundary conditions. The specific equations to treat the variables of interest in this geometry can be derived from their full forms. For example, the flux and eddy viscosity are zero in the ex-core region, leading to the precursor concentration in the ex-core region:

$$\frac{\partial C_i}{\partial t} = -\lambda_i C_i - u_x \frac{\partial C_i}{\partial x}. \quad (3.5)$$

3.2 DNP Group Generation

Generating DNP group parameter data in a reactor-agnostic manner requires a reactor-agnostic generation process. Fully detailing the flow regime with a large number of parameters is computationally infeasible. Instead, the flow is simplified to two parameters: an in-core average residence time, τ_{in} , and an ex-core residence time, τ_{ex} .

This approach captures fuel salt advection around the primary loop and approximates turbulent and diffusive mixing. The terms derive from the averaged behavior of a fuel salt control volume, treating thermal-hydraulic effects, such as vortices, as changes in residence times. This method allows for approximate fitting of any circulating fuel reactor, requiring only the generation of relevant parameters for determining the associated DNP group parameters.

The other required parameters describe chemical removal, which varies by fission product and reprocessing rate. Refueling could also be of interest, but only a single sample of fissile material is considered during irradiation. The introduction of fresh fuel would affect the incident neutron energy spectrum on the fissile sample, and is considered in Section 5.4. An irradiation time of approximately seven minutes suffices to reach equilibrium DNPs concentrations, as the saturation time frame is on the order of multiple half-lives of the longest-lived DNP. Elements with cycle times on the order of months to years exhibit negligible changes, as discussed in Section 4.2.1. Elements with changes on the order of seconds to days contribute sufficiently and must be included for accurate data generation.

The MSBR reprocessing scheme serves as the basis for analyzing the most extreme cycle times in this work. This is because the MSBR presents very large chemical removal rates, providing useful conservative estimates as to the impact these large removal rates could have on DNP group parameters. These chemical removal rates are given in Table 4.3. Rare earths, semi-noble metals, volatile fluorides, discard, and higher nuclides have small removal rates relative to the time scales of interest for saturation irradiations. Section 4.2.1 examines these longer cycle times, the inverse of the removal rates, to confirm their negligible impact. Section 4.2.1 scales the cycle times to analyze how reprocessing rates influence the group parameters.

Table 3.1: MSBR Online Reprocessing Removal Rates from [146].

Reprocessing Group	Element(s)	Removal Rate [s^{-1}]
Rare Earths	Y, La, Ce, Pr, Nd, Pm, Sm, Gd	2.32×10^{-7}
Rare Earths	Eu	2.32×10^{-8}
Noble Metals	Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te	5.00×10^{-2}
Seminoble Metals	Zr, Cd, In, Sn	5.79×10^{-8}
Gases	Kr, Xe	5.00×10^{-2}
Volatile Fluorides	Br, I	1.93×10^{-7}
Discard	Rb, Sr, Cs, Ba	3.37×10^{-9}
Salt Discard	Th, Li, Be, F	3.37×10^{-9}
Protactinium	Pa	3.86×10^{-6}
Higher Nuclides	Np, Pu	1.98×10^{-9}

With the chemical reprocessing scheme in place, the next step to generate the improved DNP group parameters is to decide on the level of fidelity. The three levels of fidelity are the 0D scaled model, the 0D flow model, and the spatially resolved model. The 0D scaled model is completed, the 0D flow model is planned, and the spatially resolved model is a potential future development.

A Monte Carlo sampling approach is used to propagate uncertainties in the various inputs, such as in emission probabilities, half-lives, fission yields, reprocessing terms, or residence times. The sampling approach simply repeats the process for generating the DNP group yields, but with randomly sampled input data. The data itself must be perturbed, and not the delayed neutron count rate itself. This is because the uncertainties in the delayed neutron count rate are not independent from time step to time step, but instead they all depend on the underlying data which changes the shape of the delayed neutron count rate at all times.

For generation of the improved DNP group parameters, the main goal is to formulate them for individual fissile nuclides. For some of the analyses presented in this work, the fissions do not come from a single fissile nuclide. For these analyses, a fission-weighted approach is used, where the DNP group parameters are scaled based on the fraction of fissions that lead to those group parameters. Equations (2.13) and (2.14) show how the fission-weighted DNP group parameters can be calculated. The fission-weighted values can then be multiplied by an importance term to account for the difference between the delayed and prompt neutron fission spectra, as shown in Equations (2.15) and (2.16) [141].

3.2.1 0D Scaled Model

This is the lowest fidelity of the three models, and it primarily serves as a way to verify the microscopic approach is working as expected by comparing with existing literature. This approach can only model saturation concentrations, as there is no decay chain tracking nuclides decaying into each other. It is called the 0D scaled model because it uses the scaled flux method, discussed in Section 2.2.2. The cumulative fission yield is used to calculate the equilibrium DNP concentrations, with a scaling term based on the amount of fuel salt in the core. An example solution with no reprocessing is shown in Equation (3.6) as [79]:

$$N_i = \frac{f_{in} C F Y_i}{\lambda_i}, \quad (3.6)$$

where,

f_{in} = the fraction of fuel-salt in the core.

The derivation for this equation is given in Appendix B. It is important to note that the flux is scaled when calculating the fission rate as well, by the same f_{in} term that scales the cumulative fission yield. Effectively, this cancels the term out, resulting in no deviation in the results based on the in-core fuel salt fraction by reprocessing time.

No parasitic absorption is accounted for in this model, nor is time dependent decay of nuclides into DNPs. There is a possible approach to include the effects of parasitic absorption, but this approach is not included [164, Benedict et al.].

Figure 3.1 shows the how the different data is loaded into MoSDeN, including external data and user defined input data. The green circles represent input data required for generating the DNP group parameters. The red diamond represents MoSDeN, the tool takes in the data to simulate irradiation of the target fissile sample to generate the irradiated concentrations, the delayed neutron count rate, and finally the DNP group parameters that fit that count rate. The blue squares represent the outputs, in this case the DNP group parameters: the DNP group yields

and the DNP group half-lives.

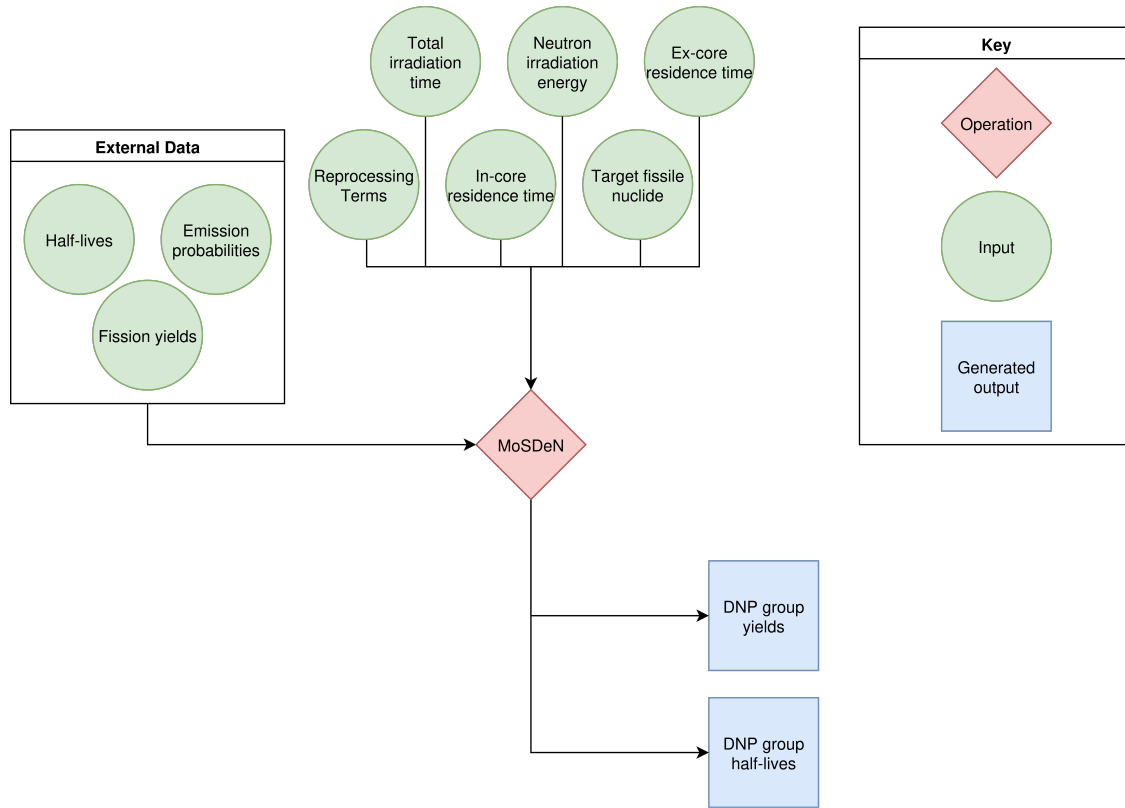


Figure 3.1: Flow diagram of 0D scaled model used to generate improved DNP group parameters.

After all this data is loaded into MoSDeN, it generates the concentration of each DNP. This concentration is then combined with emission probabilities and decay constants to form a delayed neutron count rate. A non-linear least squares solve then fits the improved DNP group parameters to the delayed neutron count rate, generating the group yields and half-lives, given as blue squares in Figure 3.1.

This model is only valid for saturation irradiation simulations, as the concentration calculated in Equation (3.6) is the equilibrium concentration. For a pulse irradiation, this approach is not valid. There are four different options available to handle pulse irradiations: reduce net decay measurement time, pull only the shortest lived group(s) with independent fission yields, use OpenMC depletion, or simply do not use the shortest lived groups typically generated from a pulse irradiation since they are unlikely to be changed by MSR physical phenomena. The first approach, reducing the net decay time, will result in overfitting of the short lived groups, leading to less accurate results. The second approach, using only the shortest lived groups with independent fission yields means that the decay chains will not be handled, but the initial pulse concentration should be accurate. However, the concentrations will quickly become less accurate as the decays of various nuclides into DNPs will not be treated, so this approach could only be potentially viable for the shortest lived group. The third approach, using OpenMC depletion, would be accurate and easily achievable. However, it would greatly increase computational cost, and the use of OpenMC in such a case is already handled by the 0D flow model, making it redundant. The final approach would entail only using the longer lived DNP group parameters from a saturation irradiation, while recognizing that the short lived groups are less accurate. Because the MSR physical phenomena that cause non-negligible influence on the DNPs take at least several seconds to come into effect, it is reasonable to focus on the longer lived groups.

3.2.2 0D Flow Model

The 0D flow model enables improved analysis of the general trends in nuclide evolutions. It treats the flow in a zero-dimensional manner using in-core and ex-core residence times, and uses OpenMC to track the full depletion matrix. The fission history function is a square wave, with a constant fission rate during the in-core residence time and no fission during the ex-core residence time. This approximation simplifies the analytical form of the fission history for creating the non-linear least squares problem, as shown in Appendix A.

Figure 3.2 shows the flow diagram for a fission history to simulate irradiation on a fissile sample circulating in the primary loop. The model passes eight parameters into OpenMC [149]: in-core residence time, ex-core residence time, neutron irradiation energy, target fissile nuclide, target density, total irradiation time, temperature, and reprocessing terms. The external data passed into OpenMC includes the emission probabilities and the depletion matrix. The depletion matrix contains information on half-lives, fission yields, cross-sections, branching ratios, and other useful data needed for OpenMC depletion. Section 3.3 provides further details on this process.

After OpenMC completes the fixed-source simulation, the final nuclide concentrations in the irradiated sample are saved for analysis in MoSDeN. MoSDeN combines the concentrations with data from the IAEA beta-delayed neutron emission database to generate a delayed neutron count rate [49]. A non-linear least squares fitting function from SciPy then calculates the group half-lives and yields using the Trust Region Reflective algorithm [188, 26]. These group parameters correspond to a specific nuclide, neutron energy, in-core residence time, ex-core residence time, reprocessing scheme, and reprocessing scalar. Section 3.4 provides further details on this block. Some additional factors to consider include self-shielding due to sample density (while avoiding self-multiplication), temperature-induced cross-section variations, and under- or over-irradiation, all of which can affect the results [87].

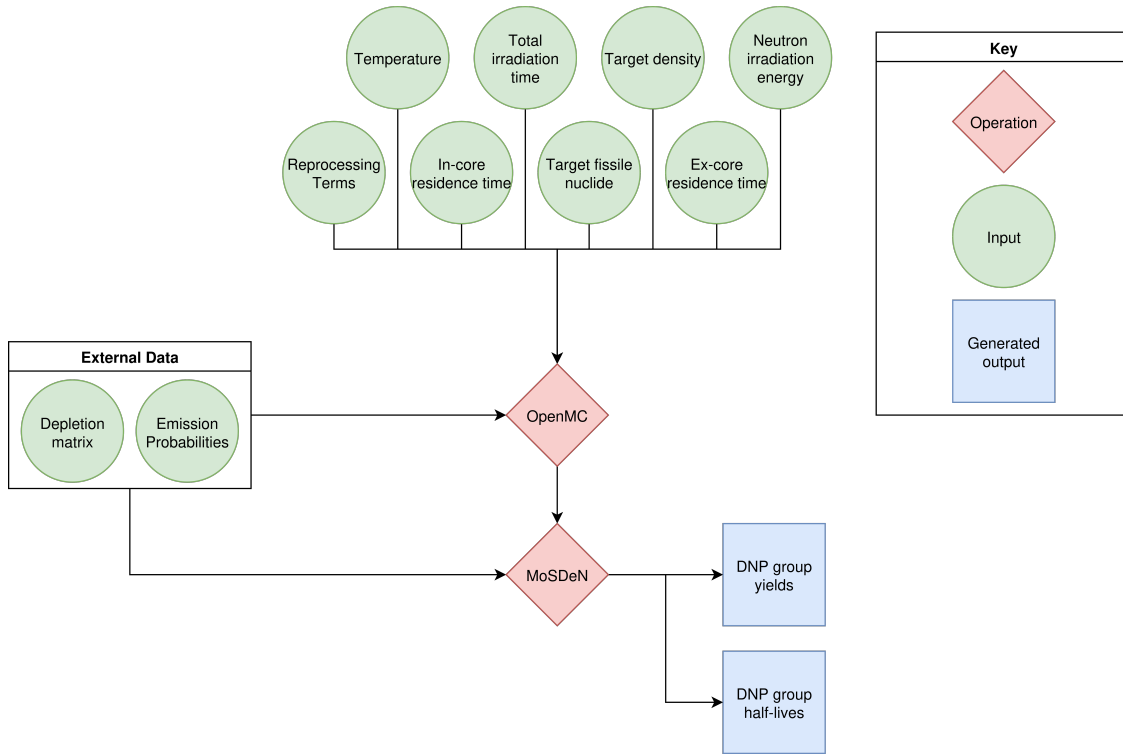


Figure 3.2: Flow diagram of 0D flow model used to generate improved DNP precursor group parameters.

The OpenMC model uses a three-gram spherical sample with vacuum boundary conditions. It treats the

problem as a fixed-source simulation, with a uniform neutron source distributed evenly throughout the sphere. This approach is not physical, as a real neutron source would have a spectrum and originate externally. However, the exact neutron spectrum for irradiating samples to generate DNP group parameters is not provided in the literature. The source distribution also has negligible impact, as the sample is small and exhibits minimal self-multiplication. Thus, this approximation does not limit the effectiveness of the 0D flow model.

Another consideration for the 0D flow model is maintaining a consistent source particle rate. While not strictly necessary for group parameter generation since the values are per-fission, it simplifies the data analysis, as the tallies depend on the source particles. Equation (3.7) provides the source rate for the irradiation simulations:

$$\dot{S} = S_{net} \frac{\tau_{in} + \tau_{ex}}{\tau_{in} T}, \quad (3.7)$$

where

- $\dot{S}$ = the source particle rate [$n \cdot s^{-1}$]
- S_{net} = the total number of source particles [n]
- τ_{in} = the in-core residence time [s]
- τ_{ex} = the ex-core residence time [s]
- T = the net irradiation time [s].

3.2.3 Spatially Resolved Model

The 0D flow model rapidly generates improved DNP group parameters, but it simplifies the treatment of flow. Instead of modeling the actual fuel salt flow, it estimates fuel circulation using residence times for each region. A spatially resolved model is useful for capturing the effects that the 0D flow model only capture with residence times for specific MSR geometries. Figure 3.3 shows the flow diagram for the spatially resolved model, which models the flow using Moltres.

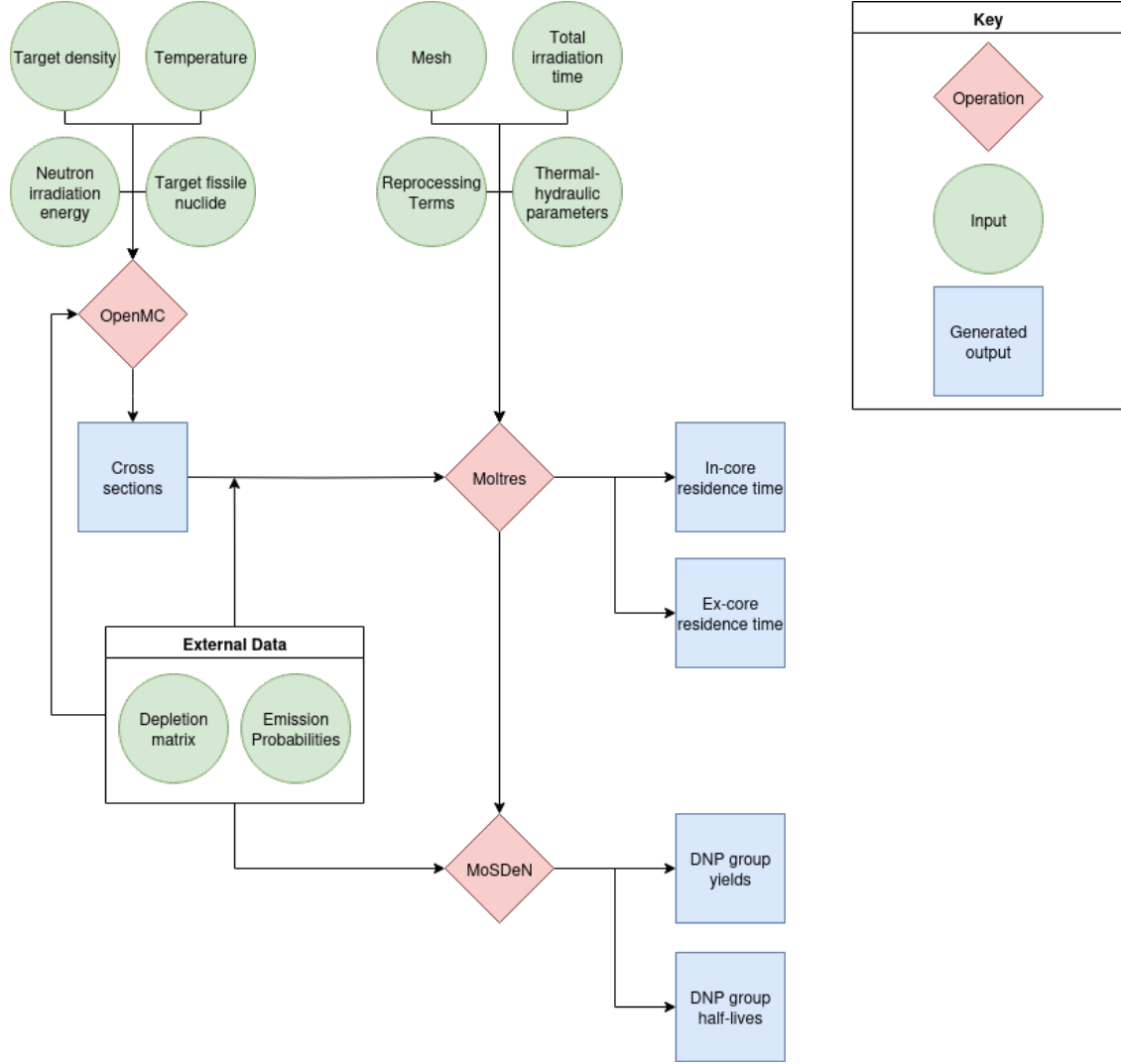


Figure 3.3: Flow diagram of spatially resolved flow model used to generate improved DNP group parameters.

The spatially resolved flow model introduces two new parameters absent in the 0D flow model. The first is the mesh, which defines the problem geometry. MOOSE provides built-in meshing tools, though complex geometries may require external tools such as Gmsh [65]. The second is the thermal-hydraulics parameters, which capture physical phenomena enabled by spatial resolution, as discussed in Section 3.1.3. Additionally, since the depletion happens in Moltres instead of OpenMC, OpenMC does not need as much data.

In the spatially resolved model, the in-core and ex-core residence times are outputs rather than inputs. These times derive from the movement of the irradiated sample over time in the fuel salt. The in-core and ex-core residence times vary as the irradiated sample diffuses and mixes further into the fuel salt. The net residence time, τ_{net} , remains constant unless the model changes, such as an increased temperature difference or induced flow rate. Equation (3.8) defines the in-core residence time as:

$$\tau_{in}(t) = \frac{\int_{V_{in}} \int_0^t N(\vec{x}, \tau) d\tau d\vec{x}}{\int_V \int_0^t N(\vec{x}, \tau) d\tau d\vec{x}} \tau_{net}, \quad (3.8)$$

where

$$\begin{aligned}
V_{in} &= \text{the in-core volume [cm}^3\text{]} \\
V &= \text{the total primary volume [cm}^3\text{]} \\
N(\vec{x}, t) &= \text{the time and space resolved sample concentration [atoms} \cdot \text{cm}^{-3}\text{]} \\
\tau_{net} &= \text{the net residence, or circulation, time of the fuel salt [s].}
\end{aligned}$$

The ex-core residence time can be calculated in the same way, or alternatively as shown in Equation (3.9) as:

$$\tau_{ex} = \tau_{net} - \tau_{in}. \quad (3.9)$$

3.3 Delayed Neutron Count Rate Generation

Although the various models have been discussed, execution details were not. The following sub-sections provide specific information on the methodology behind the generation of the improved DNP group parameters. This includes a detailed discussion on the concentration generation and the data used with those concentrations to produce a delayed neutron count rate.

3.3.1 DNP Concentrations

The concentrations from the 0D scaled model are calculated using Equation 3.6, while the concentrations in the spatially resolved model can be calculated using Equation 2.58. The 0D flow model will use OpenMC to handle calculation of the concentrations over time.

The 0D flow model will use a fixed-source problem on a three-gram spherical sample of a fissile target in OpenMC. A uniformly distributed source will be placed inside the target with a vacuum boundary condition at the edge of the sphere. The sample must have a sufficiently large volume to ensure a high number of fissions, but excessive volume increases self-shielding and self-multiplication. Adjusting the density mitigates this issue: higher density increases self-shielding and self-multiplication, while lower density reduces fission probability and worsens statistical accuracy. By using a uniformly distributed neutron source within the sample, these effects are lessened. Additionally, since the fission counts can be taken directly from OpenMC tallies, self-multiplication and self-shielding can be accounted for directly without introducing error.

The fixed-source in this model will consist entirely of neutrons at a single energy, rather than a spectrum. For the proposed analyses, only thermal fission of ^{235}U is studied. This energy dictates fission cross sections, transmutation cross sections, and fission yields. Traditionally, tabulated group parameter data is formatted for a single energy or energy range, such as fast or thermal. However, macroscopic experiments, discussed in Section 2.1.1, use reactors with varying neutron energy spectra, not mono-energetic neutrons. This could be addressed by modeling a reactor in OpenMC and using a k-eigenvalue solve to generate a neutron energy spectrum. This analysis is considered outside the current scope, but is suggested as future work in Section 5.4. Photons can also generate delayed neutrons through photofission, though this is also considered outside the current scope [90].

The neutron source distribution does not replicate the experimental process, where neutrons irradiate the sample externally rather than internally. However, an evenly distributed source eliminates self-shielding concerns. The exact spatial locations of fission events within the sample should negligibly affect the nuclide concentrations, which are the outputs of the OpenMC simulation. With a density of $10 \text{ g} \cdot \text{cm}^{-3}$, the spherical sample has a radius of

approximately 0.4 cm.

After running the OpenMC simulation, the concentrations are collected from the depletion results for the final irradiation time step. These final concentrations are then converted into a delayed neutron count rate.

3.3.2 Concentrations to Count Rate

After receiving the concentrations from OpenMC, MoSDeN determines the concentration of each nuclide as a function of time during the measurement period. The concentrations are calculated based on half-life, as shown in Equation (3.10), where the concentration as a function of time is:

$$N(t) = N_0 e^{-\lambda t}. \quad (3.10)$$

Equation (3.10) shows the approach used in the 0D scaled model, and does not account for decay chains between nuclides. However, the general form of the equation can still be used, but with inclusion of the various decay chains and branching ratios. A simple way of implementing decay with the decay chain is using zero-power depletion simulations in OpenMC. This is used in the 0D flow model, though the 0D flow model also offers the option of using Equation (3.10). These concentrations are available at coarse time steps and linearly interpolated using SciPy, allowing MoSDeN to access more accurate concentrations at any time after irradiation [186]. The specific time steps at which concentrations are written will be investigated after development of the 0D flow model.

The concentrations are combined with Equation (2.2), recreated here:

$$\dot{n}_d(t) = \epsilon \sum_{k=1}^K P_{n,k} \lambda_k N_k(t).$$

The efficiency term, ϵ , is set to 1 in this work, as the simulated method includes every produced delayed neutron in the count rate. The contributions of each DNP over time are summed to generate the delayed neutron count rate.

3.4 DNP Group Reconstruction

Once the delayed neutron count rate is generated, it is used to generate the improved DNP group parameters, $\bar{v}_{d,k}$ and λ_k . Fitting the DNP group parameters to the delayed neutron count rate curve requires a non-linear least squares method, as both the group half-lives and yields must be determined. The equation describing the problem appears in Equation (2.3), recreated here:

$$\dot{n}_d(t) \approx \epsilon F_s \sum_{k=1}^K \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}.$$

An important consideration for this process is the selection of time values for generating the DNP group parameters. Increasing the number of time nodes raises computational costs, but too few nodes result in a poorer fit [126]. Section 4.2.2 shows the sensitivity study for the time nodes, for both the node spacing and the maximum time to measure delayed neutrons.

Another consideration for the non-linear least squares solve is the covariance, which is larger for the pulse irradiation than for the saturation irradiation. This results from multiplying the group yield by the group decay constant. Attempts to replace this multiplication with an independent term produces worse fits, as the multiplication is not independent and is necessary for accurate modeling.

The residual of the solve is given in Equation (2.10), recreated here as:

$$r = \min_x \left\| \frac{b - Ax}{b + \epsilon} \right\|_2^2.$$

Half-life bounds are set from 1 millisecond to 1000 seconds, and the yield bounds are from 0 to 1. Convergence tolerances for the cost function, change in independent variables, and gradient norm are all set to 1×10^{-12} to ensure tight convergence.

The final consideration in DNP group reconstruction is uncertainty. This work tracks uncertainties using the uncertainties package [55]. These uncertainties include uncertainties in fission yields, half-lives, and emission probabilities. The uncertainties are propagated through the count rate calculation, but for the non-linear least squares solve a Monte Carlo sampling based approach is used. Each count rate is generated using randomly sampled fission yields, half-lives, and emission probabilities within their uncertainty bounds for each DNP. This enables formation of a normal distribution of group parameters, from which the mean and standard deviation of the mean can be extracted.

Once the DNP group parameters and their uncertainties are generated, the final step to obtain the finalized group parameters is to combine the saturation and pulse irradiation results. Saturation irradiation provides better statistics for longer-lived groups, while pulse irradiation is better for shorter-lived groups. Brady combines the results by using four long-period groups from saturation irradiation and four short-period groups from pulse irradiation. The groups are renormalized at the second group, where the two longest-lived group half-lives and yield ratios use the saturation results [25]. The remaining groups use the pulse irradiation results.

This work uses a similar but slightly simplified approach. The group parameters are first calculated for both pulse and saturation irradiation. Then, the two longest-lived groups use the saturation irradiation results, while the four shortest-lived groups use the pulse irradiation results.

3.5 Impact of DNP Groups on Transients

The impact of the DNP groups on transients will be evaluated using Moltres with fluid-flow-velocity field coupled multi-group diffusion kinetics, as discussed in Section 2.1.3. This approach provides high-fidelity results by coupling thermal-hydraulics and neutronics, a necessity for MSR modeling. Modeling every DNP in space and time would be prohibitively expensive, so they are instead modeled using DNP group parameters. Moltres allows direct comparison between static DNP group parameters, which do not account for MSR phenomena, and parameters that do.

The results of interest from this analysis include the reactivity and power responses to various initiating events. The initiating events and transients vary, as different reactor responses are studied. The events of interest depend on the perspective from which the reactor response is analyzed.

Specifically, there are four different perspectives of interest in this work: validation, safety, security, and safeguards. Each perspective involves a unique combination of motivation, proposed initiating event, and investigated response. These are discussed in detail in their respective sections.

3.5.1 Validation

The motivation for validation is to confirm that the proposed methodology and results align with experimental data. This ensures the demonstrated improved DNP group parameters are applicable to actual MSRs. Discrepancies between experimental results and the model highlight potential weaknesses in the approach. If no discrepancies exist, confidence in the results increases for problems similar to those validated.

The validation of the DNP group parameters follows a multi-stage approach. The first stage compares the irradiated sample's delayed neutron count rate to the Keepin et al. experiment from 1957 [87]. If the count rates do not match, the issue lies in the data or the aggregation of individual DNPs into the count rate. If the count rate is accurately predicted, the formulation is considered valid. The second stage validates the improved DNP group parameters using MSRE pump start-up and coast-down experimental data to assess their accuracy in modeling reactivity changes from DNPs [139]. This experiment has been studied using Moltres and Quasimolto, allowing comparison with their results [142, 128]. However, previous models excluded the inlet and outlet plena, leading to discrepancies with experimental data. A model including the plena, combined with updated DNP group parameters, should yield results closer to the experimental data.

3.5.2 Safety

The motivation for studying safety is to protect the public and meet regulatory requirements. Safety is discussed in detail in Section 2.3.1, though the general takeaway is that many safety-related transients exist. The DNP group parameters are critical in these transients, as they influence the reactor period, power, and temperature.

The transients investigated are ULOHS, ULOF, UTOP, and FSOC. For each transient, the reactor period, power response, maximum temperature, minimum temperature, and reactivity response are recorded. The reactor period is critical, as it reflects the reactor's controllability. The power response shows how changes in the DNP group parameters affect other responses, such as temperature. It is important to make sure differences in the DNP parameters don't lead to temperature changes or oscillations which result in unsafe reactor conditions.

3.5.3 Security

Security is discussed in detail in Section 2.3.2, but the general takeaway is that bad actors may attempt to sabotage the reactor. This ties directly into the safety design study, as sabotage aims to create unsafe reactor conditions. Ensuring the MSR operates safely after sabotage is critical. If static DNP group parameters predict safe operation while improved parameters predict unsafe operation, correcting this discrepancy in the reactor design is essential.

Since the safety study covers most transients relevant to sabotage, additional considerations can be incorporated into these transients. As sabotage may initiate multiple safety-related transients simultaneously, it is reasonable to investigate such scenarios. With four unique transients, this results in six double-transient combinations, three triple-transient combinations, and one quadruple-transient combination, totaling ten security-related sabotage-initiated transients.

3.5.4 Safeguards

Safeguards are discussed in Section 3.5.4, but the main takeaway is the importance of preventing SNM diversion in MSRs. Detection methods, such as neutron detectors and surveillance systems, play a key role in safeguarding the reactor [200]. The improved DNP group parameters in this work affect the effective delayed neutron fraction and reactivity response. Using static DNP group parameters to detect SNM diversion predicts less accurate reactivity responses than the improved parameters. If the difference is significant, relying on static parameters may reduce the detection probability of a diversion scenario.

To investigate this, diversion scenarios are studied with a single significant quantity of material removed over the course of one year, simulated using OpenMC. The change in the fuel salt composition affects the DNP group parameters, as discussed in Section 2.1.2. The importance and fission-weighted DNP group parameters can be used in simple step reactivity insertion transients, where the following four sets of data are compared: static

group parameters without diversion, static group parameters with diversion, improved group parameters without diversion, and improved group parameters with diversion. If the difference between the static group parameters with and without diversion is smaller or similar to the difference between the static and improved group parameters, then this means detection of that diversion scenario requires models to account for the MSR physical phenomena when generating DNP group parameters.

Chapter 4

Preliminary Results

The preliminary results use the lowest fidelity model, the 0D scaled model, to generate improved DNP group parameters. Specifically, thermal fission of ^{235}U is used, since there is plenty of data in the literature to verify that the stationary case with no chemical removal matches expected values. These preliminary results investigate individual DNPs, the sensitivity of the DNP group parameters to different variables, and verification of the DNP group parameters.

Of the MSR physical phenomena considered, only the chemical removal will have an impact for the 0D scaled model. This is because, although the delayed neutron count rate varies for different combinations of in-core and ex-core residence times, the number of fission events which led to that count rate also varies. Specifically, for the scaled flux method applied to the cumulative fission yield, the scaling term for both is the same. Although this approach is rapid and enables more simulations to be executed, it also does not allow for the impact of circulation of the fuel salt to be incorporated into the improved DNP group parameter generation.

An additional consideration for these preliminary results is that pulse irradiation results are not included. Whether the results from the pulse irradiation simulation are combined via simultaneous solution of a larger matrix or separately solved and then combined, they improve the DNP group parameters for the shorter lived groups [87]. This means that the longer lived groups can be expected to be more accurate, while the longer lived groups may have a larger difference. This is discussed further in Section 5.1.1.

These results, unless otherwise specified, use the parameters given in Table 4.1. The sampled simulations indicates the number of non-linear least squares solves that generate DNP group parameters. For each sampled simulation, the emission probability, half-lives, and fission yields are sampled within their uncertainties assuming a normal distribution.

4.1 Important DNPs

Because the microscopic approach uses individual DNPs, it becomes possible to analyze the contribution of each DNP to the total delayed neutron yield. This is given in Equation (2.28), recreated here as:

$$\bar{\nu}_d = \sum_{i=1}^I P_{n,i} CFY_i.$$

Another quantity of interest is, rather than the number of delayed neutrons per fission, is the number of delayed neutrons from the existing concentration of nuclides. This can be calculated by integrating the delayed neutron

Table 4.1: Default parameters for 0D scaled model simulations.

Parameter	Value
Fissile Nuclide	^{235}U [-]
Incident Neutron Energy	0.0253 [eV]
Fission Yields	ENDF/B-VII.1 [34]
Half-lives	IAEA [49]
Emission Probability	IAEA [49]
Final Decay Time	600 [s]
Decay Time Spacing	Logarithmic
Decay Time Nodes	300
Sampled Simulations	1000

count rate, as shown in Equation (2.27), recreated here as:

$$\int_0^\infty \dot{n}_{d,i}(t) dt = P_{n,i} N_i(0).$$

Figure 4.1 shows both of these approaches and how they differ. For the delayed neutron yield contributions, shown in sub-figure (a), ^{86}Ge has the largest contribution, which means that for each thermal fission event in ^{235}U , approximately 12.5% of the delayed neutrons come from ^{86}Ge . ^{86}Ge is very short lived, with a half-life of approximately 200 milliseconds. This is why, in sub-figure (b), it contributes less than 4% of the total delayed neutrons once the sample has been irradiated for some time. Since it has been irradiated, the longer lived nuclides have accumulated, and the delayed neutron count rate is dominated by them. This reinforces the justification of why pulse irradiation data is important, as the saturation irradiation does not fully capture the relative effects of the short-lived DNPs. Finally, the relative concentration of each DNP is shown in sub-figure (c). These values are generally consistent with the literature, though ^{141}Cs is an outlier [98]. This is likely due to the simplifications made in the 0D scaled model, such as exclusion of parasitic absorption.

Figure 4.2 shows the delayed neutron count rate in sub-figure (a), as well as the integrated form in sub-figures (b) and (c) which represents the total delayed neutron counts at each time. Sub-figure (a) shows how the count rate of ^{86}Ge dominates at short time scales due to its short half-life, while after a tenth of a second it becomes less important. This again demonstrates how the saturation irradiation enables better analysis of the longer-lived DNPs, but at the cost of less relative importance of the shorter-lived DNPs. Sub-figure (b) shows how the total number of counts vary over time on a log scale, so the initial contribution of ^{86}Ge is initially dominant but quickly becomes relatively unimportant compared to the other DNPs. Sub-figure (c) reflects what was shown in Figure 4.1 sub-figure (b), where ^{86}Ge is minimal and the total delayed neutron count contribution is dominated by longer lived DNPs like ^{137}I and ^{87}Br .

Another analysis of interest with the individual DNP data is calculation of the delayed neutron yield and average half-life, shown in Table 4.2. These values are calculated and then compared with the six group DNP group parameters formulated from those DNPs. Ideally these values should be identical, but because the six-group fit cannot fully capture the behavior of all the decaying DNPs, there are some small differences. These differences are small relative to the uncertainty bounds for each term.

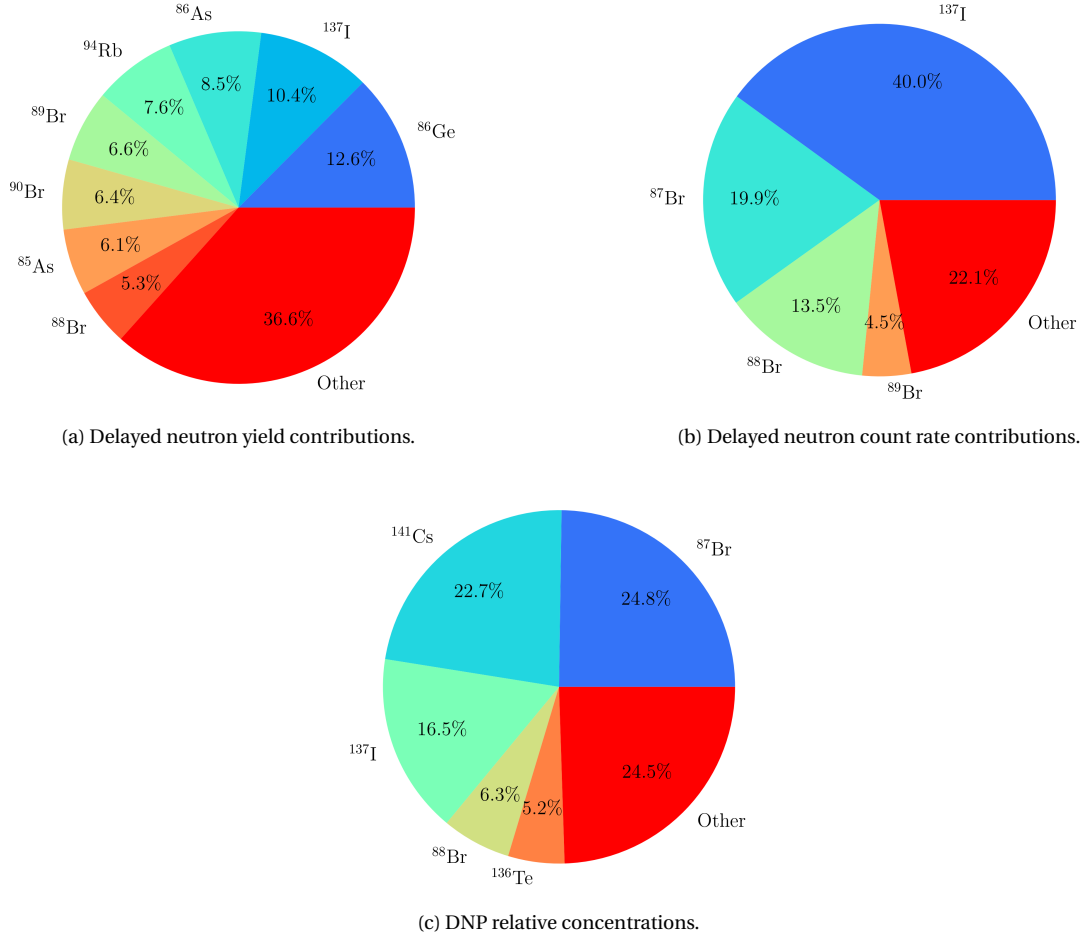


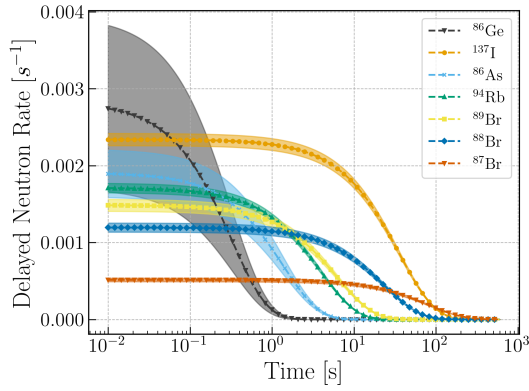
Figure 4.1: Delayed neutron contributions from each DNP.

Table 4.2: The delayed neutron yield and average half-life calculated by summing individual DNPs and from the 6 DNP group parameters.

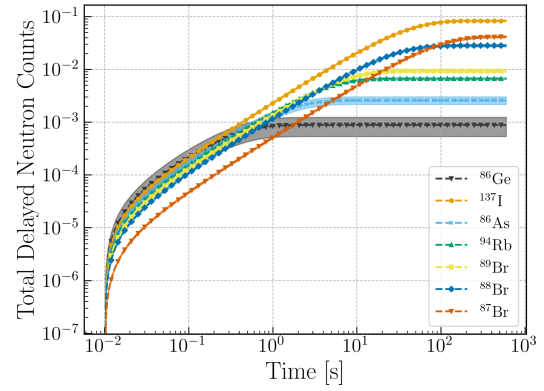
Term	DNP	Group
$\bar{\nu}_d$	0.0225 ± 0.0015	0.0224 ± 0.0013
$\bar{\tau}[\text{s}]$	6.3852 ± 0.3801	6.4029 ± 0.3365

4.2 Sensitivity Studies

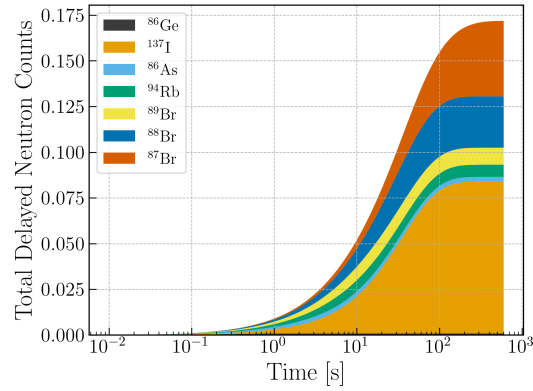
This section covers the sensitivity of the DNP group parameters to various simulation parameters, including: chemical removal, time nodes at which to evaluate the non-linear least squares problem, and uncertainties in the data. For the chemical removal analysis, the MSBR chemical reprocessing scheme is used. This is because it has large chemical removal rates, which will therefore provide a conservative estimate as to the impact these chemical removal rates may have on the DNP group parameters. The evaluation of the time nodes include the spacing between each node, the number of nodes, the total amount of time the sample should decay, and a combination of the total time and number of nodes. Finally, the data uncertainties include uncertainty in the fission yields, half-lives, and emission probabilities. To propagate these uncertainties to the DNP group parameters, the non-linear least squares solver is run 1000 times with a random normal distribution sample taken from the fission yield, half-life,



(a) The delayed neutron count rate.



(b) The total delayed neutron counts.



(c) A stack plot of the total delayed neutron counts.

Figure 4.2: Delayed neutron count rate contributions from the top two most contributing DNPs at each time step.

and emission probability of each DNP.

4.2.1 Chemical Removal

As shown in Appendix B, the fraction of fuel salt in the ex-core region determines the scaling of the chemical removal rate. For this analysis, the ex-core fraction will be assumed to be 0.5. This value is based on the approximate core zone I, core zone II, and annulus salt volume relative to the primary system salt volume in the MSBR [146]. The chemical reprocessing scheme from the MSBR used in this analysis is given in Table 4.3.

Table 4.3: MSBR Online Reprocessing Removal Rates from [146].

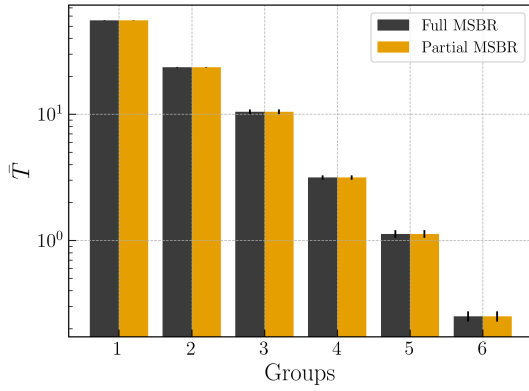
Reprocessing Group	Element(s)	Removal Rate [s^{-1}]
Rare Earths	Y, La, Ce, Pr, Nd, Pm, Sm, Gd	2.32×10^{-7}
Rare Earths	Eu	2.32×10^{-8}
Noble Metals	Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te	5.00×10^{-2}
Seminoble Metals	Zr, Cd, In, Sn	5.79×10^{-8}
Gases	Kr, Xe	5.00×10^{-2}
Volatile Fluorides	Br, I	1.93×10^{-7}
Discard	Rb, Sr, Cs, Ba	3.37×10^{-9}
Salt Discard	Th, Li, Be, F	3.37×10^{-9}
Protactinium	Pa	3.86×10^{-6}
Higher Nuclides	Np, Pu	1.98×10^{-9}

One consideration is that the 0D scaled model will underestimate the effect of the chemical removal rate. This is because it only chemically removes from the cumulative fission yield of the element in the reprocessing scheme, and does not account for decay chains which may have chemical removal rates included. For example, ^{137}I has a cycle time of 60 days, or a chemical removal rate of $1.9 \times 10^{-7} [s^{-1}]$. However, Sb and Te, which both decay into I, also have a chemical removal rate of $0.05 [s^{-1}]$, many orders of magnitudes larger than the I removal rate. However, using the 0D scaled model, with cumulative fission yields and no decay chain, means that this effect is not captured.

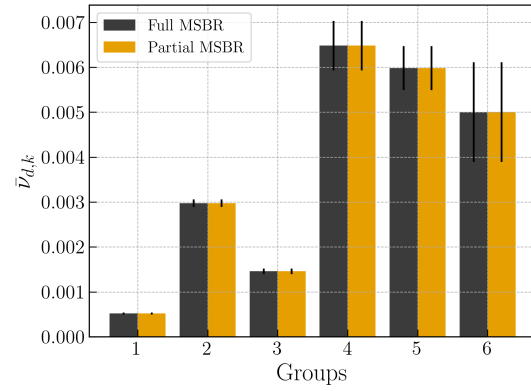
Long Cycle Time Evaluation

Based on the chemical removal rates provided in Table 4.3, there are many elements to consider. However, since the sample is only irradiated over the course of several minutes, it is of interest to determine the significance of the smaller removal rates, such as the 500 day cycle time of rare earths. For this analysis, only 20 second cycle time elements, the noble metals and gases, are considered. The difference between the full MSBR reprocessing scheme and the partial MSBR reprocessing scheme, with only 20 second cycle time elements considered, is shown in Figure 4.4.

The difference between every DNP group parameter is less than 0.001%, so for all intents and purposes, it is sufficient to call them equivalent.



(a) Delayed neutron yield contributions.

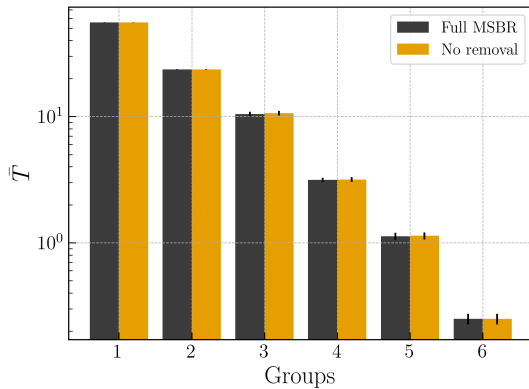


(b) Delayed neutron count rate contributions.

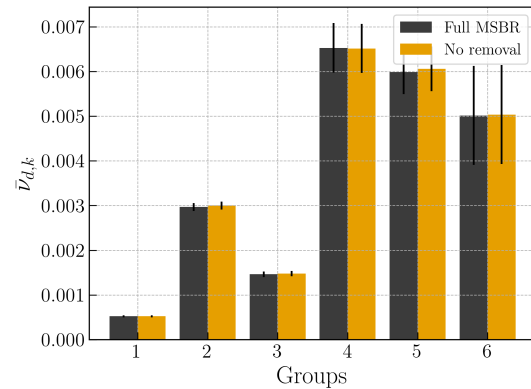
Figure 4.3: Delayed neutron contributions from each DNP.

Removal Rate Evaluation

It is also of interest to determine the sensitivity of the DNP group parameters to the chemical reprocessing scheme. The 0D scaled model will underestimate the effect of chemical removal due to the use of cumulative fission yields without explicit nuclide tracking, but this analysis can still give a useful conceptual understanding of the chemical reprocessing impact. The comparison of DNP group parameters with and without the MSBR reprocessing scheme is given in Figure 4.4. This figure shows that the half-lives are very similar, while the yields without removal are slightly higher.



(a) Delayed neutron yield contributions.



(b) Delayed neutron count rate contributions.

Figure 4.4: Delayed neutron contributions from each DNP.

The difference between the average half-lives is negligible, while the difference between the total delayed neutron yields is 0.57%, or approximately 12 *pcm*. This is a small difference, which is likely due to the nature of the 0D scaled model not treating the decay chain explicitly. However, since the effects of chemical reprocessing will consistently reduce the total delayed neutron yield due to removal of DNPs, it is important to account for this factor.

4.2.2 Time Nodes

The time node sensitivity is important, as many time nodes may be required to capture the physics fully, but at an increasing cost for each additional time node. Of interest are the distribution of nodes, the number of nodes, the amount of time the sample decays, and the combined decay time and number of nodes.

Distribution

For the distribution of time nodes, two different approaches are compared: a logarithmic spacing between 0.01 and 600 seconds, and a linear spacing between 0 and 600 seconds. Because 300 time nodes are being used, this means that the linear spacing will not capture behavior which occurs in the first 2 seconds, but instead will have a higher resolution at the longer time scales. This may not be as useful, as at longer time scales the number of DNPs that are relevant drop away quickly, as discussed in Section 4.1.

Figure 4.5 shows how the DNP group parameters compare between the logarithmic and linear spacing. The half-lives of the shorter lived groups are measured as being larger for the linear spacing, which is consistent with expectations. The linear spacing does not have any data between zero and two seconds, and there is greater weighting on the longer lived groups due to the data distribution. This also leads to the shorter lived groups capturing a larger yield, with the yields being pulled from groups two, three, and four to be added to groups five and six. The difference between the total DNP yields is approximately 0.1%, so changing the spacing of the time nodes doesn't change the macroscopic behavior of the DNP groups. Instead, it affects the distribution of delayed neutrons over time, where the logarithmic distribution is more accurate due to capturing the effects of shorter lived DNPs.

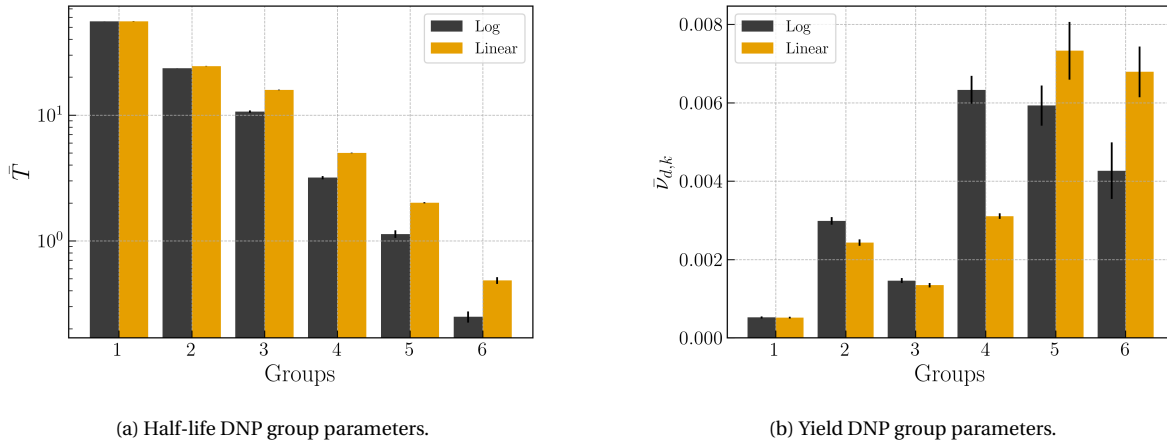


Figure 4.5: DNP group parameters for linear and logarithmic spacings of the time nodes.

Number of Nodes

Figure 4.6 shows how varying the number of nodes affects the results. It appears that there is no significant difference between the results when varying the number of nodes. The main difference appears in the yields of groups four and five, with increasing nodes increasing the yield of group four and decreasing the yield of group five. There are also some smaller differences, such as the yield of group six decreasing and the yields of groups two and three increasing with an increasing number of nodes.

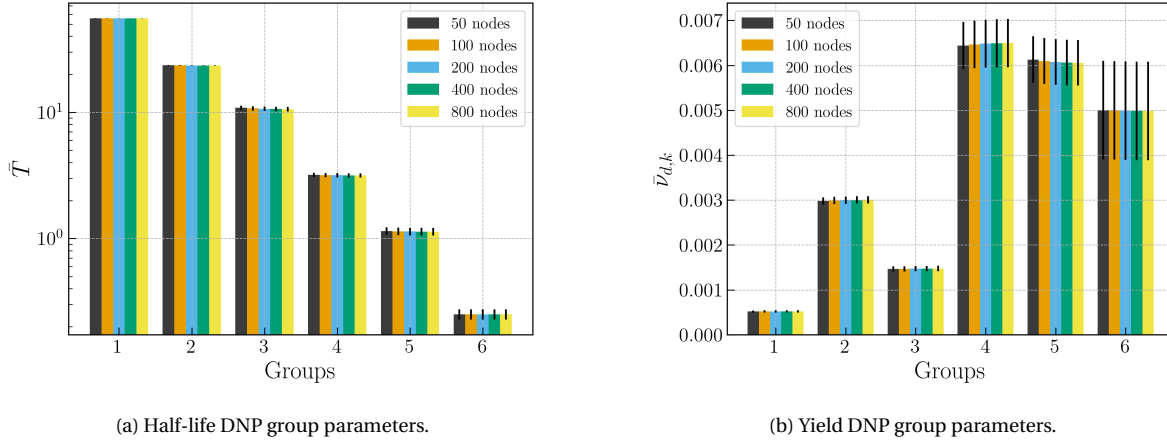


Figure 4.6: DNP group parameters for varying numbers of time nodes.

The simulation with the most nodes is the most accurate, so the percent difference of the DNP group yields for each group can be calculated, as well as the computational cost associated with the increase in accuracy. This is shown in Table 4.4, where using 50 nodes results in the group five yield being off by approximately 1%, and using 200 nodes keeps differences below 0.02%. The computational cost is also fairly constant for each of these different numbers of nodes. Based on these results, the selected default of 300 nodes appears to offer sufficiently high accuracy yields, within 0.1%, while also residing in the constant time cost regime.

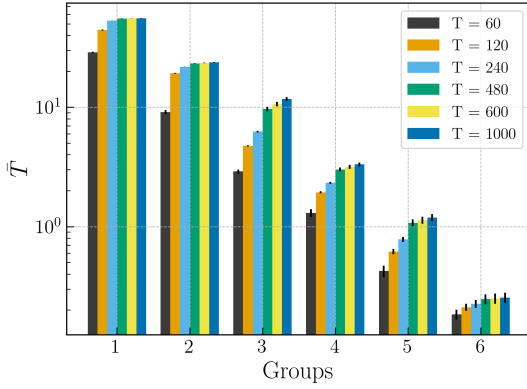
Table 4.4: Comparison of how the number of nodes changes the computational cost and relative difference in yields compared to the 400 node case.

Nodes	Yield Difference [%]						Cost [s]
	1	2	3	4	5	6	
50	0.30	0.90	0.53	0.75	0.99	0.59	297
100	0.17	0.49	0.25	0.39	0.46	0.77	297
200	0.01	0.02	0.00	0.01	0.01	0.09	300
400	0.00	0.00	0.00	0.00	0.00	0.01	304
800	-	-	-	-	-	-	310

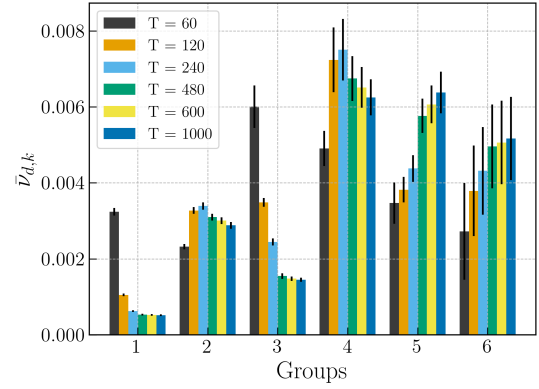
Total Decay Time

As the total decay time increases, there are two effects occurring at the same time: a decrease in the time node density, discussed in Section 4.2.2, and an increase in the number of nodes evaluated at longer times. Because the longest lived DNP, ^{87}Br , has a half-life of approximately one minute, it is reasonable to assume that the default final time of ten minutes offers sufficient data on all DNP decays.

Figure 4.7 shows that this is true, where the differences in DNP group parameters between a 600 second decay time and a 1000 second decay time is very small relative to the uncertainty bounds. The short decay times do not accurately model the DNPs, with underestimations of the half lives. The half-life underestimations are then correlated with overestimation of some of the longer lived DNP group yields. The DNP yield of groups two and four have an interesting trend, where they underestimate for short decay times and overestimate for intermediate decay times. The cause of this is likely due to the simultaneous fitting of the surrounding groups, as the raw data itself remains unchanged.



(a) Half-life DNP group parameters.



(b) Yield DNP group parameters.

Figure 4.7: DNP group parameters for varying total decay times.

Unfortunately, it cannot be claimed that increasing the net decay time strictly improves the accuracy of the DNP group parameters. Excessively short decay times would not properly capture long-lived DNPs, while excessively long decay times would overfit the long-lived DNPs. One way to check these values is to compare the total delayed neutron yields and average half-lives with values from the literature. These values are compared in Table 4.5. From this table, it can be seen that the differences in these values are small and within uncertainty bounds of each other.

Table 4.5: Comparison of how the net decay time, T changes the total delayed neutron yield and average half-life.

T [s]	$\bar{v}_d$	$\bar{\tau}$ [s]
60	0.0227 ± 0.0016	6.1926 ± 0.4058
120	0.0227 ± 0.0015	6.3308 ± 0.3804
240	0.0227 ± 0.0015	6.3581 ± 0.3651
480	0.0226 ± 0.0013	6.3647 ± 0.3432
600	0.0226 ± 0.0013	6.3648 ± 0.3424
1000	0.0226 ± 0.0013	6.3647 ± 0.3426

Time Node Density

Section 4.2.2 shows how varying the total decay time affects the results, and Section 4.2.2 shows how varying the number of time nodes affects the results. This section evaluates the sensitivity of the DNP group parameters to the node density, number of nodes, and decay time by comparing varying numbers of nodes and decay times. Figure 4.8 shows how the DNP group parameters vary for varying node densities.

If the node density was the driving factor in the DNP group parameters, then the black bar, with a final time of 600 seconds and with 400 time nodes, would be equivalent to the teal bar, which has a final time twice as long and with twice as many time nodes. Instead, the results are approximately constant with varying time nodes, and the main difference is in the final decay time. This means that the sensitivity analysis discussed in Section 4.2.2 is valid, and does not need to also account for the varying time node density.

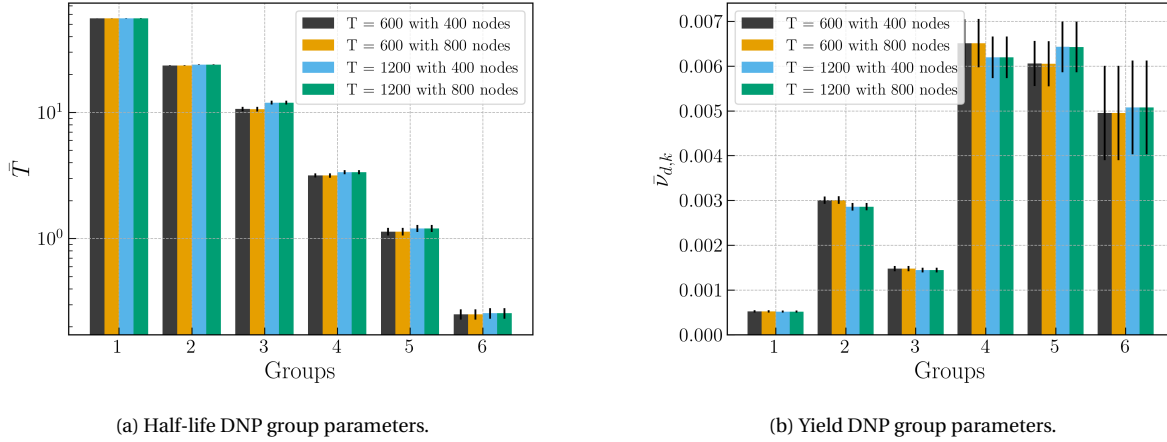


Figure 4.8: DNP group parameters for varying node densities.

4.2.3 Nuclear Data Uncertainties

To find the sensitivity of the DNP group parameters to the uncertainties in the nuclear data, a Monte Carlo approach is used. From each piece of data used in the 0D scaled model, such as the half-life of ^{87}Br , a random value would be sampled assuming a normal distribution. This is repeated for every fission yield, half-life, and emission probability for every DNP.

Count Rate

The count rate, normalized to the Keepin et al. count rate, is given in Figure 4.9. The red circles indicate the count rates from various sampled simulations, while the blue dashed line indicates the DNP group parameters used and the uncertainty based on the 1000 Monte Carlo sampled non-linear least squares solves. The black X's indicate the mean count rate at each point. The other six group DNP parameters plotted include Keepin, which is 1.0 since the plot is normalized to the Keepin values, the Brady group parameters, and the Synetos group parameters [87, 23, 175].

This figure shows that beyond roughly three seconds, there is agreement within the uncertainty bounds between the current DNP group parameters and most of the other results in the literature, with the Brady group parameters having a noticeable difference at longer times. For the short time scales, as previously discussed in the introduction to this chapter, are inaccurate due to the lack of pulse irradiation data. It is expected that once this data is added, the earlier times will fit more closely. To demonstrate this, the first two groups are used while the shorter-lived groups three through six are replaced with the Keepin et al. DNP group parameters [87]. This modified set of DNP group parameters can be seen in Figure 4.10.

The modified DNP group parameters are not a perfect comparison, as the non-linear least squares solve adjusts the yields of each group relative to the others and based on the half-lives of the others. This means that not including pulse irradiation data will also affect the longer-lived groups, though by a smaller amount than the effects on the shorter-lived groups. However, there is generally better agreement when adjusting the shorter lived groups, particularly during the times less than one second.

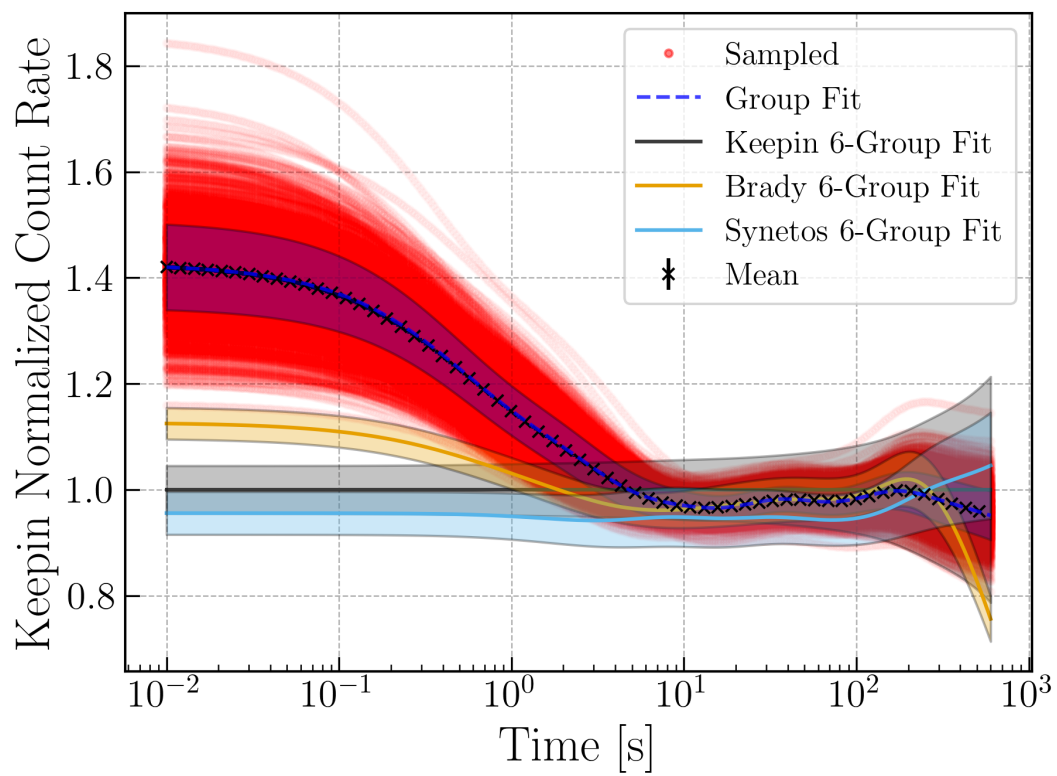


Figure 4.9: The delayed neutron count rate for thermal fission of ^{235}U normalized to the Keepin et al. six-group DNP group parameters.

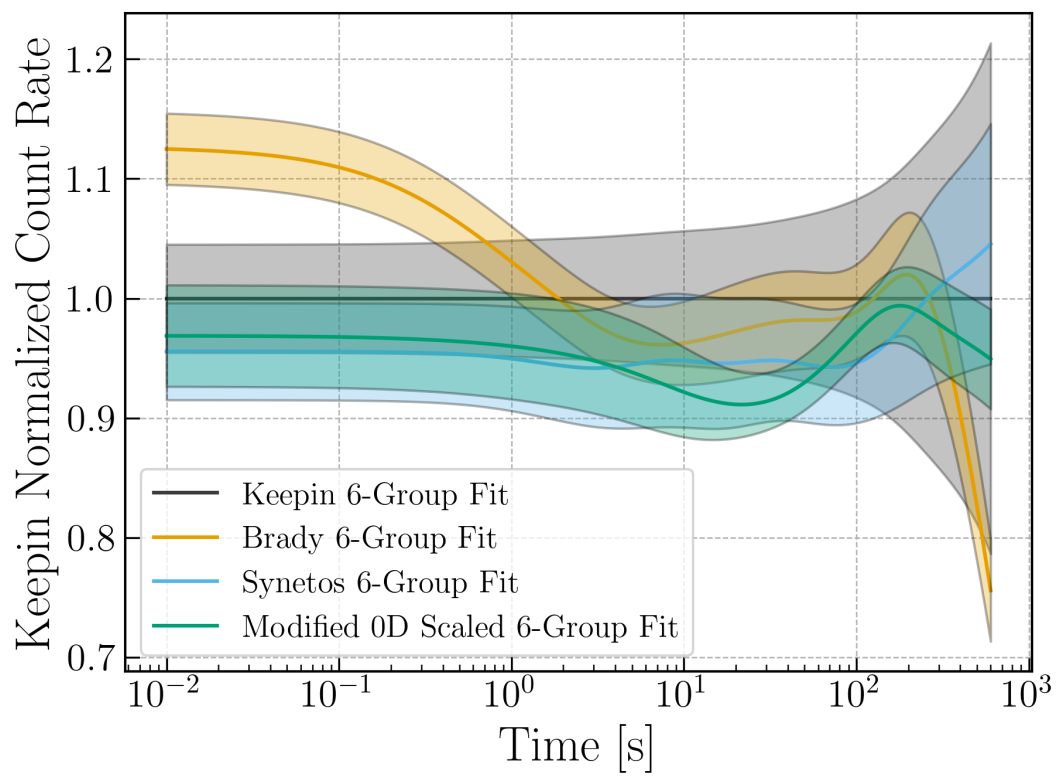


Figure 4.10: The delayed neutron count rate for thermal fission of ^{235}U normalized to the Keepin et al. six-group DNP group parameters.

Individual DNPs

The number of Monte Carlo simulations is increased to 5000 for this analysis, as the contribution of each DNP needs sufficient data to ensure differences in DNP group parameters are caused by its own sampled variation. To find the nuclides with the most impactful sensitivities, a linear regression fits the slope of the relative change in DNP group parameters to the relative change to the individual DNP nuclear data, while also calculating the Pearson correlation coefficient (PCC). The PCC describes the linear relationship between two variables on a scale from -1, representing a perfect negative correlation, +1 is a perfect positive correlation, and 0 is no correlation [61]. The changes in nuclear data include changes to emission probability, P_n ; fission yields, which affect the concentration, N ; and half-lives, τ .

Sensitivities of interest in this analysis are for absolute values of the PCC greater than 0.2. This value is selected somewhat arbitrarily, but gives a useful indication that a nuclide has a non-negligible correlation with the DNP group parameter of interest. The results of this analysis are given in Table 4.6.

Many of the correlations in Table 4.6 are intuitively understandable. One of the most predictable sensitivities is the effect of ^{87}Br on the group one half-life and yield, as ^{87}Br dominates this group. Specifically, the sensitivities that affect the DNP group parameters with a PCC greater than 0.2 are the emission probability on the group yield, the concentration on the group yield, and the half-life on the group half-life. Figure 4.11 shows how the group one parameters vary with changes in the ^{87}Br data, as well as less strongly correlated data, such as the emission probability of ^{87}Br on the group three half-life. Although it is not intuitive that changes to the emission probability of ^{87}Br would affect the half-life of group three, there is a non-zero correlation. This is because the DNP group parameters cannot be taken in isolation, but as a collective set of parameters that best fit the delayed neutron count rate curve. Thus, changes to that curve propagate to all group parameters, albeit with varying magnitudes to different groups and different parameters.

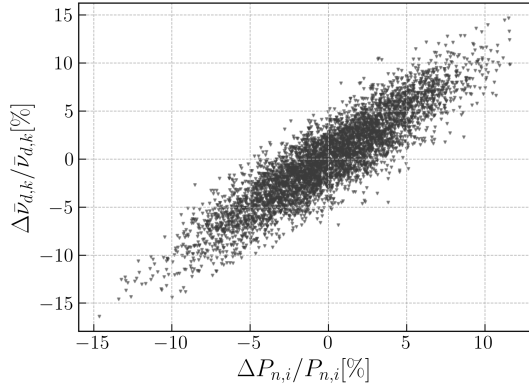
Another nuclide of interest in Table 4.6 is ^{86}Ge , as it is also an important nuclide for calculating delayed neutron yield contributions as shown in Figure 4.1. It can be seen that ^{86}Ge has an impact on many DNP group parameters as there is variation in different parameters, especially its half-life. Figures 4.12 and 4.13 show the sensitivity to each DNP group parameter to each variation in data for ^{86}Ge .

Figure 4.12 has the largest sensitivities for the group half-lives when there are changes in the half-life of ^{86}Ge . It is interesting to note that, for increasingly longer-lived groups, the impact of the half-life of ^{86}Ge on the group half-life becomes less and less. This is intuitive, as a short-lived nuclide such as ^{86}Ge should have much more impact on short-lived groups than on long-lived groups. Another interesting pattern is that the group six half-life of ^{86}Ge sensitivity appears to have a line below which there are no data points, indicating a lack of symmetry. This shape is likely due to the fact that there are many longer-lived DNPs, whereas ^{86}Ge is one of the few very short-lived DNPs that dominates in this time regime.

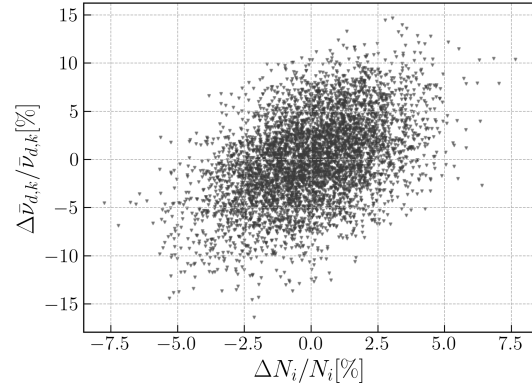
Figure 4.13 shows there are changes in the group yields based on changes in the half-life of ^{87}Br . There is not a clear trend moving from shorter to longer-lived groups, though there are some negative correlations for groups three and four, which may indicate that a longer-lived ^{87}Br leads to a decreased yield in those groups while also increasing their half-lives, as shown in Figure 4.12. This indicates those groups capture fewer short-lived DNPs as the half-life of ^{87}Br increases both by increasing their group half-lives and decreasing their yields. Another interesting sensitivity correlation is the emission probability of ^{87}Br on the group six yield, which appears to have a narrow spread of values for small emission probabilities but then spreads out for larger emission probabilities. The trend is positive, which is because the half-life of ^{87}Br is very short and is contained in group six. One contributing factor to this spreading out of the yield is likely due to the lack of pulse irradiation data. Because the short-lived groups are less accurate without this data, removing ^{87}Br leads to a more consistent yield in group six. However, as ^{87}Br becomes more impactful, the possible value for the yield spreads out due to the lack of pulse irradiation data.

Table 4.6: PCCs with a magnitude larger than 0.2 for variation in DNP data which correspond to changes in DNP group parameters.

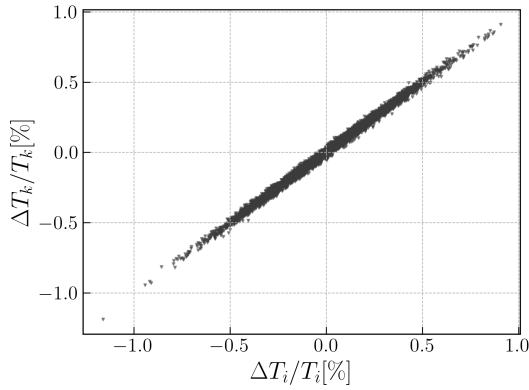
Nuclide	Group Value	DNP Value	PCC
⁸⁴ Ga	$\tau_3[s]$	$N_i[-]$	-0.211724
⁸⁴ Ga	$\tau_4[s]$	$N_i[-]$	-0.222001
⁸⁴ Ga	$\tau_5[s]$	$N_i[-]$	-0.203050
⁸⁵ As	$\bar{v}_{d,4}[-]$	$N_i[-]$	0.757084
⁸⁵ As	$\tau_4[s]$	$N_i[-]$	-0.411166
⁸⁵ As	$\bar{v}_{d,5}[-]$	$N_i[-]$	0.718393
⁸⁵ As	$\tau_5[s]$	$N_i[-]$	0.448911
⁸⁸ Br	$\bar{v}_{d,3}[-]$	$P_{n,i}[-]$	0.276294
⁸⁸ Br	$\bar{v}_{d,3}[-]$	$\tau_i[s]$	-0.221999
⁸⁸ Br	$\tau_3[s]$	$P_{n,i}[-]$	0.379647
⁸⁸ Br	$\tau_3[s]$	$N_i[-]$	0.264940
¹³⁸ I	$\bar{v}_{d,3}[-]$	$P_{n,i}[-]$	0.211254
¹³⁷ I	$\bar{v}_{d,2}[-]$	$P_{n,i}[-]$	0.478664
¹³⁷ I	$\bar{v}_{d,2}[-]$	$\tau_i[s]$	-0.270646
¹³⁷ I	$\bar{v}_{d,2}[-]$	$N_i[-]$	0.746893
¹³⁷ I	$\tau_2[s]$	$\tau_i[s]$	0.813782
¹³⁷ I	$\tau_2[s]$	$N_i[-]$	0.252540
⁸⁶ As	$\bar{v}_{d,5}[-]$	$N_i[-]$	0.512737
⁹⁸ Y	$\bar{v}_{d,4}[-]$	$P_{n,i}[-]$	0.243371
⁸⁶ Ge	$\bar{v}_{d,2}[-]$	$\tau_i[s]$	-0.238309
⁸⁶ Ge	$\tau_2[s]$	$\tau_i[s]$	0.359997
⁸⁶ Ge	$\bar{v}_{d,3}[-]$	$\tau_i[s]$	-0.637401
⁸⁶ Ge	$\tau_3[s]$	$\tau_i[s]$	0.656167
⁸⁶ Ge	$\bar{v}_{d,4}[-]$	$\tau_i[s]$	-0.407397
⁸⁶ Ge	$\tau_4[s]$	$\tau_i[s]$	0.673363
⁸⁶ Ge	$\bar{v}_{d,5}[-]$	$\tau_i[s]$	0.263800
⁸⁶ Ge	$\tau_5[s]$	$\tau_i[s]$	0.662307
⁸⁶ Ge	$\bar{v}_{d,6}[-]$	$P_{n,i}[-]$	0.830397
⁸⁶ Ge	$\bar{v}_{d,6}[-]$	$N_i[-]$	0.464433
⁸⁶ Ge	$\tau_6[s]$	$P_{n,i}[-]$	-0.306136
⁸⁶ Ge	$\tau_6[s]$	$\tau_i[s]$	0.893433
⁸⁷ Br	$\bar{v}_{d,1}[-]$	$P_{n,i}[-]$	0.896423
⁸⁷ Br	$\bar{v}_{d,1}[-]$	$N_i[-]$	0.456307
⁸⁷ Br	$\tau_1[s]$	$\tau_i[s]$	0.997442
¹³⁷ Sb	$\bar{v}_{d,3}[-]$	$N_i[-]$	0.229445
¹³⁷ Sb	$\tau_3[s]$	$N_i[-]$	-0.229578
¹³⁷ Sb	$\tau_4[s]$	$N_i[-]$	-0.250857
¹³⁷ Sb	$\tau_5[s]$	$N_i[-]$	-0.299691



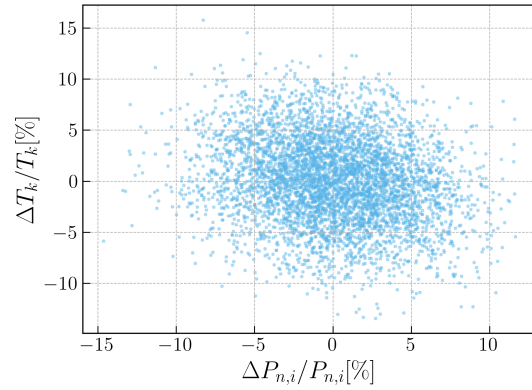
(a) DNP group 1 yield variation due to emission probability variation with a PCC of 0.90.



(b) DNP group 1 yield variation due to fission yield variation with a PCC of 0.46.



(c) DNP group 1 half-life variation due to half-life variation with a PCC of 1.00.



(d) DNP group 3 half-life variation due to emission probability variation with a PCC of -0.19.

Figure 4.11: DNP group parameter sensitivity to changes in ^{87}Br nuclear data.

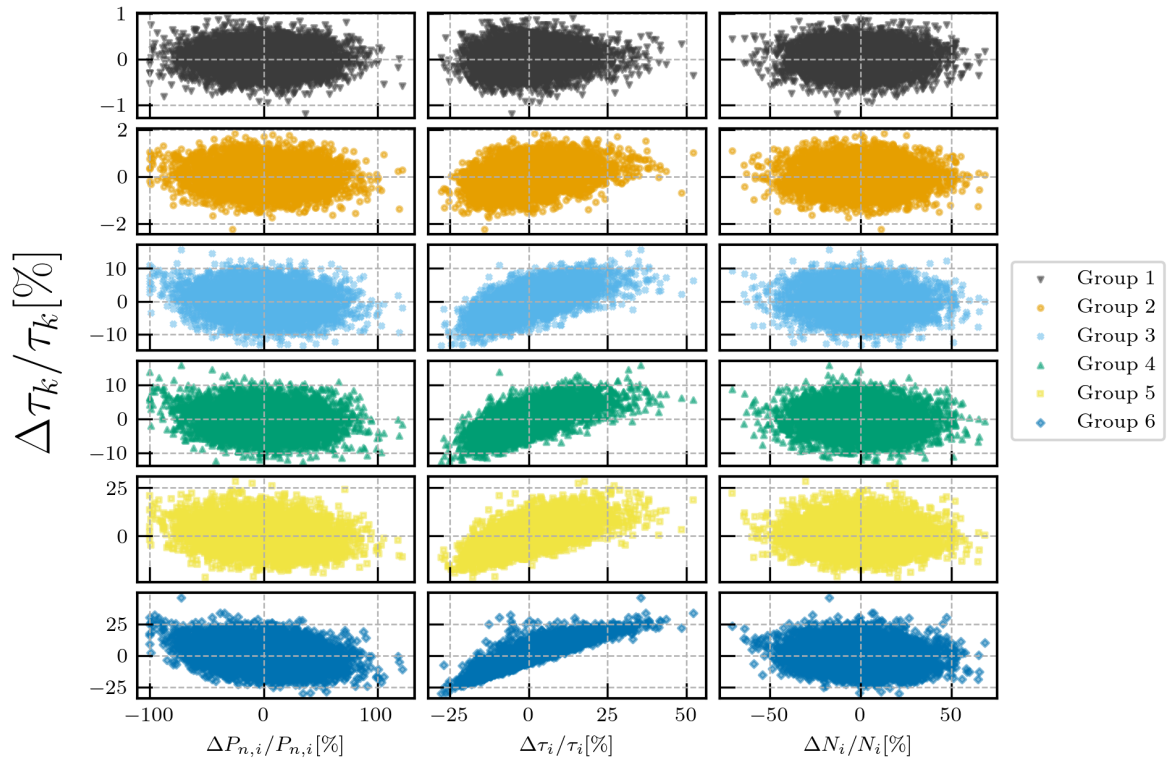


Figure 4.12: The difference in each DNP group half-life from the ^{86}Ge relative changes in emission probability, half-life, and concentration.

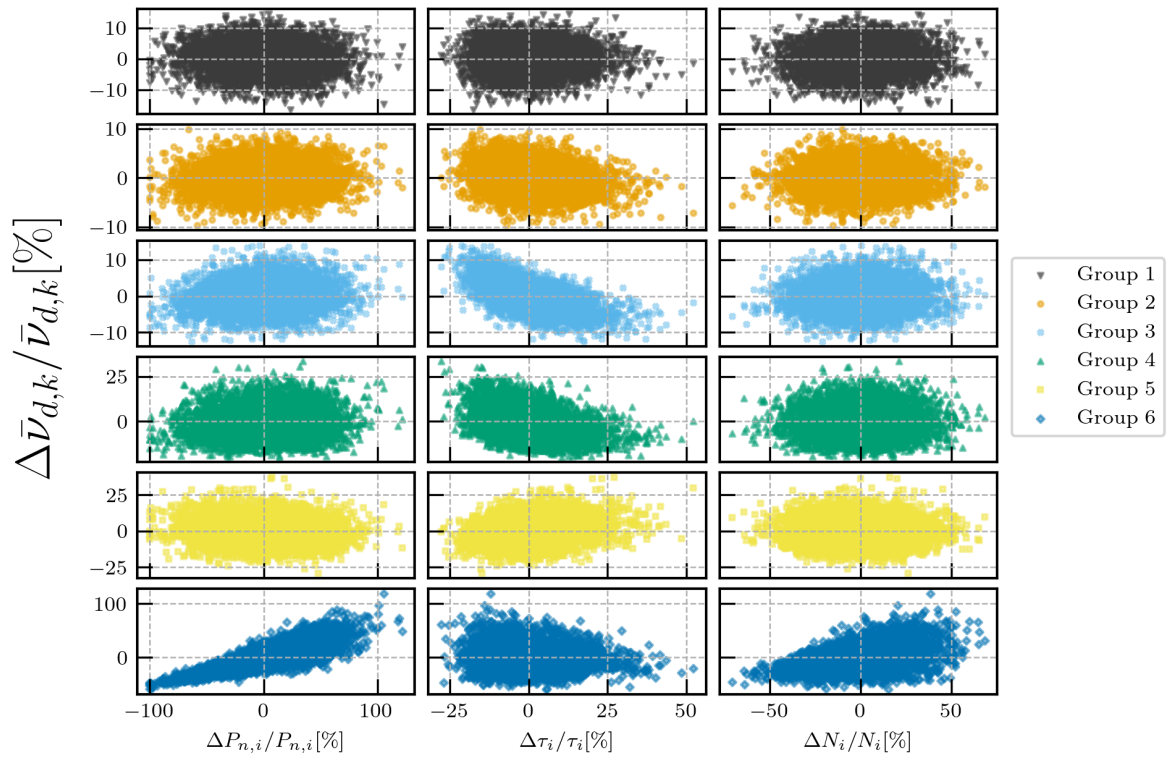


Figure 4.13: The difference in each DNP group yield from the ^{86}Ge relative changes in emission probability, half-life, and concentration.

4.3 Verification

The 0D scaled model is verified using DNP group parameters from thermal fission of ^{235}U . This verification uses the data from literature, introduced in Section 2.1.1, combined and recreated here with the results from this work included.

Tables 4.7, 4.8, 4.10, and 4.10 show the DNP group parameters from literature and how they compare to the 0D scaled model DNP group parameters. As previously discussed, because the 0D scaled model does not include the pulse irradiation data, the shorter-lived groups are expected to be less accurate. This can be seen in the yields, where the groups 5 and 6 yields are significantly different, while groups 1 and 2 are much more similar to results from the literature. The group half-lives are similar to the literature values, though group three is an outlier. This is again likely due to the pulse irradiation data missing, as it is noticeably longer than the other values in the literature.

Table 4.7: DNP group yields from thermal fission of ^{235}U (recreated from [126]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [87]	0.00052	0.00346	0.00310	0.00624	0.00182	0.00066
Charlton et al. [36]*	0.00054	0.00344	0.00319	0.00607	0.00186	0.00066
ENDF/B-VI.8 [151]	0.00058	0.00302	0.00288	0.00646	0.00265	0.00111
Mills et al. [118]	0.00056	0.00325	0.00196	0.00658	0.00287	0.00122
Brady and England [23]	0.00068	0.00341	0.00292	0.00611	0.00310	0.00158
Saleh et al. [158]	0.00055	0.00363	0.00297	0.00593	0.00168	0.00063
Synetos and Williams [175]*	0.00051	0.00328	0.00214	0.00583	0.00270	0.00063
Tuttle [182]	0.00065	0.00362	0.00320	0.00692	0.00218	0.00044
Waldo et al.[189]	0.00055	0.00366	0.00327	0.00660	0.00192	0.00070
0D scaled model	0.00052	0.00297	0.00146	0.00650	0.00597	0.00500

* Modified to include a sixth group by Parish et al. [126].

Table 4.8: DNP group yield uncertainties from thermal fission of ^{235}U (recreated from [126]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [87]	0.00006	0.00025	0.00041	0.00047	0.00019	0.00013
Charlton et al. [36]*	0.00003	0.00005	0.00024	0.00013	0.00014	0.00014
ENDF/B-VI.8 [151]	-	-	-	-	-	-
Mills et al. [118]	0.00004	0.00023	0.00020	0.00051	0.00035	0.00012
Brady and England [23]	0.00004	0.00019	0.00016	0.00034	0.00017	0.00009
Saleh et al. [158]	0.00005	0.00032	0.00027	0.00104	0.00015	0.00012
Synetos and Williams [175]*	0.00004	0.00020	0.00036	0.00037	0.00029	0.00012
Tuttle [182]	0.00007	0.00013	0.00041	0.00019	0.00021	0.00007
Waldo et al.[189]	0.00006	0.00041	0.00041	0.00038	0.00017	0.00009
0D scaled model	0.00002	0.00009	0.00006	0.00056	0.00051	0.00111

* Modified to include a sixth group by Parish et al. [126].

Table 4.11 shows how the total delayed neutron yield for 100 fissions compares between the literature and for the 0D scaled model. The differences between the microscopic and macroscopic delayed yields for the same dataset exist for many fissile nuclides, not only ^{235}U [112]. The 0D scaled model yield is noticeably larger than the other values in the literature, but the raw data is also larger. This indicates not an issue with the DNP group parameter

Table 4.9: DNP group half-lives from thermal fission of ^{235}U (recreated from [126]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [87]	55.89897	22.72614	6.24457	2.30281	0.60802	0.23028
Charlton et al. [36]*	55.45177	22.43195	6.02737	2.18658	0.49159	0.23028
ENDF/B-VI.8 [151]	51.72740	21.19716	5.73797	2.28913	0.81595	0.24295
Mills et al. [118]	55.27489	22.80089	7.67860	2.77148	1.07381	0.28177
Brady and England [23]	52.11633	21.32761	5.68620	2.18727	0.70114	0.23462
Saleh et al. [158]	55.45177	22.65187	6.24457	2.31049	0.63013	0.23028
Synetos and Williams [175]*	56.35343	22.35959	7.00149	2.72893	0.77016	0.23028
Tuttle [182]	54.57852	21.86584	6.02737	2.22877	0.49511	0.17911
Waldo et al.[189]	54.57852	21.86584	6.02737	2.22877	0.49511	0.17911
0D scaled model	55.52977	23.54278	10.43942	3.14523	1.12586	0.25015

* Modified to include a sixth group by Parish et al. [126].

Table 4.10: DNP group half-life uncertainties from thermal fission of ^{235}U (recreated from [126]).

Source	Group					
	1	2	3	4	5	6
Keepin et al. [87]	1.35239	0.67061	0.22503	0.09181	0.08000	0.02219
Charlton et al. [36]*	0.44361	0.50817	0.10482	0.06208	0.03138	0.02219
ENDF/B-VI.8 [151]	-	-	-	-	-	-
Mills et al. [118]	0.09918	0.11723	0.37861	0.11259	0.08521	0.01020
Brady and England [23]	-	-	-	-	-	-
Saleh et al. [158]	3.99253	0.14805	0.39380	0.03851	0.01719	0.02219
Synetos and Williams [175]*	0.45816	0.28851	0.77794	0.21488	0.09413	0.02219
Tuttle [182]	1.28926	0.82773	0.20965	0.08600	0.04244	0.02545
Waldo et al.[189]	1.28926	0.82773	0.20965	0.08600	0.04244	0.02545
0D scaled model	0.15676	0.12560	0.44990	0.12771	0.07629	0.02325

* Modified to include a sixth group by Parish et al. [126].

formulation, but rather, that the pulse irradiation data is necessary to capture the smaller yields of the shorter-lived DNP groups. This can be seen in Table 4.7, where the group 5 and 6 yields are greatly over-valued relative to other DNP group yields in the literature. Using the modified 0D scaled model, where groups three through six use the Keepin et al. DNP group parameters, results in a yield which is more consistent with the other values in the literature.

Table 4.12 shows how the average half-lives compare between the 0D scaled model and the literature. The 0D scaled model average half-life is too short, which is primarily due to the overvaluing of the group 5 and 6 yields, which weighs those half-lives more than they should be. Using the modified 0D scaled model with groups three through six replaced by the Keepin et al. DNP group parameters once again results in the model aligning much closer with the literature. This indicates, once again, the importance of the pulse irradiation data to properly capture the short-lived DNP behavior.

Table 4.11: Total delayed neutron yields of 100 fissions from thermal fission of ^{235}U (based on [79]).

Approach	Data	Delayed Yield
Microscopic	JEF-2.2 [11]	1.708 ± 0.12
Microscopic	JEF-3.1.1 [159]	1.477 ± 0.08
Microscopic	ENDF/B-VII.1 [34]	1.905 ± 0.10
Microscopic	ENDF/B-VI.8 [151]	1.670
Microscopic	Mills et al. [118]	1.645 ± 0.07
Microscopic	Brady and England [23]	1.780 ± 0.05
Microscopic	0D Scaled Model Data	2.254 ± 0.15
Microscopic	0D Scaled Model	2.256 ± 0.13
Hybrid	Modified 0D Scaled Model	1.531 ± 0.09
Macroscopic	JEF-2.2 [11]	1.655
Macroscopic	JEF-3.1.1 [159]	1.620
Macroscopic	ENDF/B-VII.0 [33]	1.585
Macroscopic	Keepin et al. [87]	1.580 ± 0.07
Macroscopic	Charlton et al. [36]	1.577 ± 0.03
Macroscopic	Saleh et al. [158]	1.645 ± 0.11
Macroscopic	Synetos and Williams [175]	1.510 ± 0.06
Macroscopic	Tuttle [182]	1.700 ± 0.05
Macroscopic	Waldo et al.[189]	1.670 ± 0.07

Table 4.12: Average half-life from thermal fission of ^{235}U .

Approach	Data	Average half-life [s]
Microscopic	ENDF/B-VI.8 [151]	7.66
Microscopic	Mills et al. [118]	8.63 ± 0.52
Microscopic	Brady and England [23]	7.90 ± 0.34
Microscopic	0D Scaled Model Data	6.39 ± 0.38
Microscopic	0D Scaled Model	6.39 ± 0.34
Hybrid	Modified 0D Scaled Model	8.74 ± 0.46
Macroscopic	Keepin et al. [87]	9.03 ± 0.63
Macroscopic	Charlton et al. [36]	8.92 ± 0.28
Macroscopic	Saleh et al. [158]	9.49 ± 0.90
Macroscopic	Synetos and Williams [175]	8.96 ± 0.56
Macroscopic	Tuttle [182]	8.84 ± 0.45
Macroscopic	Waldo et al.[189]	8.72 ± 0.73

Chapter 5

Proposed Work

Completion of the proposed work will result in a complete analysis, from start to finish, of improved DNP group parameters for thermal fission of ^{235}U and the effects of these improved parameters on transient simulations. MoSDeN, which will be completed with this work, will serve as a tool to investigate the DNP group parameters with MSR physical phenomena included. The proposed work is not exhaustive, and will not investigate every fissile nuclide, nor every irradiation energy. Work which is deemed interesting yet not critical is included in Section 5.4.

The critical work includes developing the 0D flow model, validating and verifying the results of the 0D flow model, generating improved group spectra along with the DNP group parameters, and finally using the DNP group parameters in transient simulations. The non-critical work includes development of the spatially resolved model, uncertainty tracking of MSR physical phenomena in the DNP group parameters, and investigation into the effects of varying the incident neutron energy spectrum on the DNP group parameters.

5.1 0D Flow Model

The new model which will be developed beyond the 0D scaled model is the 0D flow model. The 0D scaled model only generates saturation irradiation results, scales the flux and chemical removal based on the fraction of salt in-core and ex-core, and does not track decays of nuclides into DNPs during the decay of the sample after irradiation. However, the 0D scaled model offers a fast simulation, which does enable rapid development and simulation of many different parameters with ease.

The 0D flow model improves the fidelity of the 0D scaled model by offering both saturation and pulse irradiation results, handles in-core and ex-core regions in time rather than simply scaling the flux, and enables tracking of decay parents for DNPs. There is also the option of using the 0D scaled model with the scaled flux approximation but with the decay chain modeled. The cost of this improved fidelity is an increased computational cost, but it is still not exceedingly expensive and can be run on a desktop. The neutron transport solve is relatively inexpensive, as the model only requires a fixed-source irradiation of the sample.

5.1.1 Least Squares Residual Treatment

A key aspect of the methodology is combining pulse and saturation irradiation DNP group parameters. One approach, used by Brady and Keepin et al., is combining two different fits after they have been generated. Brady uses the group 5 and 6 parameters from the pulse irradiation with the remaining parameters from the saturation

irradiation to form the six DNP group parameters [25]. Keepin et al. fit four groups with each irradiation, and then renormalize at the second group [87].

Another approach is the Levenberg-Marquardt method, which combines pulse and saturation irradiation data [102, 111, 117]. This method of fitting DNP group parameters to both datasets simultaneously may yield more accurate results. This can also be accomplished using the trust region reflective (TRF) approach, which allows for bounds to be included for the parameters [26].

The proposed methods to compare are the current method, the Levenberg-Marquardt method, and the TRF method. The new methods minimize the residual of the form:

$$r = \min_x \left\| \frac{A_p x - b_p}{b_p + \epsilon} + \frac{A_s x - b_s}{b_s + \epsilon} \right\|_2^2, \quad (5.1)$$

where,

- r = the residual of the problem [-]
- A_p = the pulse irradiation coefficient matrix [s^{-1}]
- b_p = the pulse irradiation delayed neutron count rate [$n \cdot s^{-1}$]
- A_s = the saturation irradiation coefficient matrix [s^{-1}]
- b_s = the saturation irradiation delayed neutron count rate [$n \cdot s^{-1}$]
- x = the solution vector containing group yields [-]
- ϵ = a small value to avoid division by zero
- $= 1 \times 10^{-12}$.

This formulation simultaneously treats the pulse and saturation irradiation data, generating improved DNP group parameters which fit both sets of delayed neutron count rates.

The results of this analysis will be used to evaluate which method can provide a solution with the least computational cost while also generating accurate group parameters. The group parameters will be evaluated by calculating the ℓ^2 -norm of the residual for the least squares problem using the improved DNP group parameters over all times for both pulse and saturation irradiations.

5.1.2 Data Source Analysis

Huynh et al. compare the effects of cumulative fission yields and emission probabilities from Joint Evaluated Fission and Fusion (JEFF), ENDF, and Japanese Evaluated Nuclear Data Library (JENDL) on the total delayed neutron yield [79]. A similar comparison can also be found in the JEFF report 17 [11]. Mathieu et al. investigate additional datasets, including the National Nuclear Data Center (NNDC) website (which sources data from Evaluated Nuclear Structure Data File (ENSDF)) and various data compilations [112, 134, 8, 153].

These comparisons can be expanded to include the IAEA database, NNDC website, JEFF, ENDF, JENDL, ENSDF, and other data compilations for emission probabilities (delayed neutron branching ratios), half-lives, fission yields, and cross sections. Comparing these data sources clarifies the uncertainty caused by differences in the data. Table 5.1 shows the data sources used in the comparison. The "Concentration" column refers to data including fission yields and cross-sections that affect the DNP concentration term. Although half-lives also influence concentration, data sources without a concentration contribution provide only half-life and emission probability data. Audi et al., Pfeiffer and Kratz, and Rudstam et al. present data from the literature rather than official databases. Audi et al. use

ENSDF, Atomic Mass Evaluation (AME), experimental data from the literature, and trends in neighboring nuclides. Pfeiffer and Kratz present experimental data, an update to the Kratz Herrmann formula (KHF), and data from a unified macroscopic-microscopic model within the quasi-particle random-phase approximation (QRPA). Rudstam et al. present experimental data.

Table 5.1: Various data sources as well as the data which they contain.

Data name	Year	Source	Half-life	Emission probability	Concentration
IAEA database	2018	[49]	Yes	Yes	No
JEFF 3.3	2017	[137]	Yes	Yes	Yes
ENDF/B-VIII.0	2018	[29]	Yes	Yes	Yes
JENDL-5	2021	[82]	Yes	Yes	Yes
ENSDF	2025	[54]	Yes	Yes	No
Audi et al.	2003	[8]	Yes	Yes	No
Pfeiffer and Kratz	2002	[134]	Yes	Yes	No
Rudstam et al.	1993	[153]	Yes	Yes	No

The expected outcome is that recently updated data and official databases will be more accurate than collections from the literature. Recent data should be more accurate due to improved equipment and measurement techniques. Official databases are expected to be more reliable, as they have many users reporting errors and have undergone extensive development over time. Investigating these data sources and comparing the DNP group parameters generated from each will be useful when analyzing uncertainty. If the differences are large, understanding the uncertainty propagation will be important. If the differences are negligible, this will simplify further uncertainty analysis.

5.1.3 Verification and Validation

Verifying and validating the DNP group parameters is essential to ensure their practical utility. Verification with the literature is important to ensure that the fundamental approach is working as expected. Since the literature does not contain DNP group parameters which include MSR physical phenomena, this verification with the literature will only be for the stationary solve.

After this, although the group parameters are verified with the literature, that does not necessarily mean they capture all physical phenomena of interest. Validation identifies potential weaknesses in the approach by highlighting discrepancies between predicted and experimental results.

Verification of the DNP group parameters will be of the same form as in Section 4.3. It has already been shown that these results should have a percent difference of 9% for thermal fission in ^{235}U for JEFF-3.1.1 data, so a similar percent difference can be expected for other datasets [112]. Results will vary based on the source of the data, but the DNP group parameters should closely match the results from the literature.

Validation of the DNP group parameters is conducted through the MSRE pump start-up and coast-down experimental data. This data assesses the accuracy of the group parameters in modeling reactivity changes from DNPs [139]. The validation requires modeling the MSRE core and upper and lower plena in Moltres [60]. The reactivity effects of the DNPs are calculated for both transients using static and improved DNP group parameters. The improved group parameters are expected to predict reactivity more accurately as they account for more physical phenomena in the MSRE. If neither set of reactivity responses aligns with the experimental results, then there is an issue in the kinetics simulation, as Moltres has previously demonstrated the ability to model these transients

[142, 128]. These transients have also been evaluated using other tools, providing additional comparison [56, 46]. If the static group parameters predict reactivity more accurately than the improved group parameters, this suggests an issue with the improved group parameter formulation. The issue must be resolved before proceeding, as the improved DNP group parameters should provide higher accuracy in MSR modeling.

5.2 Improved Group Spectra

The delayed neutron energy spectra, χ_d is important for accurate prediction of reactor kinetics [43]. This can be seen in the equation for the effective delayed neutron fraction in Equation (2.32), recreated here as:

$$\beta_{eff} = \frac{\int \Phi^\dagger \chi_d \beta \bar{v} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}{\int \Phi^\dagger \chi \bar{v} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}.$$

Because the DNP group yields and half-lives are affected by the MSR physical phenomena, it follows that the DNP group spectra are also affected.

The methodology to evaluate the delayed neutron spectra is discussed in Section 2.1.2. The DNP group spectra will be generated using the half-life grouping and fractional grouping methods. The DNP group spectra will then be compared with the spectra generated from a stationary problem with no chemical removal to determine the effect of MSR phenomena on the spectra.

5.3 Transient Simulations

Transient simulations are critical for validating the DNP group parameters using MSRE transients and for investigating transients affecting security, safety, and safeguards design decisions. Section 3.5 discusses the proposed simulations in detail. For safety design, the ULOF, UTOP, ULOHS, and FSOC are investigated for various reactor parameters to assess the impact of the improved DNP group parameters. For security design, sabotage-initiated transients are assumed to build off those studied for safety design. For safeguards design, comparisons are made between diversion scenarios with different PR times and material quantities removed. These diversion scenarios alter the fuel composition, affecting the effective delayed neutron fraction and reactor behavior during transients [190]. The improved DNP group parameters will be used to investigate the SNM removal which could remain undetected while using previous DNP group parameters.

The kinetics simulations will use Moltres and the MSRE to demonstrate how the DNP group parameters affect reactivity and temperature responses in various scenarios. These results can be compared with other analyses from the literature [73, 171]. The results will identify which transients are more sensitive to the improved data and highlight potential concerns. A PRK model can also be used for simple reactivity insertion tests with uncertainty tracking to more straightforwardly demonstrate differences between the DNP group parameters.

5.4 Future Work

5.4.1 Spatially Resolved Model

Although the 0D flow model provides useful insights, it does not provide a high level of fidelity for any specific geometry. A spatially resolved model will be developed to evaluate the accuracy of the 0D flow model. This model verifies whether the 0D flow model produces sufficiently accurate DNP group parameters. Section 3.2.3 discusses

the spatially resolved model in detail. This spatially resolved model will compare multiple levels of fidelity for generating DNP group parameters: 0D flow model using cumulative fission yields, 0D flow model with depletion, spatially resolved flow model using cumulative fission yields, and spatially resolved flow model with depletion. Each step increases computational cost, so if a simpler method generates DNP group parameters with negligible differences, it will be used preferentially.

The process for comparing the 0D flow model and the spatially resolved model is as follows: the spatially resolved model runs and generates DNP group parameters and residence times, the residence times are used to feed into the 0D flow model, the resulting DNP group parameters are compared between the two models. If the results are within the uncertainty bounds of each other, then this indicates that the 0D flow model only needs residence time data for a given MSR to generate the DNP group parameters. If they do not match, then a spatially resolved approach will be necessary to properly generate the MSR DNP group parameters.

Sensitivity to Number of DNPs

The number of DNPs to track is of particular interest for the spatially resolved flow model. The 0D flow model's computational cost increases negligibly with more DNPs, while the spatially resolved flow model requires significantly more memory and operations to track each DNP over space and time. To improve computational efficiency, minimizing the number of DNPs is desirable. If the 0D flow model generates DNP group parameters with negligible differences compared to the spatially resolved flow model, minimizing the number of DNPs becomes less critical. However, it is of interest to optimize the spatially resolved flow model for potential future analyses which could include spatially resolved reprocessing and other effects.

The impact of each DNP is given in Equation (2.27), replicated here:

$$\bar{v}_{d,i} = P_{n,i} \lambda_i N_{i,0}.$$

Dividing $\bar{v}_{d,i}$ by the total delayed neutron yield, $\bar{v}_d$, provides the relative contribution of each DNP to the total delayed neutron yield. The DNPs with the most impact can be modeled preferentially to determine the minimum number of DNPs to model with negligible changes to the DNP group parameters. A full sensitivity study will be conducted to determine a reasonable number of DNPs to include in the spatially resolved flow model.

Reactor Generalization

One of the desired outcomes for this work is formulation of DNP group parameters for MSRs which can be applied generally. If the DNP group parameters can be applied generally to different systems with similar residence times, then the amount of computational work required is drastically reduced. However, it is necessary that sufficient evidence is provided demonstrating that the improved parameters are identical for different MSRs.

Specifically, the MSRE and the MSFR will be simulated using the spatially resolved model [147, 28]. The models will be adjusted such that the in-core and ex-core residence times are identical for each. The result will be a set of group parameters for two distinct reactors but with the same residence time tabulations. Instead of calculating the neutron flux, both reactors will be given an incident neutron energy spectrum for irradiating the sample. The impact of the incident neutron energy spectra is discussed in Section 5.4.3, so differences caused by the spectra will not be studied in this section. Additionally, the fuel salt and fuel temperature in the MSFR will be replaced with those from the MSRE.

The two different reactor simulations will have all of the same input parameters aside from the geometry and thermal-hydraulic parameters. If the DNP group parameters have negligible differences, then the geometry

and thermal-hydraulic parameters are not necessary when generating the group parameters. Instead, all that is needed are the other input parameters, meaning the work does not need to be replicated for every different reactor geometry and thermal-hydraulic configuration. If the DNP group parameters have non-negligible differences, then each MSR needs to have DNP group parameters generated for its own specific geometry and flow conditions. Further investigation may also be warranted to determine if there is potential for a reduced order model in which one or two parameters which could be used to capture these differences, thus reducing the total amount of work necessary to model all MSR designs.

5.4.2 MSR Uncertainties

Several sources of uncertainty are considered, including half-lives, emission probabilities, cross-sections, and fission yields. These uncertainties propagate into the DNP group and kinetics parameters [141]. However, uncertainties also arise from MSR physical phenomena, including flow rate and reprocessing rates, where the reprocessing rate is correlated with the flow rate. These uncertainties are important because they propagate to the DNP group parameters, which affect the reactor's response to different transients. It is important to understand this for safety, security, and safeguards design, as the design should account for these uncertainties. In particular, uncertainties of interest are in the residence times and chemical removal rates.

The residence time correlates linearly with the flow rate, which is dependent on the temperature. Although the flow rate would also have uncertainty based on variance in the pump, this uncertainty is considered outside the scope of this work. The temperature affects fuel salt density, buoyant force, and viscosity, all influencing the flow rate and residence time. The temperature is influenced by the fission rate, which has uncertainties from the flux and fission cross section. For the 0D flow model, the residence time uncertainties can be conservatively approximated by treating the difference between the inlet and outlet temperatures as the temperature uncertainty. Assuming a constant mass flow rate, the density change due to the large temperature uncertainty provides a velocity uncertainty, which determines the variance in the residence time. If this conservative approach to temperature uncertainty results in negligible residence time uncertainty, it can be neglected for the spatially resolved flow model. Otherwise, the spatially resolved flow model can account for this uncertainty using the STOCHASTIC TOOLS module to directly determine residence time uncertainties. The STOCHASTIC TOOLS module is a toolbox designed for performing stochastic analysis for MOOSE-based applications. The STOCHASTIC TOOLS module enables setting a range of temperatures and simulating each temperature, which can be used to generate a histogram of residence times as a function of core temperature.

The chemical removal uncertainty is challenging to treat, as there are different chemical processes for removing elements, such as sparging the fuel salt in the core, filtering elements with a nickel filter, and removing elements via liquid-liquid reductive extraction [156]. The removal efficiency of xenon in a gas separation system is given by [130, 156] as:

$$\epsilon = \frac{1 - e^{-k_L a A_C L \left(\frac{1}{Q_{salt}} + \frac{RT}{H} \frac{1}{Q_{He}} \right)}}{1 + \frac{RT}{H} \frac{Q_{salt}}{Q_{He}}}, \quad (5.2)$$

where,

R = the universal gas constant [$L \cdot Pa \cdot mol^{-1} \cdot K^{-1}$]

T = the salt temperature [K]

H = Henry's law constant for solute gas [$Pa \cdot mol^{-1} \cdot L$]

Q_{salt} = the volumetric salt flow rate [$m^3 \cdot s^{-1}$]

Q_{He} = the volumetric helium flow rate [$m^3 \cdot s^{-1}$]

k_L = the liquid phase mass transfer coefficient [$m \cdot s^{-1}$]

a = the gas-liquid interfacial area per unit volume [m^{-1}]

A_C = the contactor cross-section [m^2]

L = the contractor length [m].

The liquid-liquid reductive extraction, such as with bismuth, has distribution coefficients and equilibrium constants dependent on the element, temperature, and concentration [119]. In general, an increase in the temperature leads to an increased distribution coefficient and a decreased equilibrium constant. The distribution coefficient represents the ratio of the bismuth phase concentration to the salt phase concentration, and can be thought of as an efficiency term that is multiplied with the mass flow rate to determine the removal rate. However, the distribution coefficient changes with temperature and flow rate differently for each element and fuel salt used.

Propagating the uncertainty in the chemical removal of elements from the fuel salt is challenging, as there are many variables to consider. These include the fuel salt composition, flow rate, redox potential, hydrostatic pressure, interfacial tension, and temperature [165, 180]. Thus, generic equations are not formulated for each possible combination of variables. Instead, only the chemical removal of noble gases and noble metals from the MSRE are investigated [106]. The noble gases are actively pumped out via the off-gas system while noble metals, including molybdenum, technetium, ruthenium, rhodium, palladium, silver, and antimony plate out via reduction by UF_6 . Some elements, specifically selenium, niobium, and tellurium, vary in their behavior. These elements only sometimes plates out based on the redox potential of the salt; a well reduced salt causes them to plate out, otherwise they act as salt seekers [85, 106].

The removal rate, λ_r , of noble gases from the pump bowl via the off-gas system is [84, 106]:

$$\lambda_r = \frac{\dot{V}}{V} \epsilon, \quad (5.3)$$

where,

$\dot{V}$ = the salt flow rate through the gas-liquid separator loop [$m^3 \cdot s^{-1}$]

V = the total volume of salt in the primary loop [m^3]

ϵ = the removal efficiency [-].

For the MSRE, the values are a salt flow rate of 0.003153, a total salt volume of 1.996, and a removal efficiency of 0.026 [147]. For the 0D flow model, this removal occurs while the sample is in the ex-core region and stops while the sample is in the in-core region. This is because the removal occurs in the pump bowl and not inside the core. For the spatially resolved flow model, this removal also occurs in the ex-core region, treated in the same way as the 0D flow model.

The removal rate, $\lambda_{i,l,r}$, of noble metal i to waste stream l via plate-out is [106]:

$$\lambda_{i,l,r} = \frac{h_{i,l}A_l}{V}, \quad (5.4)$$

where,

$h_{i,l}$ = the mass transfer rate of element i to waste stream l [$m \cdot s^{-1}$]

A_l = the surface area of waste stream l [m^2].

The removal of noble metals does not occur in a single region, unlike the noble gas removal. Instead, there is a distribution of noble metals: 50% are in the hastalloy-N surfaces in the loop, 49% are in the ex-core region, and 1% are in the graphite surfaces in the in-core region [85]. Because 99% of the plate-out occurs in the ex-core region, the removal can be treated as occurring only in the ex-core region. The surface areas and mass transfer rates for the noble metals are given in Table 5.2, recreated from [106].

Table 5.2: Noble metal mass transfer rates, surface areas, and resulting removal rates (recreated from [106] with data from [85]).

Component	$h_{i,l}$ [$m \cdot s^{-1}$]	A_l [m^2]	$\lambda_{i,l,r}$ [s^{-1}]
HX	4.657×10^{-5}	2.93×10^1	6.827×10^{-4}
Fuel loop piping, core and pump volute	1.041×10^{-4}	6.60×10^0	3.441×10^{-4}
Core wall cooling annulus	4.318×10^{-5}	1.43×10^1	3.095×10^{-4}
Core graphite (exposed to salt)	5.334×10^{-6}	1.36×10^2	3.637×10^{-4}
Miscellaneous	-	-	1.695×10^{-4}
Bubbles	4.233×10^{-4}	3.205×10^1	6.7978×10^{-3}
Total	-	-	8.6674×10^{-3}

5.4.3 Incident Neutron Energy Spectra

The incident neutron energy spectrum directly affects the DNP group parameters [23]. However, it would be computationally expensive to generate new DNP group parameters for every unique incident neutron energy spectrum. Some coarser resolution is required, so it is of interest to determine a reasonable categorization of the DNP group parameters for different energy spectra.

This sensitivity study will investigate different levels of fidelity for organizing the energy spectra: fine, middling, coarse, and single-value spectra. The approach uses the 0D flow model, with changes only to the neutron source energy spectra. The fine, middling, and coarse spectra include the full incident neutron energy spectra but with different bin sizes. The single-value spectra include several categories: neutron energy classification, mean energy, and median energy. The neutron energy classification is provided in Table 5.3, recreated from [109]. The study will also examine hand-crafted spectra where the mean and median remain unchanged, but the spectrum itself varies.

Table 5.3: Classification of neutron energy categories by energy range from [109].

Category	Energy Range [MeV]
Cold	$< 3 \times 10^9$
Thermal	$3 \times 10^9 - 4 \times 10^7$
Epithermal	$4 \times 10^7 - 1 \times 10^4$
Intermediate	$1 \times 10^4 - 2 \times 10^{-1}$
Fast	$2 \times 10^{-1} - 1 \times 10^1$
Relativistic	$> 1 \times 10^1$

This sensitivity study will determine the necessary level of detail in the incident neutron energy spectrum for generating DNP group parameters. For example, this analysis can investigate if the spectral effects of refueling is sufficiently great to warrant updates to the improved DNP group parameters. Organizing by category or mean energy is expected to suffice for most cases. For cases where this approach is inadequate, a more robust method can be selectively applied.

Appendix A

Derivation of the Delayed Neutron Count Rate in the 0D Flow Model

This derivation begins with the Equations (AA.1) and (AA.2) provided by Wilson and England [192]:

$$R(t) = \sum_{k=1}^K \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}, \quad (\text{AA.1})$$

and,

$$\dot{n}_d(T, t) = \bar{v}_d \int_0^T F(\tau) R(T - \tau + t) d\tau, \quad (\text{AA.2})$$

where,

T = the length of the fission history [s]

t = the decay time from the end of the fission history [s]

$F(\tau)$ = the fission rate history [$\frac{\text{fissions}}{\text{s}}$].

Because the results are normalized to a single delayed neutron, the $\bar{v}_d$ term is divided out.

Although the saturation, or infinite, irradiation derivation makes use of assuming the time is infinite, due to the spatial dependence of this problem, the in-core and ex-core timing must be handled properly. For example, if the fission rate history uses a Fourier decomposition of a square wave, then it would not be possible to extract spatial information from this formulation. Specifically, it is important where in the primary loop the sample initially exists. If the sample is initially in the core inlet, then the final portion of the fission rate history would consist purely of decay. If the sample is initially in the center of the core, then the final portion of the fission rate history would consist of irradiation.

For this derivation, the sample is assumed to exist as a point at the core inlet. This means the full cycle consists of an in-core residence time of fission followed by an ex-core residence time of decay, after which the cycle repeats until the net irradiation time is met. The fission rate history is therefore given as:

$$\begin{aligned} \frac{F(\tau)}{F_0} = & H(\tau) - H(\tau - T) + \sum_{j=1}^J H(\tau - j \cdot \tau_{in} - j \cdot \tau_{ex}) \\ & - H(\tau - j \cdot \tau_{in} - (j - 1) \cdot \tau_{ex}), \end{aligned} \quad (\text{AA.3})$$

where,

H = the Heaviside function [-]

$$J = \left\lceil \frac{T}{\tau_{in} + \tau_{ex}} \right\rceil$$

= the minimum number of cycles to meet the irradiation time, T [-]

F_0 = the magnitude of the constant fission rate [$\frac{fissions}{s}$].

Plugging this into Equation (AA.2) for group k yields:

$$\begin{aligned} \dot{n}_{d,k}(T, t) = & \int_0^T H(\tau) \bar{v}_{d,k} \lambda_k e^{-\lambda_k(T-\tau+t)} d\tau - \int_0^T H(\tau - T) \bar{v}_{d,k} \lambda_k e^{-\lambda_k(T-\tau+t)} d\tau \\ & + \int_0^T \sum_{j=1}^J H(\tau - j \cdot \tau_{in} - j \cdot \tau_{ex}) \bar{v}_{d,k} \lambda_k e^{-\lambda_k(T-\tau+t)} d\tau \\ & - \int_0^T \sum_{j=1}^J H(\tau - j \cdot \tau_{in} - (j-1) \cdot \tau_{ex}) \bar{v}_{d,k} \lambda_k e^{-\lambda_k(T-\tau+t)} d\tau, \end{aligned} \quad (AA.4)$$

which can be simplified to:

$$\begin{aligned} \frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k} \lambda_k} = & \int_0^T H(\tau) e^{-\lambda_k(T-\tau+t)} d\tau - \int_0^T H(\tau - T) e^{-\lambda_k(T-\tau+t)} d\tau \\ & + \int_0^T \sum_{j=1}^J H(\tau - j \cdot \tau_{in} - j \cdot \tau_{ex}) e^{-\lambda_k(T-\tau+t)} d\tau \\ & - \int_0^T \sum_{j=1}^J H(\tau - j \cdot \tau_{in} - (j-1) \cdot \tau_{ex}) e^{-\lambda_k(T-\tau+t)} d\tau \end{aligned} \quad (AA.5)$$

Solving the integrals yields:

$$\begin{aligned} \frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k} \lambda_k} = & \frac{1}{\lambda_k} \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T)} \right) + 0 \\ & + \sum_{j=1}^J \frac{H(T - (j \cdot \tau_{in} + j \cdot \tau_{ex}))}{\lambda_k} \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T-j \cdot \tau_{in}-j \cdot \tau_{ex})} \right) \\ & - \sum_{j=1}^J \frac{H(T - (j \cdot \tau_{in} + (j-1) \cdot \tau_{ex}))}{\lambda_k} \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T-j \cdot \tau_{in}-(j-1) \cdot \tau_{ex})} \right). \end{aligned} \quad (AA.6)$$

The Heaviside functions in the summations resolve to unity while the cycle is within range of the net irradiation time, and become 0 when the cycle time goes beyond the net irradiation time. The problem then simplifies to:

$$\begin{aligned} \frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k}} = & \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T)} \right) \\ & + \sum_{j=1}^J \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T-j \cdot \tau_{in}-j \cdot \tau_{ex})} \right) \\ & - \sum_{j=1}^J \left(e^{-\lambda_k t} - e^{-\lambda_k(t+T-j \cdot \tau_{in}-(j-1) \cdot \tau_{ex})} \right). \end{aligned} \quad (AA.7)$$

If the net irradiation time is assumed to be infinite, then the delayed neutron count rate becomes:

$$\dot{n}_{d,k}(T, t) = F_0 \bar{v}_{d,k} e^{-\lambda_k t}, \quad (\text{AA.8})$$

which matches the result from Wilson and England [192]. However, as previously mentioned, this infinite approach will not work for the spatially dependent analysis. For example, if an in-core residence time is selected which is relatively small compared to the ex-core residence time, Equation (AA.8) would not vary at all, even though the fission rate history would be significantly different. Instead, the problem takes on the form:

$$\frac{\dot{n}_{d,k}(T, t)}{F_0 \bar{v}_{d,k} e^{-\lambda_k t}} = 1 - e^{-\lambda_k T} + \left(1 - e^{\lambda_k \tau_{ex}}\right) \sum_{j=1}^J e^{-\lambda_k (T - j \cdot \tau_{in} - (j-1) \cdot \tau_{ex})}. \quad (\text{AA.9})$$

By isolating the delayed neutron count rate and summing over all groups, the finalized equation can be formulated as:

$$\dot{n}_d(T, t) = F_0 \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t} - \bar{v}_{d,k} e^{-\lambda_k t} e^{-\lambda_k T} \quad (\text{AA.10})$$

$$+ \bar{v}_{d,k} e^{-\lambda_k t} \left(1 - e^{\lambda_k \tau_{ex}}\right) \sum_{j=1}^J e^{-\lambda_k (j \cdot \tau_{in} + (j-1) \cdot \tau_{ex} - T)}, \quad (\text{AA.11})$$

which can be rewritten as:

$$\dot{n}_d(T, t) = F_0 \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t} \left(1 - e^{-\lambda_k T} + \left(1 - e^{\lambda_k \tau_{ex}}\right) \sum_{j=1}^J e^{-\lambda_k (T - j \cdot \tau_{in} - (j-1) \cdot \tau_{ex})}\right). \quad (\text{AA.12})$$

If the in-core residence time is zero, then the delayed neutron count rate goes to zero, which is the expected behavior.

Appendix B

Derivation of the Equilibrium Concentration from the Cumulative Fission Yield

This derivation begins with the Bateman equation, given in Equation (2.55) and recreated here as:

$$\frac{dN_j}{dt} = \sum_{i \neq j} [(\gamma_{i \rightarrow j} \sigma_{f,i} \phi + \lambda_{i \rightarrow j} + \sigma_{i \rightarrow j} \phi) N_i] - (\lambda_j + \sigma_j \phi + \lambda_{r,j}) N_j + \dot{N}_{f,j},$$

where,

- N_j = the atomic density of isotope j $\left[\frac{\text{atoms}}{\text{cm}^3} \right]$
- $\gamma_{i \rightarrow j}$ = the fractional fission product yield of j from the fission of isotope i
- $\sigma_{f,i}$ = the microscopic fission cross-section of isotope i $[\text{cm}^2]$
- ϕ = the spectrum-averaged scalar flux in the fuel region $\left[\frac{\text{neutrons}}{\text{cm}^2 \text{s}} \right]$
- $\lambda_{i \rightarrow j}$ = the decay constant of decay $i \rightarrow j$ $[\text{s}^{-1}]$
- $\sigma_{i \rightarrow j}$ = the microscopic transmutation cross-section of reaction $i \rightarrow j$ $[\text{cm}^2]$
- N_i = the atomic density of isotope i $\left[\frac{\text{atoms}}{\text{cm}^3} \right]$
- λ_j = the decay constant of isotope j $[\text{s}^{-1}]$
- $\lambda_{r,j}$ = the removal constant for removal of isotope j $[\text{s}^{-1}]$
- σ_j = the microscopic total transmutation cross-section of isotope j $[\text{cm}^2]$
- $\dot{N}_{f,j}$ = the feed rate for isotope j $\left[\frac{\text{atoms}}{\text{cm}^3 \cdot \text{s}} \right]$.

This can be rewritten to the simplified form given in Equation (2.58), recreated here as:

$$\frac{dN_j}{dt} = S_j(t) - \lambda_{net,j} N_j(t),$$

where,

$$\begin{aligned}
 S_j &= \text{the source term for nuclide } j, \\
 &\quad \text{containing fission, decay, and transmutation sources } \left[\frac{\text{atoms}}{\text{cm}^3 \text{s}} \right] \\
 \lambda_{net,j} &= \text{the loss term for nuclide } j, \\
 &\quad \text{containing decay and transmutation losses } [\text{s}^{-1}].
 \end{aligned}$$

This can be rewritten using the scaled flux method, which is useful for flowing fuel reactors as it scales the flux term by the fraction of fuel salt in the core. This changes Equation (2.57) to Equation (BB.1), assuming chemical removal only in the ex-core region, given as:

$$\frac{dN_j}{dt} = f_{in}S_j(t) - \lambda_j N_j(t) - f_{ex}\lambda_{r,j}N_j(t), \quad (\text{BB.1})$$

where,

$$\begin{aligned}
 f_{in} &= \text{the fraction of fuel salt in the core} \\
 f_{ex} &= \text{the fraction of fuel salt not in the core} \\
 \lambda_{r,j} &= \text{the chemical removal rate of } j.
 \end{aligned}$$

The source term, S_j , can then be written using the cumulative fission yield as shown in Equation (BB.2). The cumulative fission yield is the total number of some nuclide formed per fission event, including decay chains that lead to that nuclide. Because the cumulative fission yield captures decays into the nuclide j , the $\lambda_{i \rightarrow j}$ term is not included. Additionally, parasitic absorption is neglected in this derivation, which leads to:

$$S_j(t) = CFY_j \phi \Sigma_f, \quad (\text{BB.2})$$

where,

$$CFY_i = \text{the cumulative fission yield of nuclide } i.$$

Finally, the fission reaction rate term, $\phi \Sigma_f$, is assumed to be equal to 1. Equation (BB.3) can be used to calculate the concentration as a function of time using cumulative fission yields, given as:

$$\frac{dN_j}{dt} = f_{in}CFY_j - \lambda_j N_j(t) - f_{ex}\lambda_{r,j}N_j(t) \quad (\text{BB.3})$$

Assuming a stationary fuel with no recirculation or chemical removal, the equation simplifies greatly. The solution to that equation, assuming no initial concentration of nuclide j , is given as:

$$N_j(t) = \frac{CFY_j}{\lambda_j} (1 - e^{-\lambda_j t}) \quad (\text{BB.4})$$

As previously mentioned, the cumulative fission yield gives the total amount of the nuclide generated including from decay chains, so the time dependent result is non-physical. For a saturation irradiation however, the sample is irradiated for a sufficiently long time such that all DNPs reach equilibrium. This means that the equation should be

taken as time approaches infinity, which leads to the form [79]:

$$N_j(t) = \frac{CFY_j}{\lambda_j}. \quad (\text{BB.5})$$

For the case of a nuclide in a circulating fuel reactor with a non-zero chemical removal rate, the concentration as a function of time comes out as:

$$N_j(t) = \frac{f_{in}CFY_j}{f_{ex}\lambda_{r,j} + \lambda_j} \left(1 - e^{-f_{ex}\lambda_{r,j}t - \lambda_j t} \right) \quad (\text{BB.6})$$

Equation (BB.6) can also be taken as time goes to infinity, resulting in:

$$N_j(t_\infty) = \frac{f_{in}CFY_j}{f_{ex}\lambda_{r,j} + \lambda_j} \quad (\text{BB.7})$$

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